

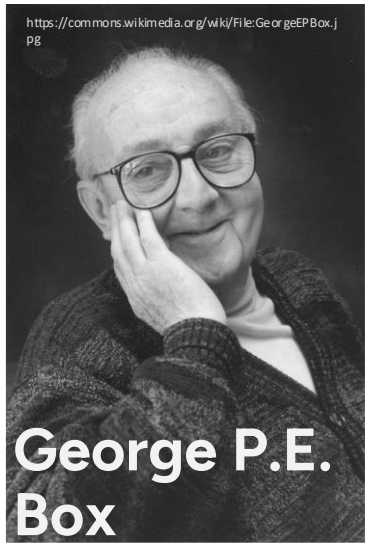
Lecture Series 1: An Introduction to Statistics for Medical Engineering CM2018

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L1-1 An Introduction to Statistics

The Iterative Learning Process



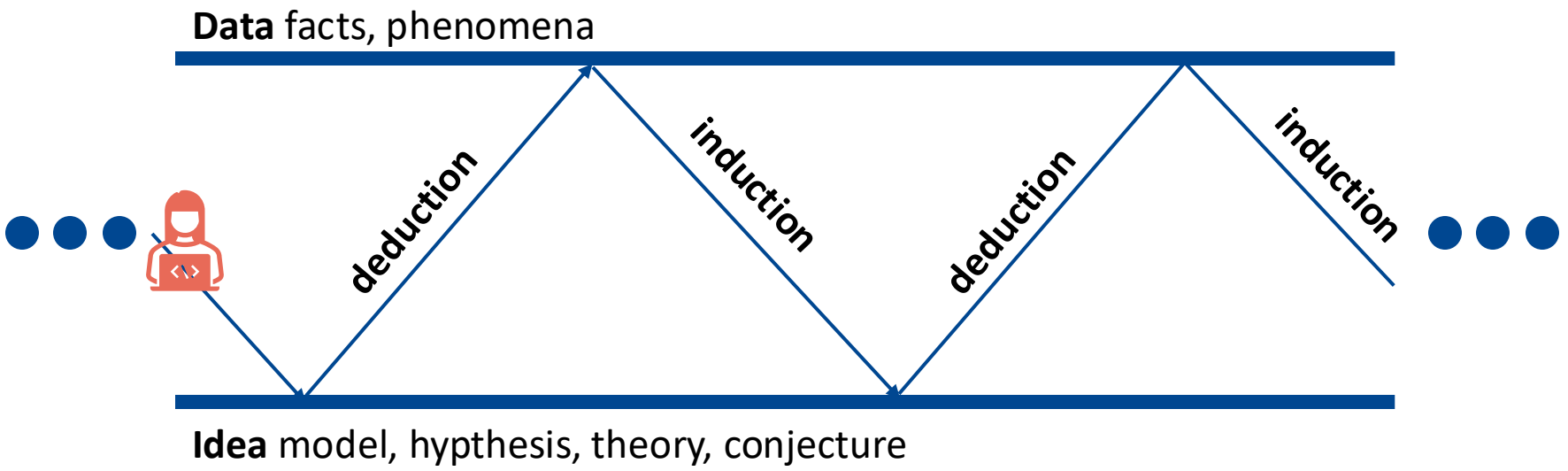


Figure adapted from Box GEP, *et al*. 2005

Iterative Problem Solving

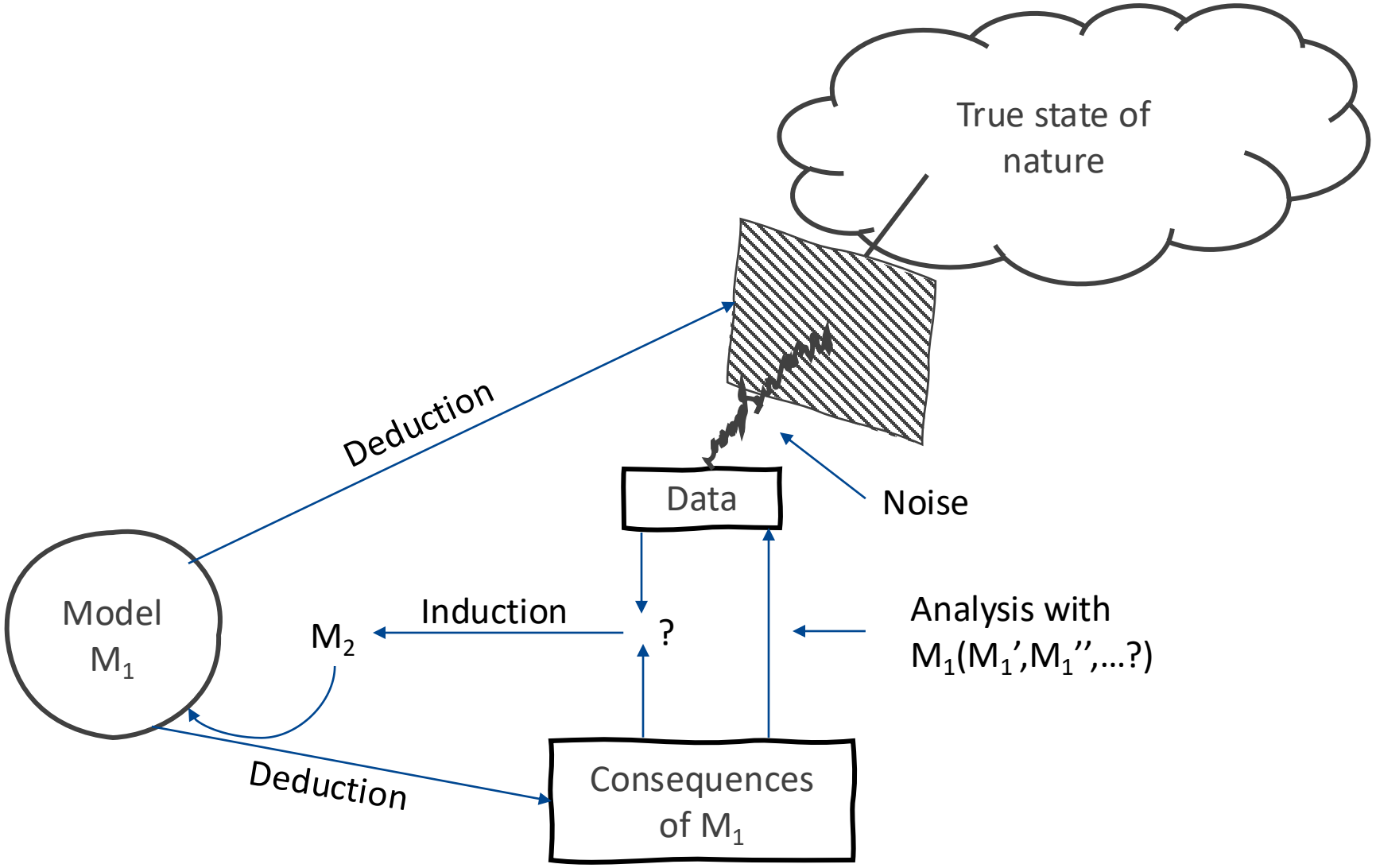
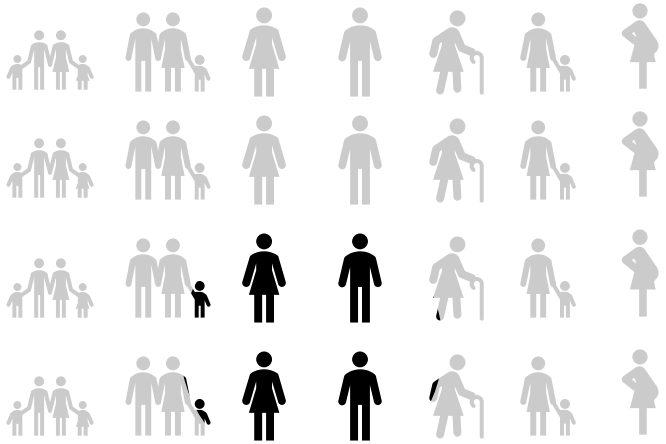
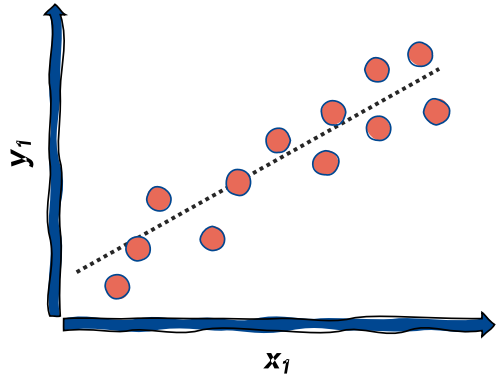


Figure adapted from Box GEP, *et al*. 2005

Why do we need statistics?

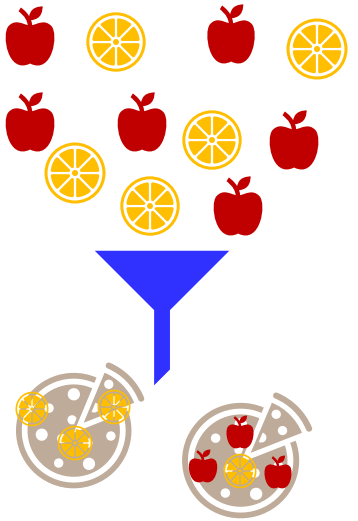


Make inference about a population based on a sample



Measure and control uncertainty

- Clarify question
- Identify relevant variables
- Disentangle evidence from a mix of influential factors
- Reduce the risk of mistaken conclusions from research data

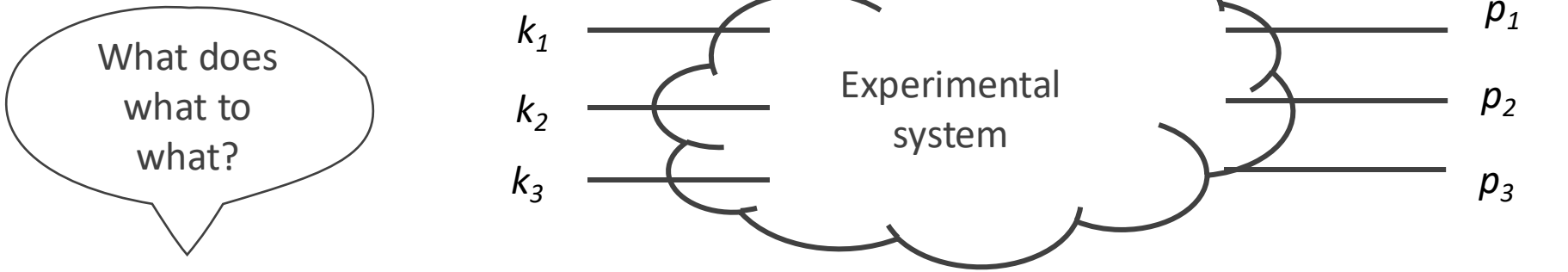


Limitations

- Will not eliminate uncertainty
- Will not provide credible conclusions from poor data.
- Onus on user to apply credible analysis on credible data.

Hiffenburgh 2020 Statistics in Medicine

The experimenter's problem and statistical methods
complexity



n entities $= k_3 \times p_3$

some model function of $y \leftarrow f(x)$

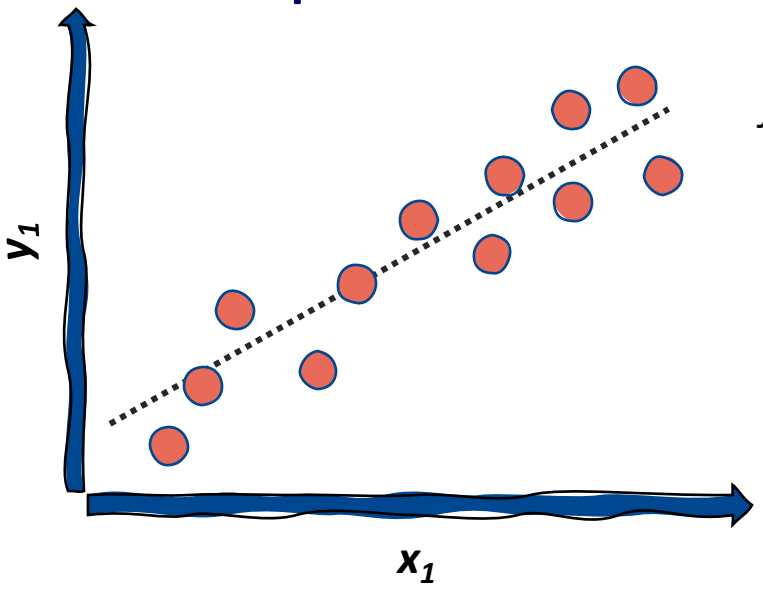
$$\hat{y}_1 = \beta_{1,1}\, x_1$$

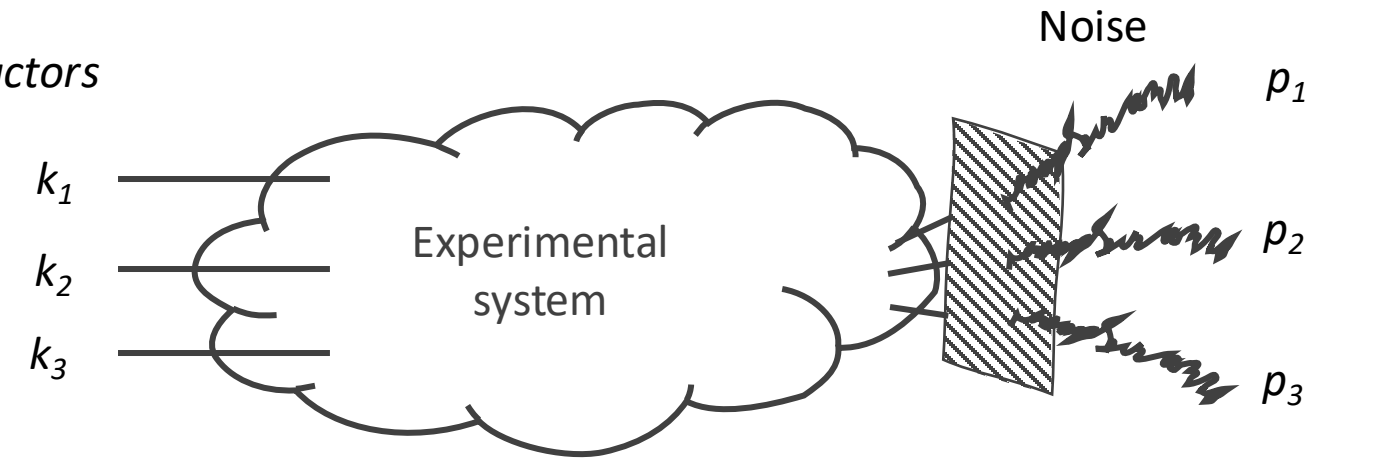
$$\hat{y}_2 = \beta_{2,1}\, x_1 + \beta_{2,3}\, x_3$$

β coefficient

Box GEP, *et al*. 2005

The experimenter's problem and statistical methods
experimental error





n entities = $k_3 \times p_3$

some model function of $y \leftarrow f(x)$

$$y_1 \sim \beta_{1,1}\, x_1 + \mathcal{E}$$

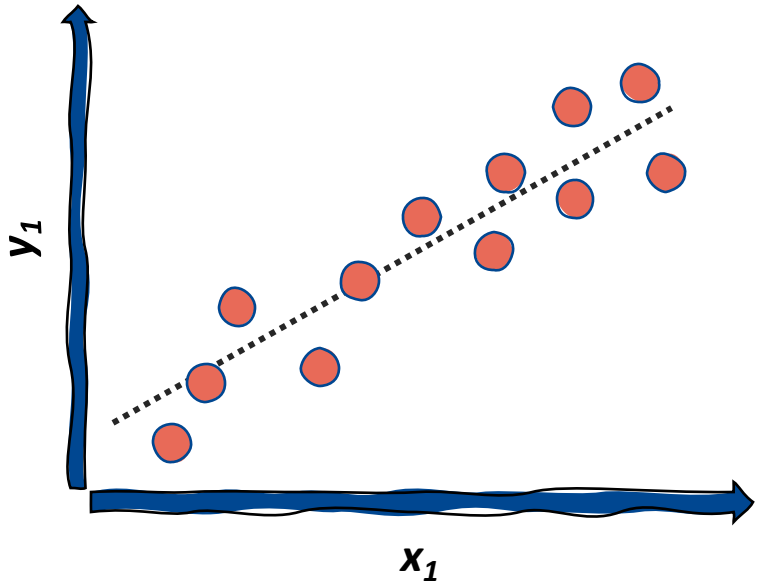
$$y_2 \sim \beta_{2,1}\, x_1 + \beta_{2,3}\, x_3 + \mathcal{E}$$

β *coefficient*

Box GEP, *et al*. 2005

The experimenter's problem and statistical methods
confusion of correlation with causation

What does
what to what?



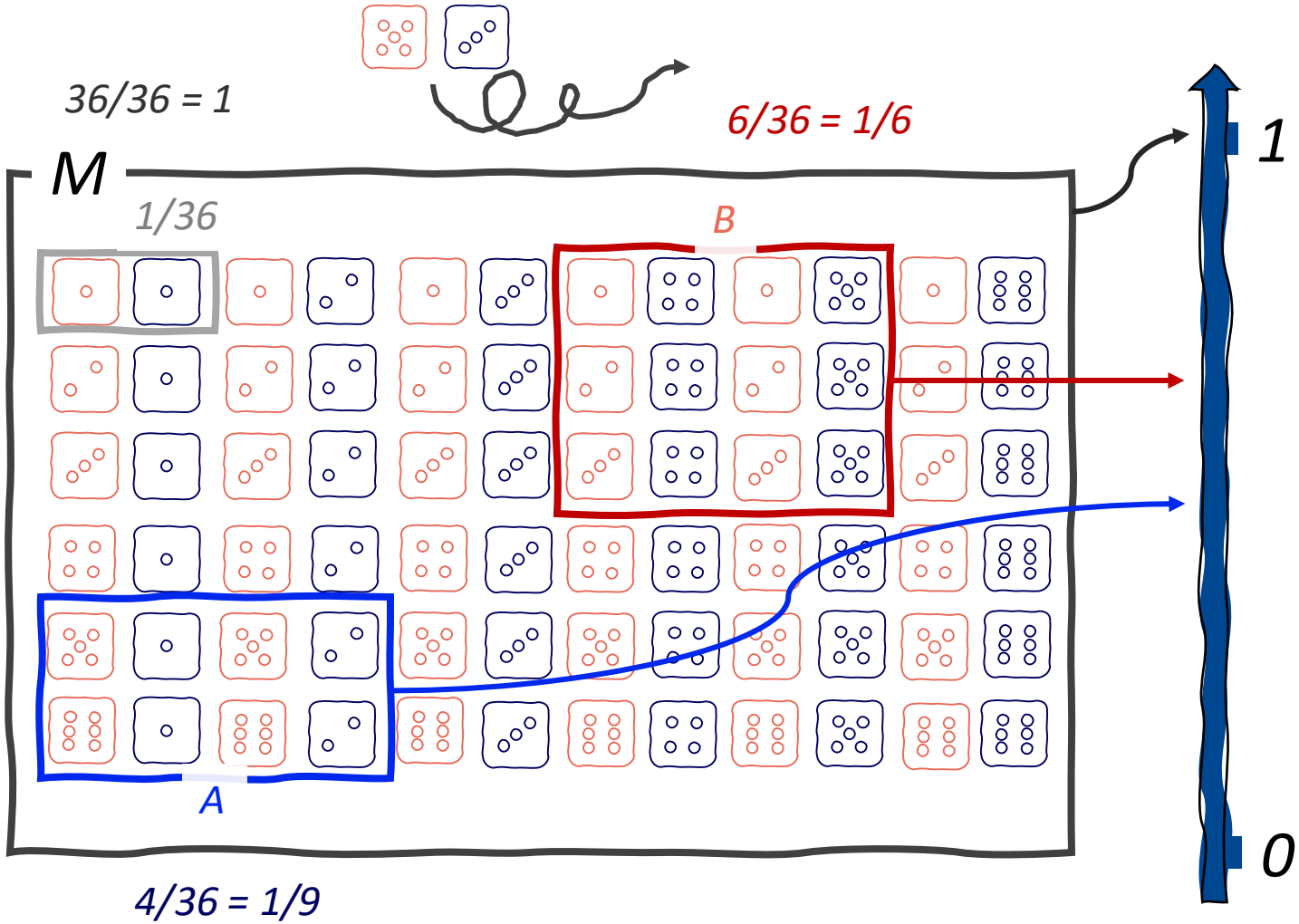
$$y_1 \sim \beta_1 x_1 + \mathcal{E}$$

$$y_1 \sim \beta_{y1,w} w + \mathcal{E}$$

$$x_1 \sim \beta_{x1,w} w + \mathcal{E}$$

Box GEP, *et al*. 2005

What is statistics about? ways of handling uncertainty, or probability (discrete probabilities)



$$P(M) = 1$$

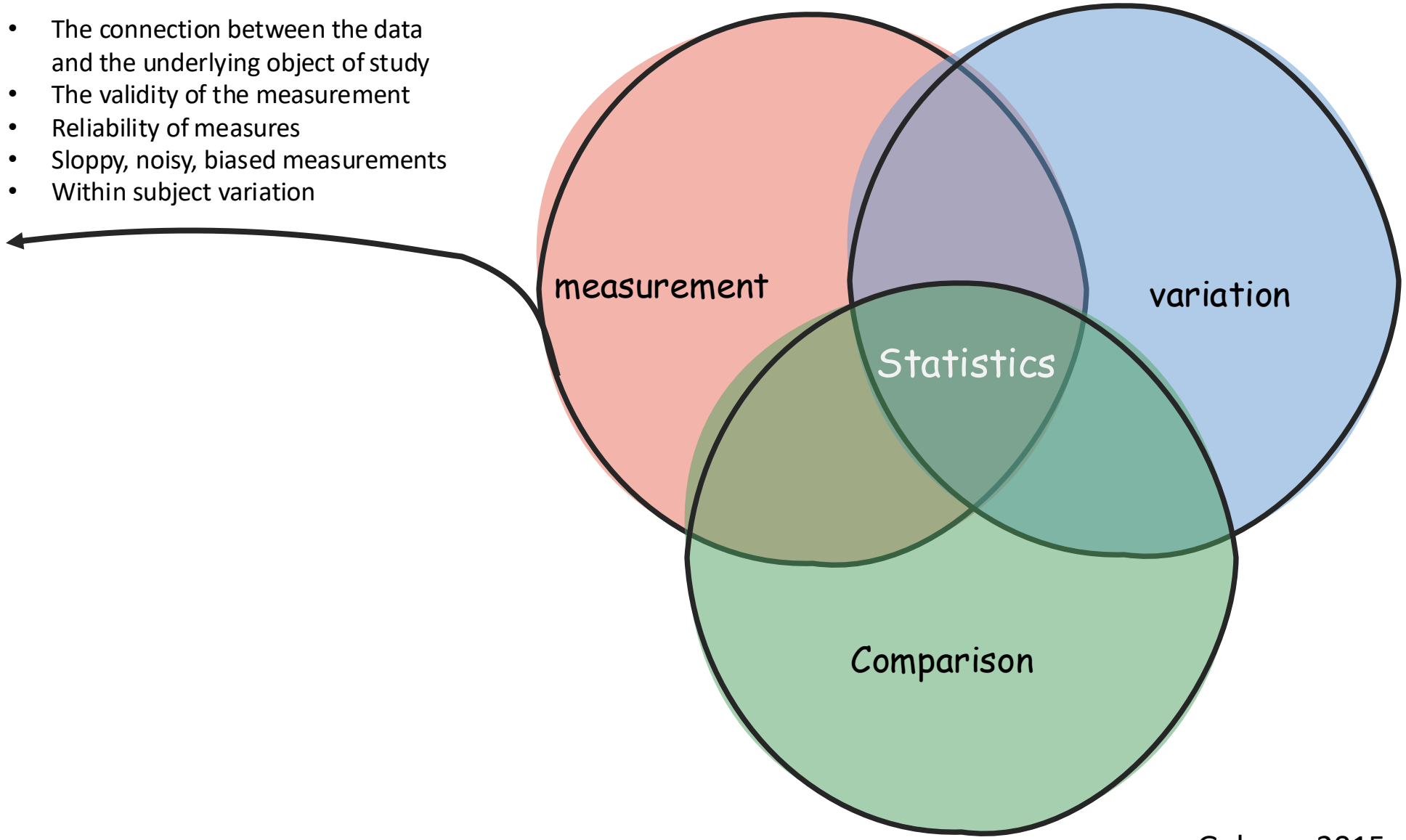
$$P(\emptyset) = 0$$

$$P(A \cup B) = P(A) + P(B),$$
$$if\ A \cap B = \emptyset$$

Lavin 2013

Why do we need statistics?

- The connection between the data and the underlying object of study
- The validity of the measurement
- Reliability of measures
- Sloppy, noisy, biased measurements
- Within subject variation



Gelman 2015

How to use statistics

Find out as much as you can about the problem

What are we investigating?
Do we have prior data?
How was the data collected?
How were responses measured?
How much theory is there on the system or phenomenon?

Don't forget about nonstatistical knowledge

"Statistical techniques are useless unless combined with appropriate subject matter knowledge and experience"

Define objectives

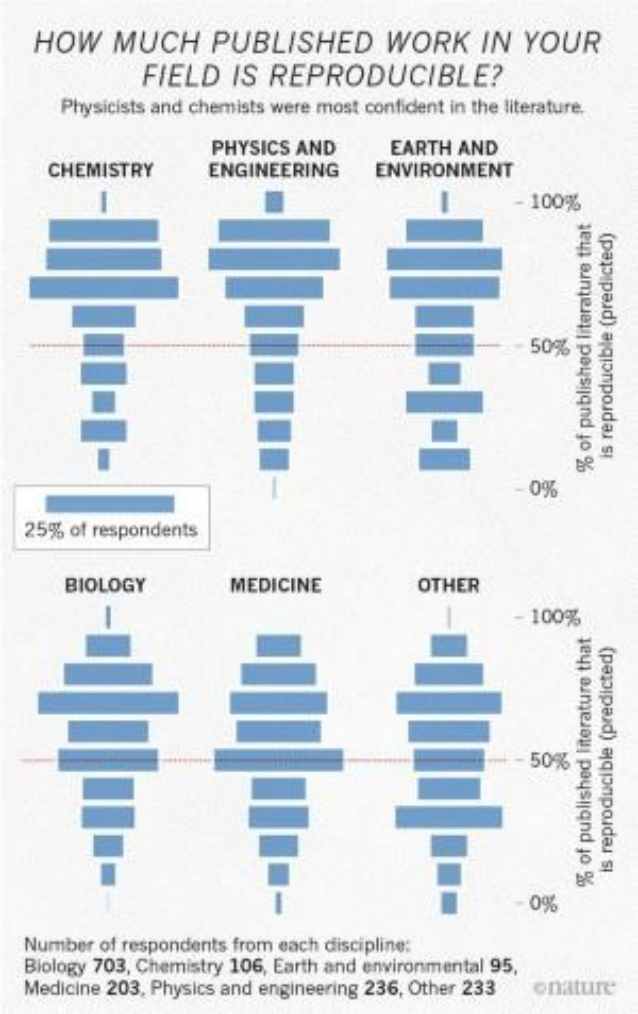
(i) clearly define an objective,
(ii) all involved parties should agree,
(iii) necessary resources,
(iv) agree on criteria when objectives have been met,
(v) all are involved if objectives change.

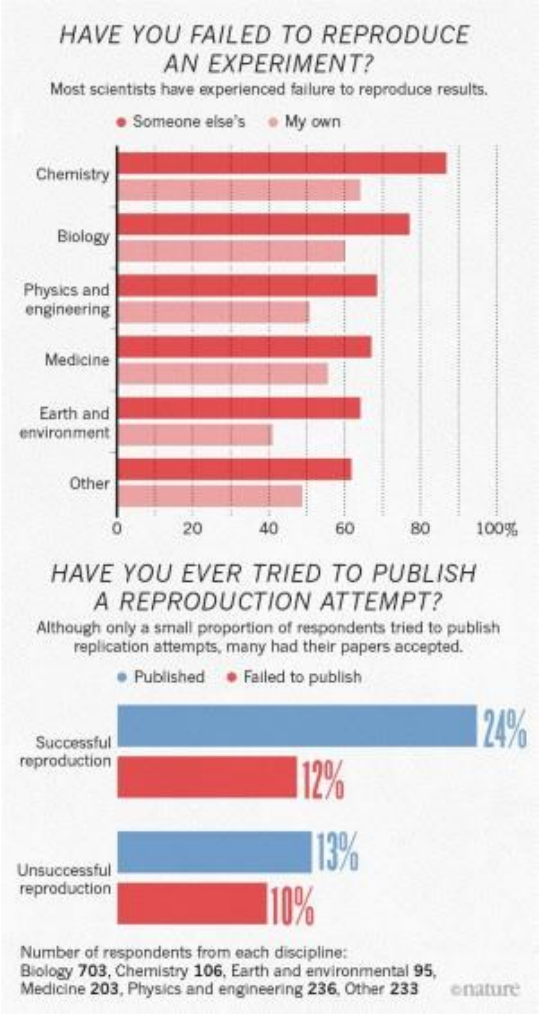
Learn from each other

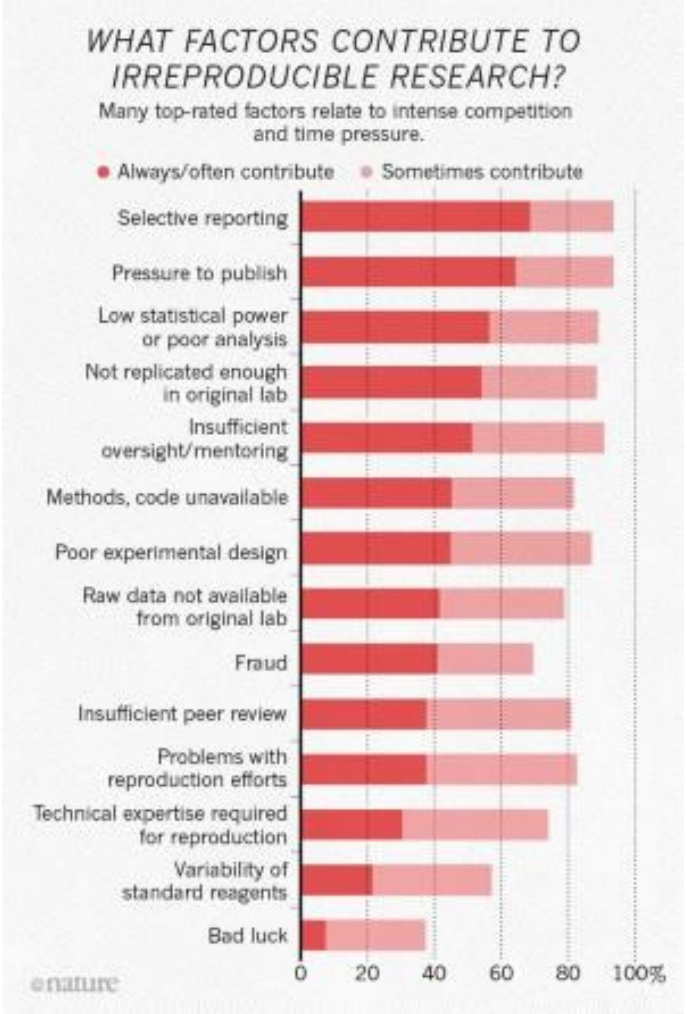
Experimenters, statisticians and data modellers learn from each other through collaboration, leading to novel approaches.

Box GEP, *et al*. 2005

The Replication Crisis







Baker 2016; Forbes *et al*. 2023

The Confirmatory Data Analysis (CDA) and Exploratory Data Analysis (EDA) Continuum

Aim: confirm or refute hypothesised patterns.

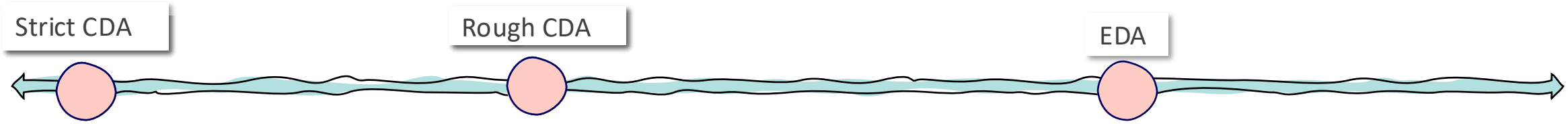
Methods: t-tests, ANOVAs, tests of regression coefficients, chi-square test. simple or multiple regression. GLM, mixed-effects, SEM (latter methods more relevant to rough CDA). Bayesian equivalents. State methods beforehand.

Aim: Validate hypothesised patterns.

Methods: Focuses on estimation rather than precise hypothesis testing. Measures of uncertainty, estimates of parameters, effect size computation.

Aim: Detect patterns in data without a priori hypothesis.

Methods: Robust procedures, data transformations, residual analysis, smoothing procedures, graphical tools (scatter plots, boxplots, stem-and-leaf plots, multi-way dot plots); machine learning. CDA methods if user is transparent in reporting. Cannot use p-values and CIs for probabilistic interpretations.



Ethics, Rules, Assumptions for p-values and confidence intervals:
- model assumptions must be met.
- Specify intended power level and sample size in advance.
- Only test one hypothesis, or adjust for multiple hypotheses testing.

Strengths:
- Give weight to conclusions.
- Emphasises and requires replication.
- Translation of data to mechanism.

Limitations:
- Not flexible.
- P-values can be misinterpreted.
- Conforms to a null hypothesis framework.

Strengths:
- Flexibility allows discovery.
- Encourages replication.
- Stronger evidence than EDA because relationships were hypothesised in advance.

Limitations:
- Conclusions are more provisional and subject to CDA.
- Higher risk of bias.
- Translational mechanism between data and decision is weaker.
- Can yield conflicting conclusions (uncertainty).

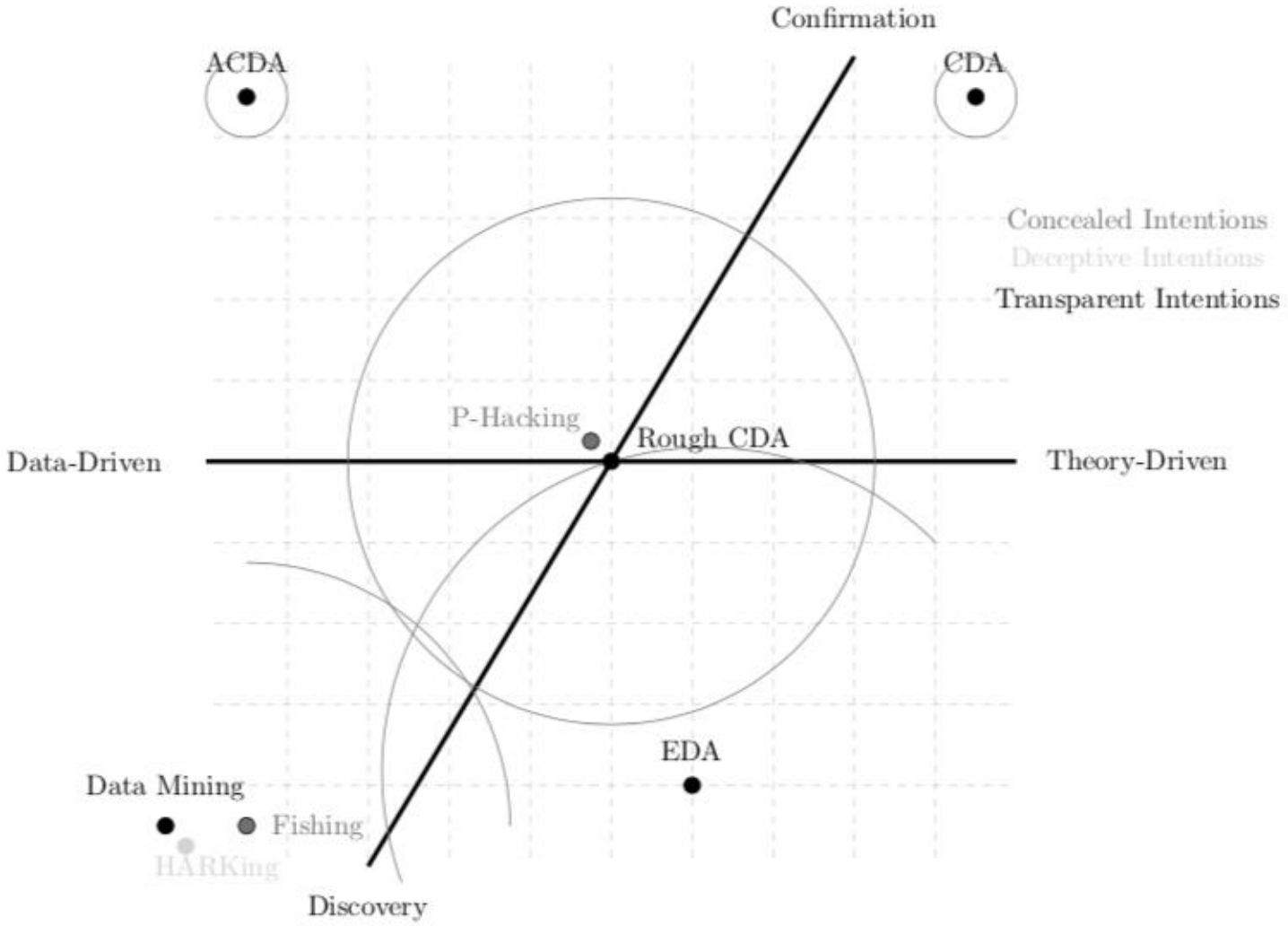
Ethics and Rules:
- Cannot do CDA on dataset that has been used for EDA.
- Cannot present EDA findings as CDA.
- All findings are provisional awaiting confirmation from independent data.

Strengths:
- Good for discovery recognising the uncertainty in findings.
- Makes full use of data.
- Can be used on historical datasets.

Limitations:
- Potentially too many options leading to analysis paralysis.
- Conclusions are provisional.
- No translational mechanism from data to conclusion.

Tukey 1980; Fife and Rodgers 2021

A CDA–EDA 3-D Space



ACDA: atheoretic CDA
HARKing: hypothesising after results are known

Fife and Rodgers 2021



L1-2 Probability

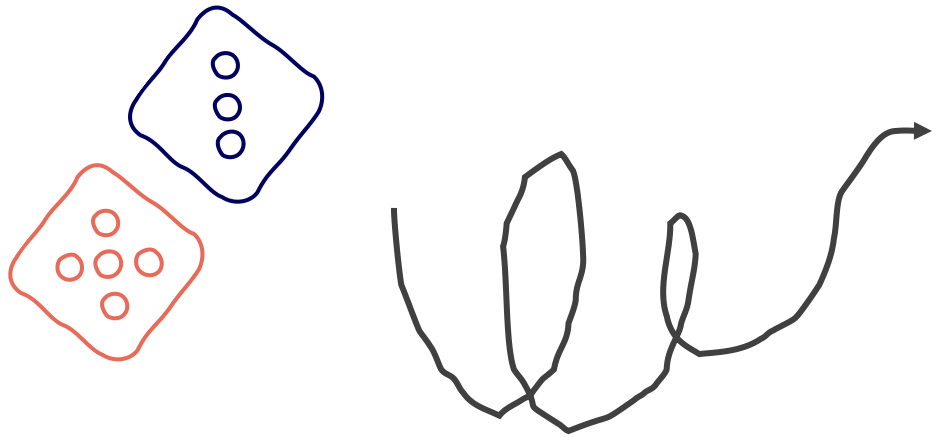
What is statistics about?

ways of handling uncertainty, or probability (discrete probabilities)

- experiment
- sample outcome (s)
- sample space (S)
- event

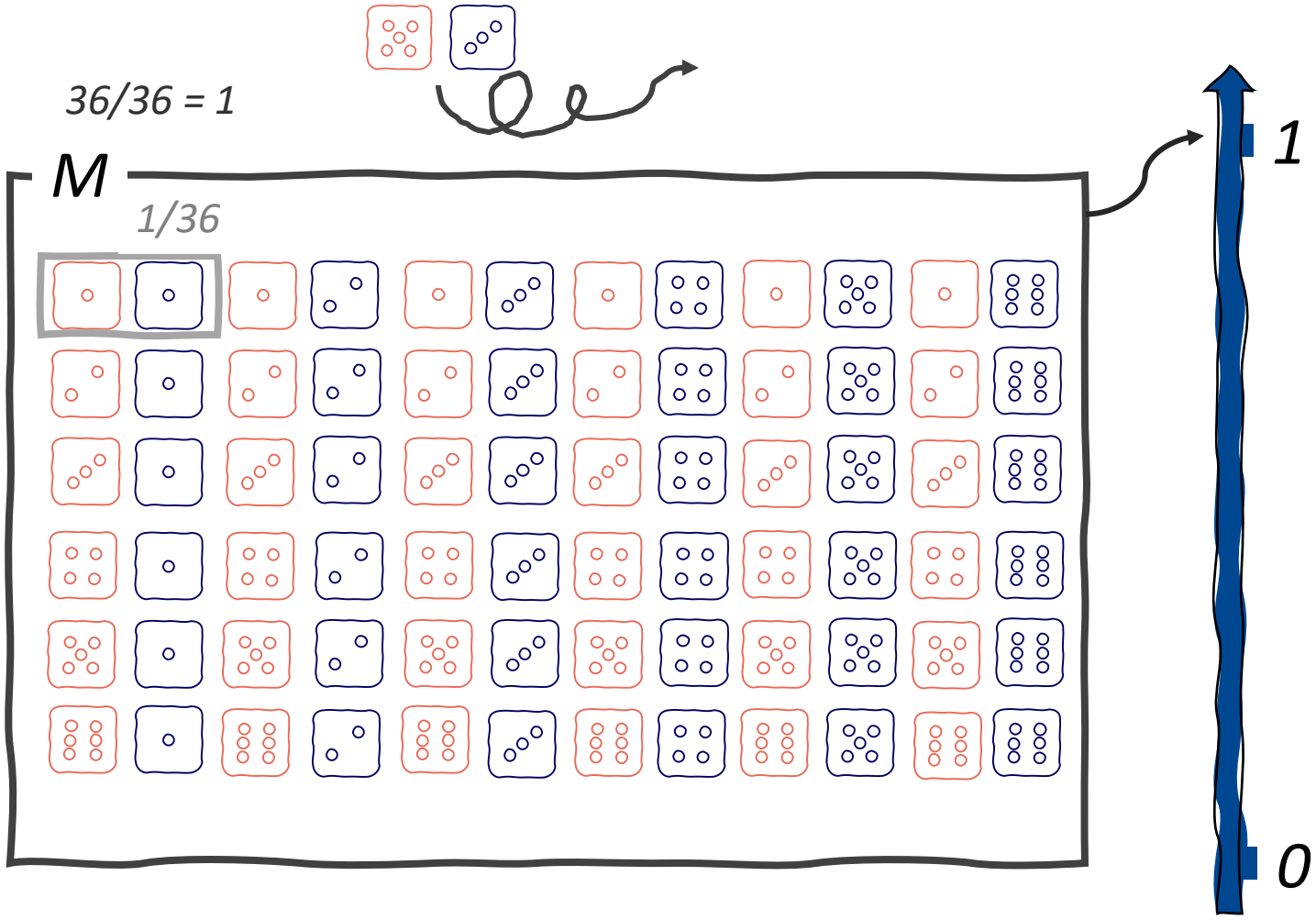
So, s is an element of S
Or, s is a member of S

$$s \in S$$



Lavin 2013; Larsen & Marx 2003

What is statistics about? ways of handling uncertainty, or probability (discrete probabilities)

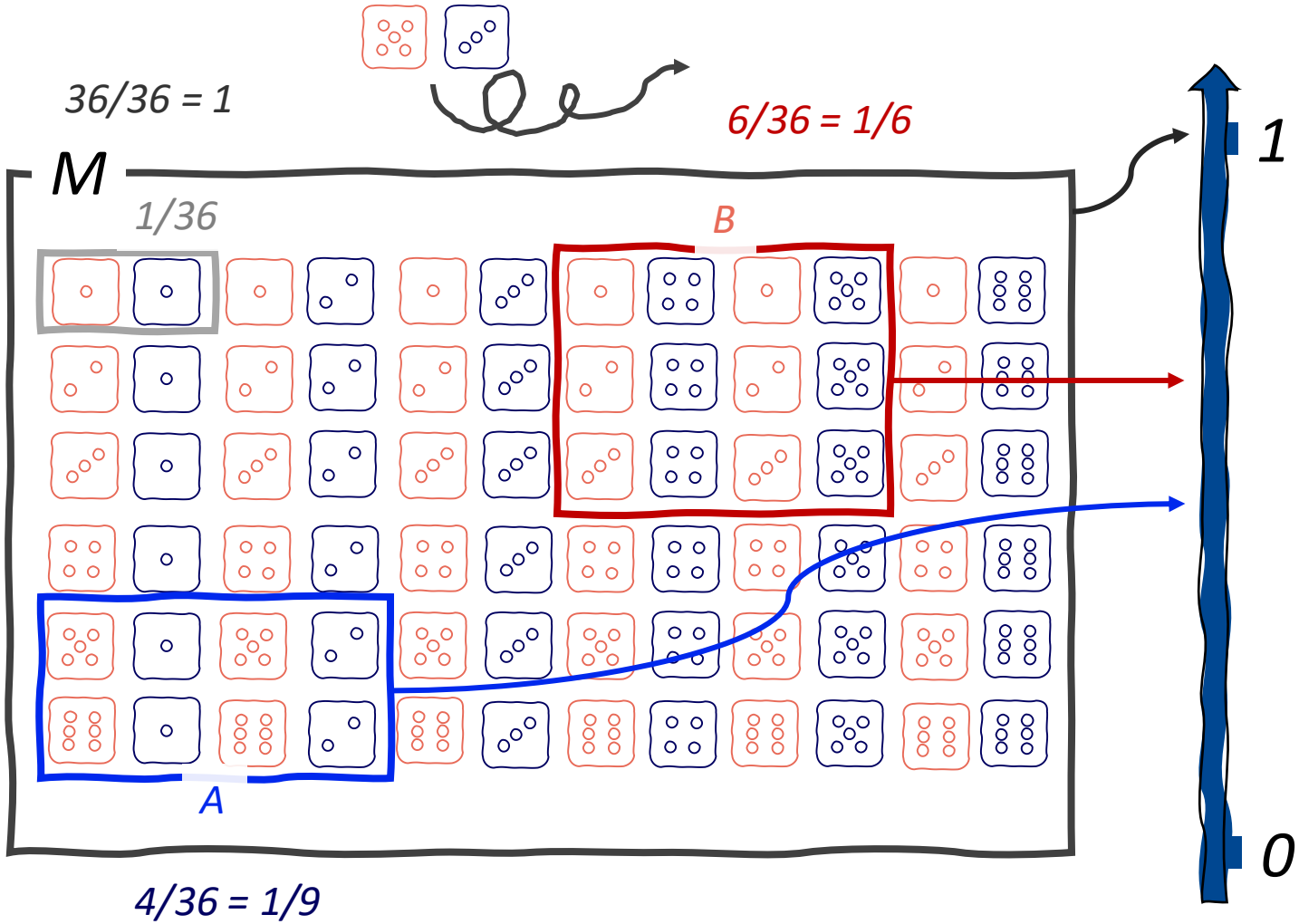


$$P(M) = 1$$

$$P(\emptyset) = 0$$

Lavin 2013

What is statistics about? ways of handling uncertainty, or probability (discrete probabilities)



$$P(M) = 1$$

$$P(\emptyset) = 0$$

$$P(A \cup B) = P(A) + P(B),$$
$$if\ A \cap B = \emptyset$$

Lavin 2013

Continuous probabilities

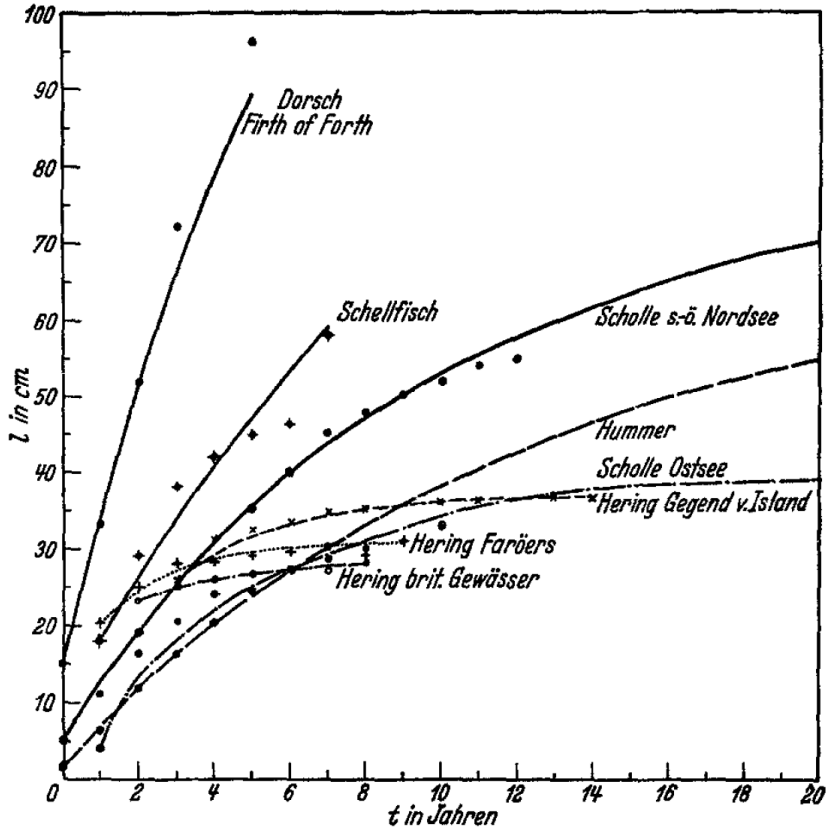
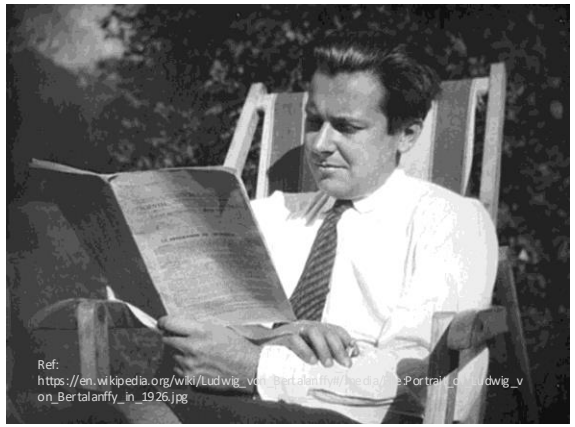


Abb. 4. Längenwachstum von Meerestieren, nach Angaben von PÜTTER (1920), die berechneten Kurven ausgezogen, mit Bezeichnung der empirischen Bestimmungen. (Einige mit anderen zusammenfallende Bestimmungspunkte wurden weggelassen.)

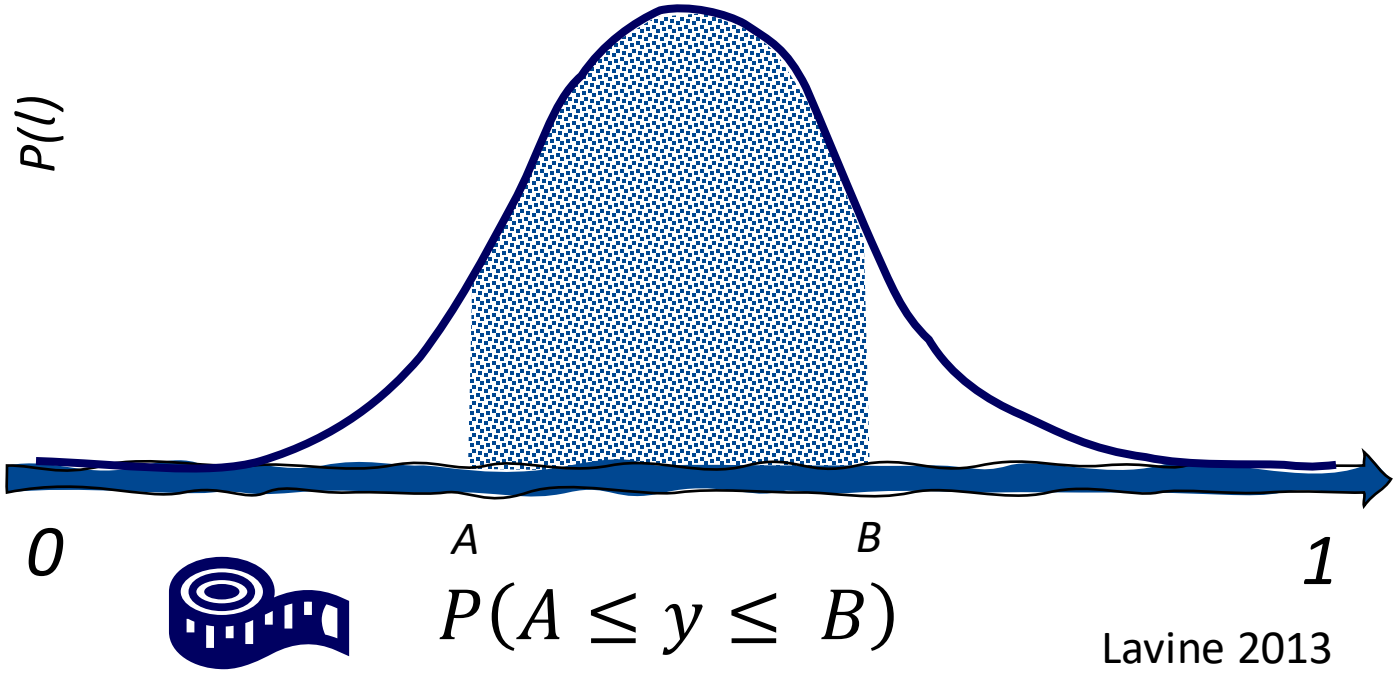


Ref: https://en.wikipedia.org/wiki/Ludwig_von_Bertalanffy#/media/File:Portrait_of_Ludwig_von_Bertalanffy_in_1926.jpg

Von Bertalanffy 1934

Continuous probabilities

Probability density function (pdf):
derivative of probabilities



Lavine 2013

$$\frac{d}{db} P[X \in (a,b]] = \frac{d}{db} \int_a^b f_X(x)dx = f_X(b)$$

Plain text explanation of pdf: The probability or relative likelihood of the random variable at a given value.

Properties of the pdf

$$P(y) \geq 0 \; for \; all \; y$$

$$if \; p(y) < 0 \; on \; interval \; (a,b):$$

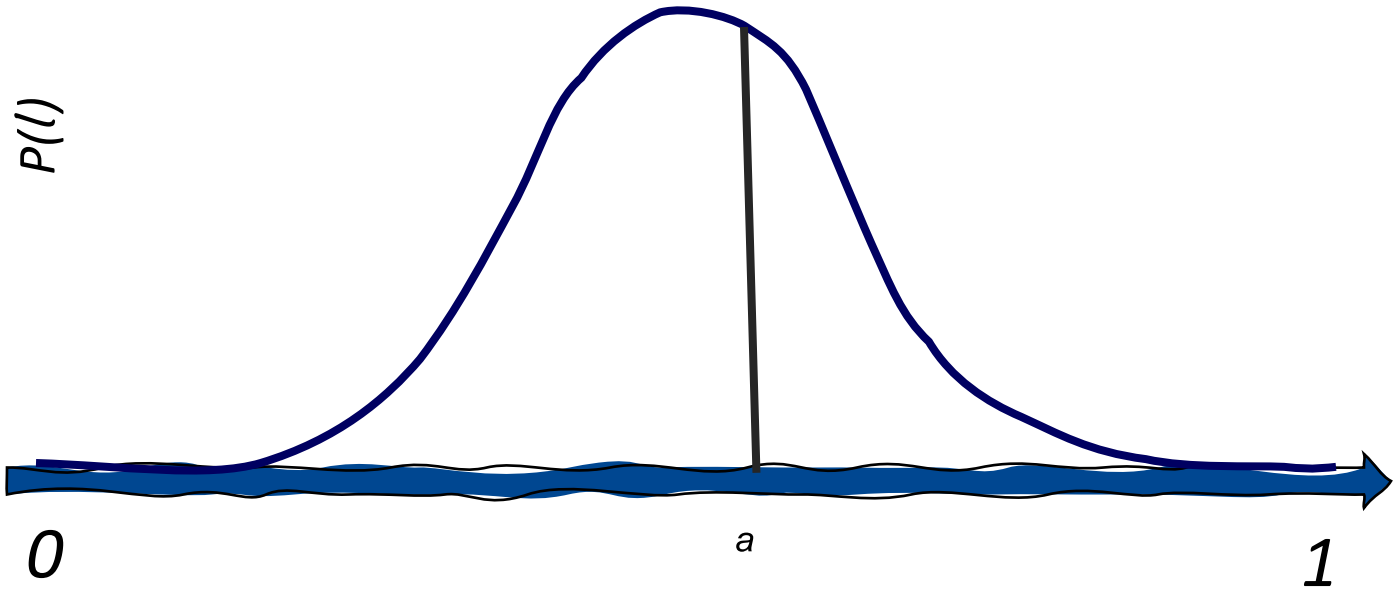
$$P[Y \in (a,b)] = \int_a^b p(y)dy < 0$$

$$\int_{-\infty}^{\infty} P(y)dy = 1$$

$$P[Y \in (-\infty,\infty)] = \int_{-\infty}^{\infty} p(y)dy = 1$$

Continuous probabilities

Probability density function (pdf)



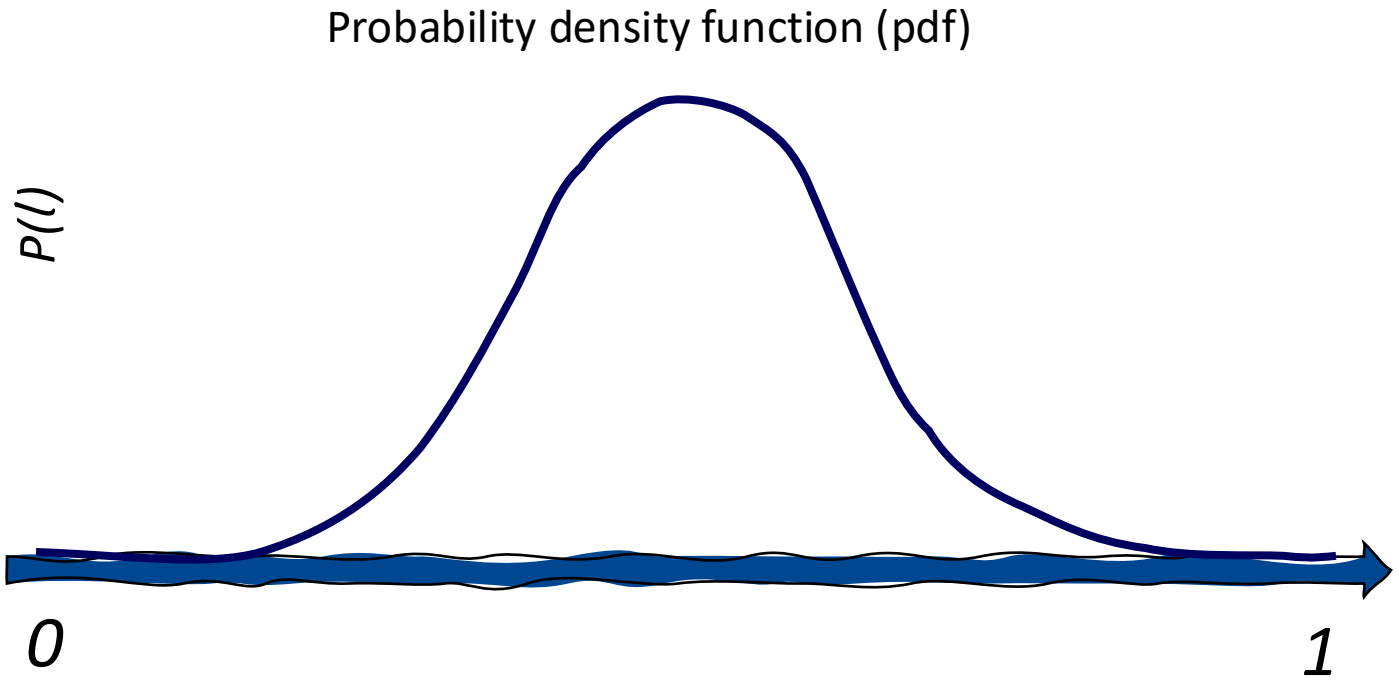
$$P[Y = a] = P(a \leq Y \leq a) = \int_a^a f(x)dx = 0$$

$$P[Y = a] = \lim_{\in \to 0} P\left[Y \in [a, a + \in]\right]$$

$$= \lim_{\in \to 0} \int_a^{a+\in} p_Y(y)dy = 0$$

Lavine 2013

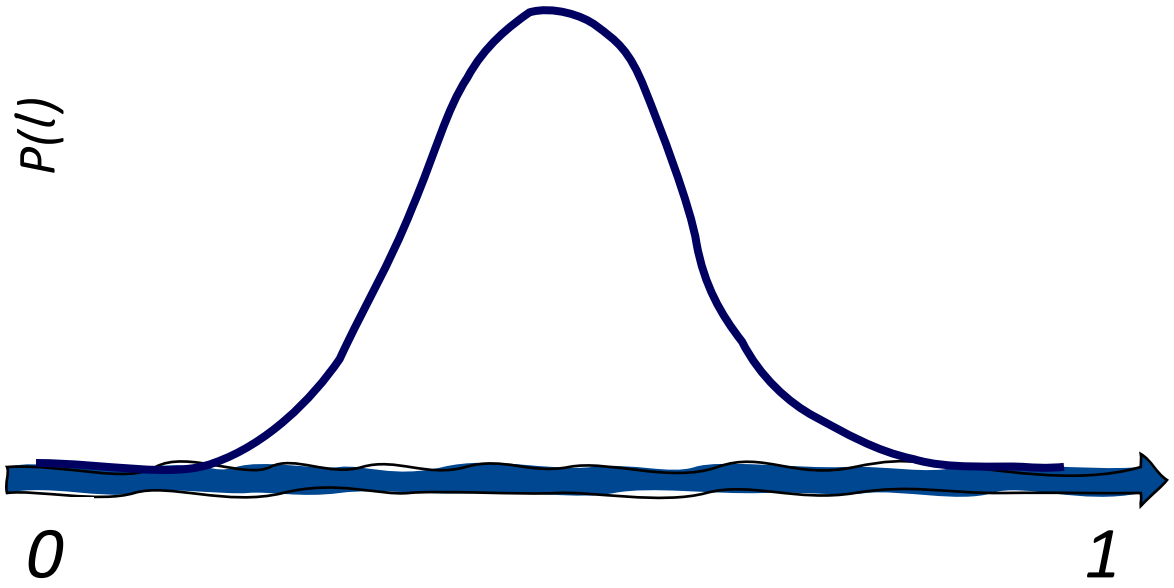
Continuous probabilities



Continuous probabilities

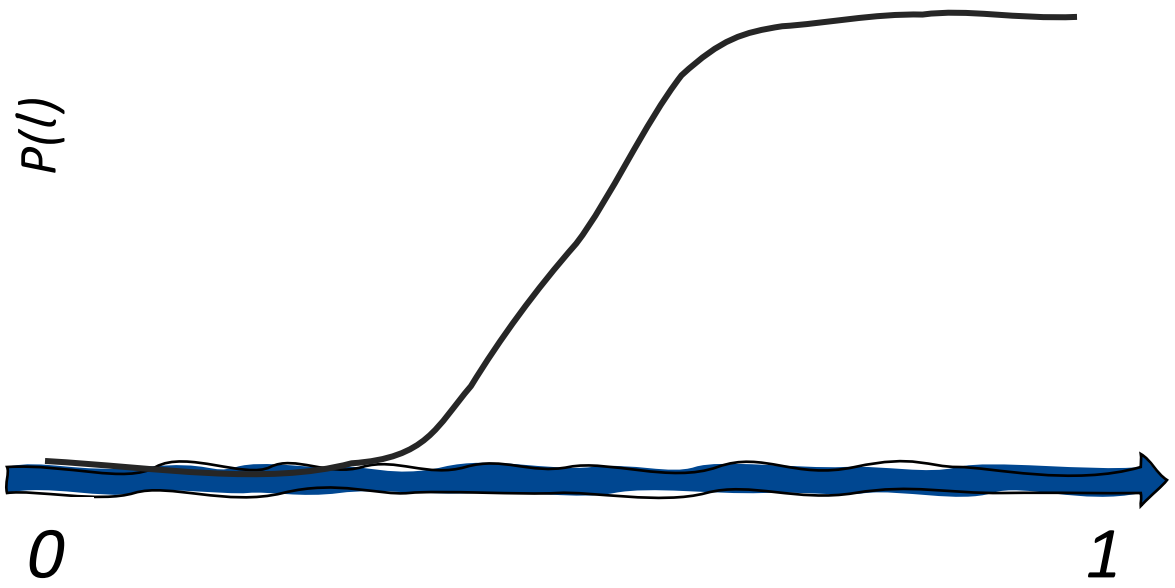
Probability density function (pdf)

$$P(A \leq y \leq B)$$



Cumulative density function (cdf)

$$P(y \leq B)$$



pdf is obtained by differentiating the cdf:

$$f(x) = \frac{d}{dx}[F(x)]$$

cdf is obtained by integrating the pdf:

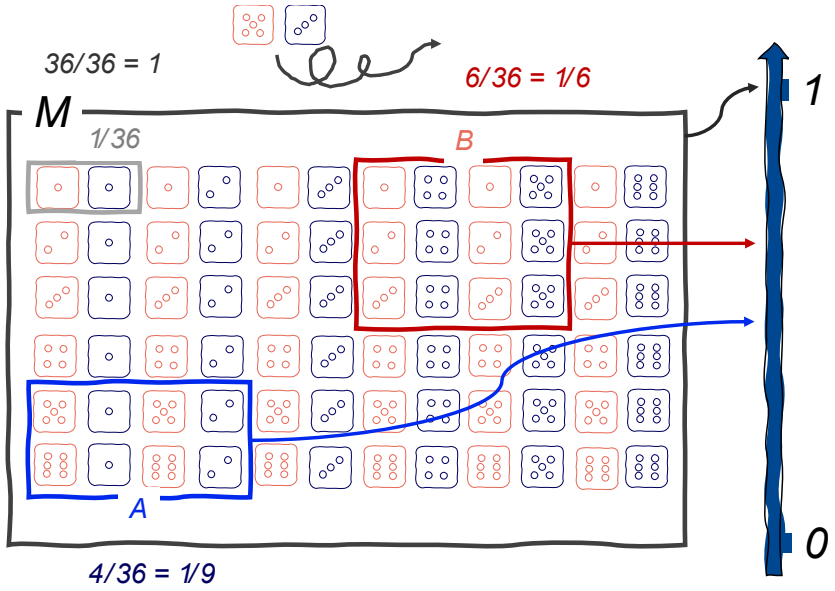
$$F(x) = \int_{-\infty}^{x} f(t)dt$$

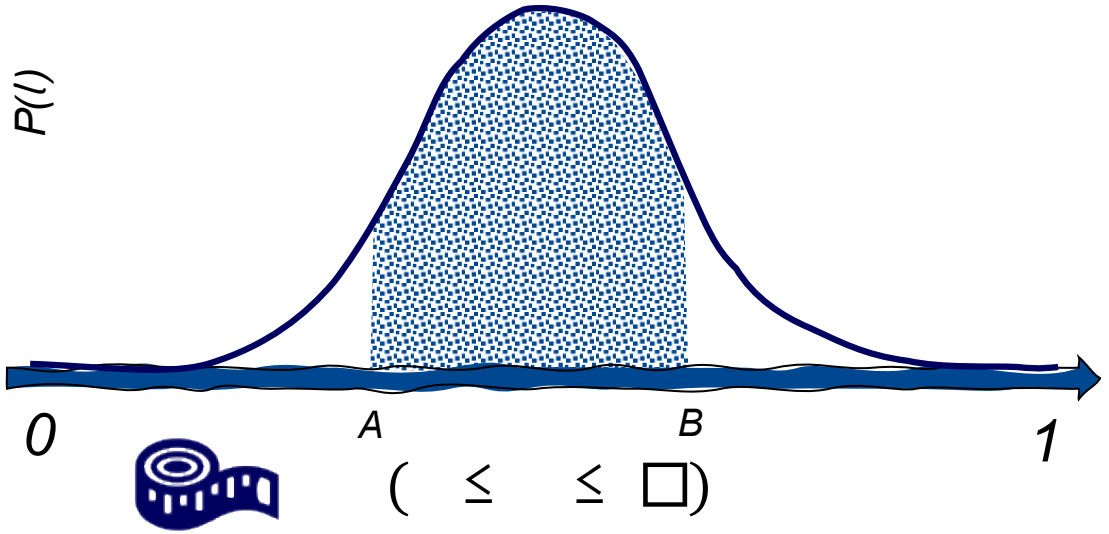
Plain text explanation of cdf: The probability of the random variable taking on a value less than or equal to a fixed real number.

Lavine 2013; Kuter 2024

L1-3 Data Types and Distributions

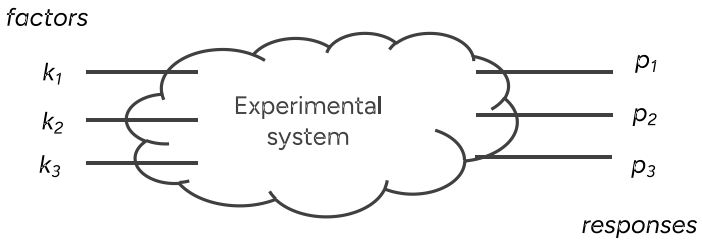
Probability distributions





Lavine 2013

Types of data

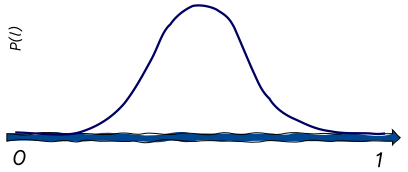


Data

quantitative and qualitative

- Numerical
  - continuous or discrete
  - Interval data — temperature: Celcius (no true zero)
  - Ratio data — temperature: Kelvin (true zero)
- Categorical
  - Ordinal data — temperature: very low, low, medium, high, very high (ordered)
  - Nominal data — status: healthy, diseased (no order)

What is a distribution?

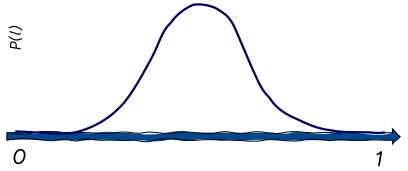


representing the distribution

Distribution or Probability Distribution
A conceptual model of a distribution.

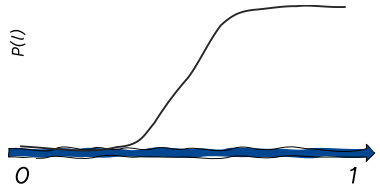
Parameters
such as mode, mean, variance, etc.
Percentiles and quartiles.

Other summary statistics.



PDF or PMF
Probability density/mass function represent a distribution of a continuous/discrete variable.

What is the probability of obtaining a value, *x*, between values *a* and *b*?



CDF
Cumulative density function represent a distribution of a continuous/discrete variable.

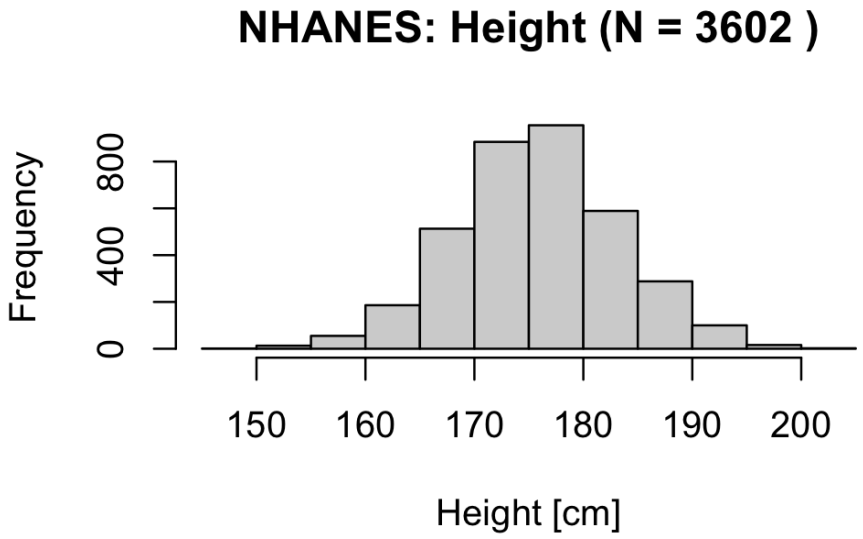
What is the cumulative probability of obtaining a value of up to *x*?

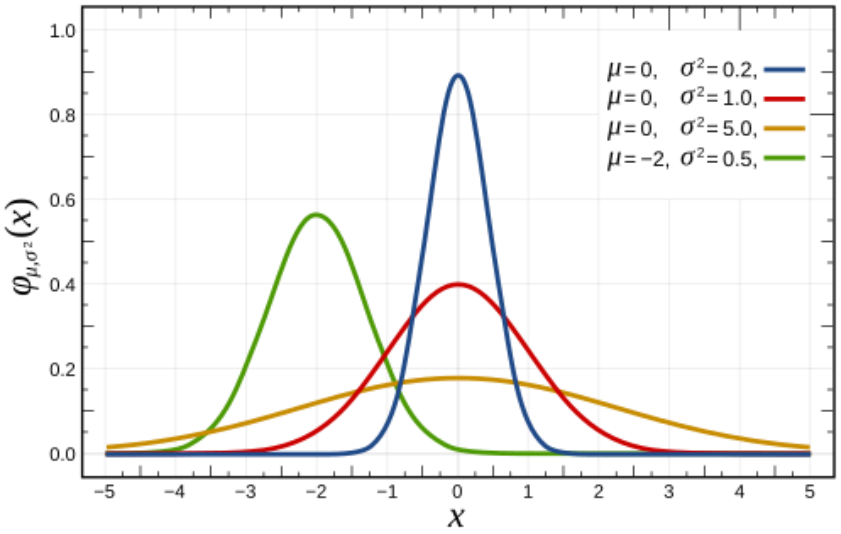
Parametric distributions

$$X \sim Distr(\mu, \sigma^2)$$

- Parametric distributions can be defined by parameters.
- You are probably familiar with many of these.
- For example, a normal distribution is defined by:
  - *mu*, the mean.
  - *sigma squared*, the variance.
- Shape parameters (*alpha*, *beta*, *etc*).
- Useful approximations for representing and interpreting data, generalizing and comparing across samples of data, simulating and modelling.

Distributions normal or Gaussian






```r
# Normal distribution
# We can load the US National Health and Nutrition Survey
library(NHANES)

# Let's filter the data for adult males (more on why later)
nhanes_subset <- NHANES %>%
  filter(Age > 18 & Gender == "male")

# It is reasonable to check N observations to make sure this is still substantial
n_height <- sum(complete.cases(nhanes_subset$Height))

# Then, we can plot the height data as a histogram
hist(nhanes_subset$Height, xlab = "Height [cm]",
ylab = "Frequency", main = paste("NHANES: Height (N=", n_height, ")"))
```

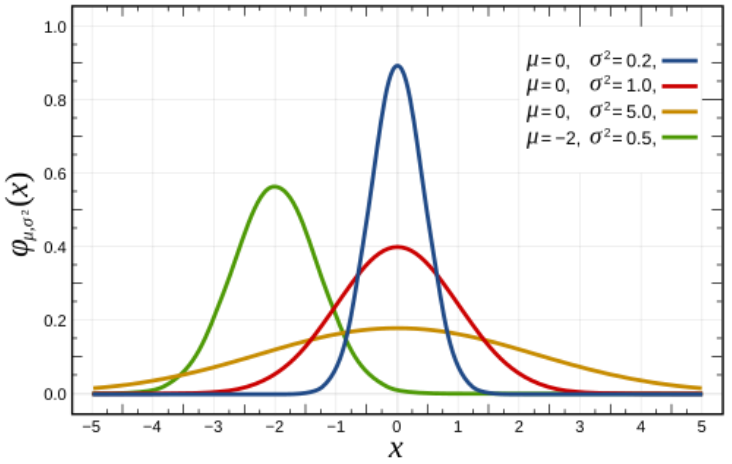
Distributions normal or Gaussian

$mean$: μ
$variance$: σ^2
$X \sim N(\mu, \sigma^2)$

The pdf is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Probability density function (pdf)

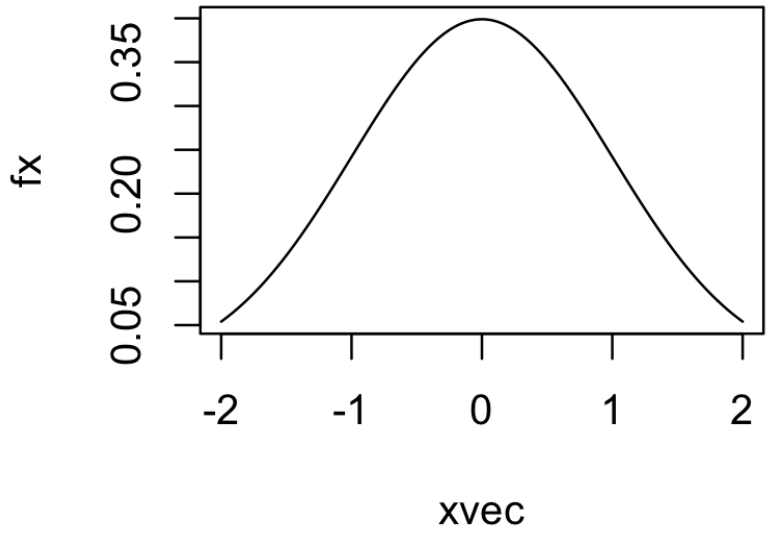



```r
# a normal distribution has two parameters
# mu, the mean, and sigma^2, the variance:
mu <- 0
sigma <- 1

# we device a vector of n x:
xvec <- seq(-5,5,0.01)

# this is the pdf of a normal distribution:
fx <- ( 1 / sqrt(2*pi*sigma^2)) * exp(-(xvec-mu)^2/(2*sigma^2))

# we can then create a very simple line plot:
plot(xvec, fx, type = "l", lty = 1)
```



Distributions normal or Gaussian

$mean$: μ
$variance$: σ^2
$X \sim N(\mu, \sigma^2)$

The cdf is:

$$\Phi\left(\frac{x-\mu}{\delta}\right) = \frac{1}{2}\left[1 + erf\left(\frac{x-u}{\sigma\sqrt{2}}\right)\right]$$

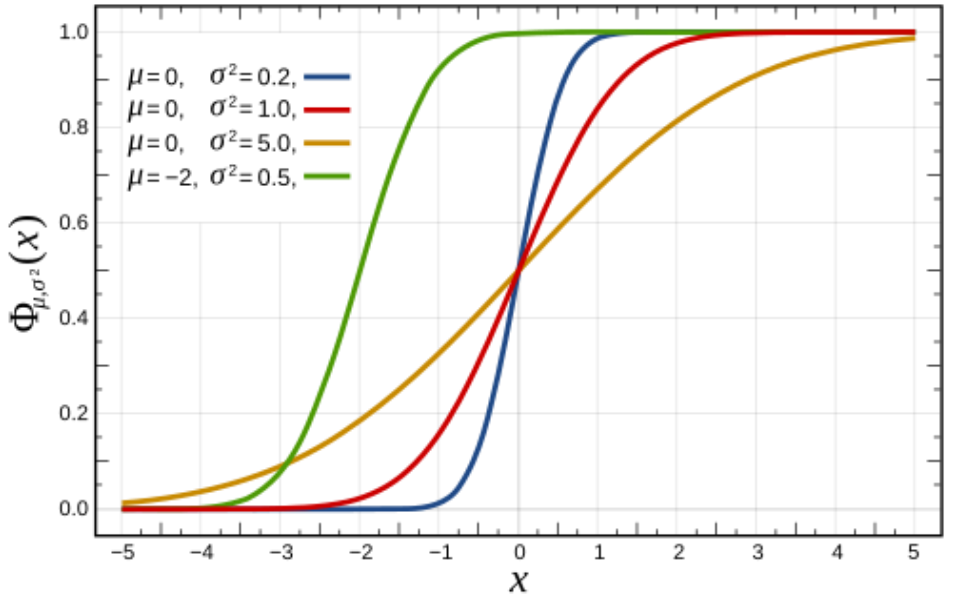
```
…
# equally, we can use the built-in functions
# the pdf function:
xpdf <- dnorm(xvec, mu, sigma)

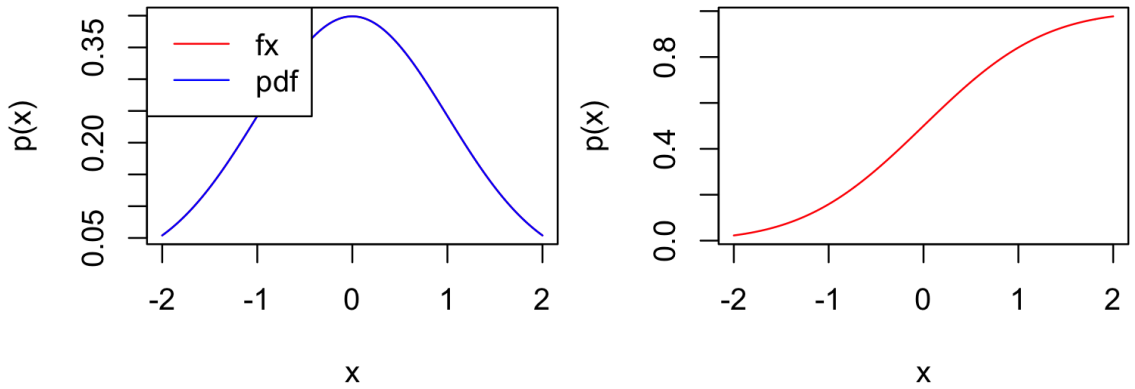
# the cdf function:
xcdf <- pnorm(xvec, mu, sigma)

# create a simple plot of these - are they consistent?
# the built-in pdf function and the "home-made" version:
plot(xvec, fx, type = "l", lty = 1, col="red", ylab = "p(x)",
xlab = "x")
lines(xvec, xpdf, type = "l", lty = 1, col="blue")
legend("topleft", legend=c("fx","pdf"), col=c("red","blue"),
lty = 1, cex = 1)

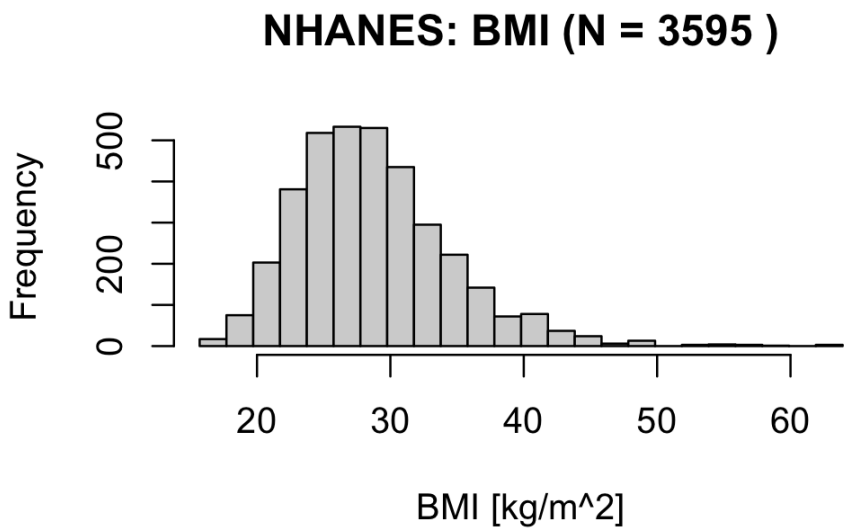
# plotting the cdf:
plot(xvec, xcdf, type = "l", lty = 1, col="red",
ylab = "p(x)", xlab = "x")
```

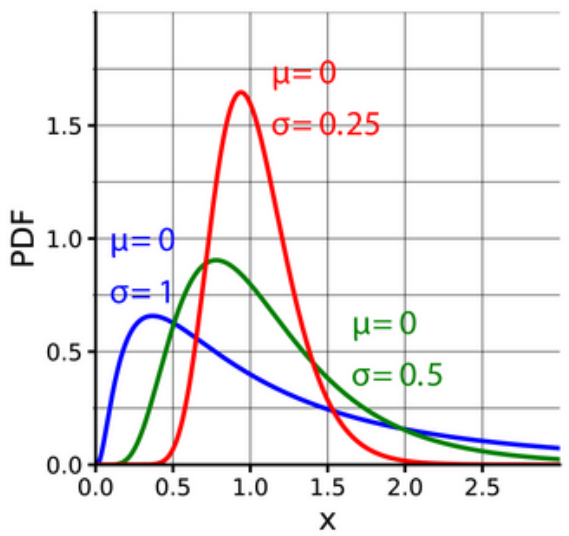
Cumulative density function (cdf)

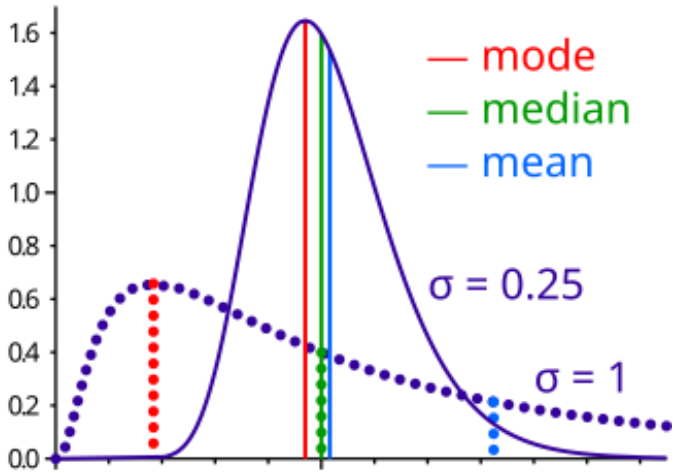




Distributions lognormal







$mean$: e^{μ}

$variance$: $\left(e^{\sigma^2}-1\right)e^{2\mu+\sigma^2}$

$X \sim Lognormal(\mu, \sigma^2)$

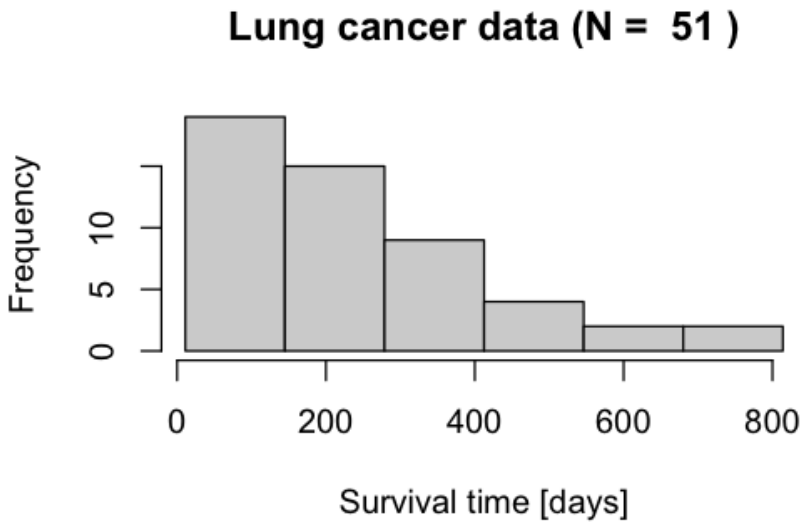
The pdf is:

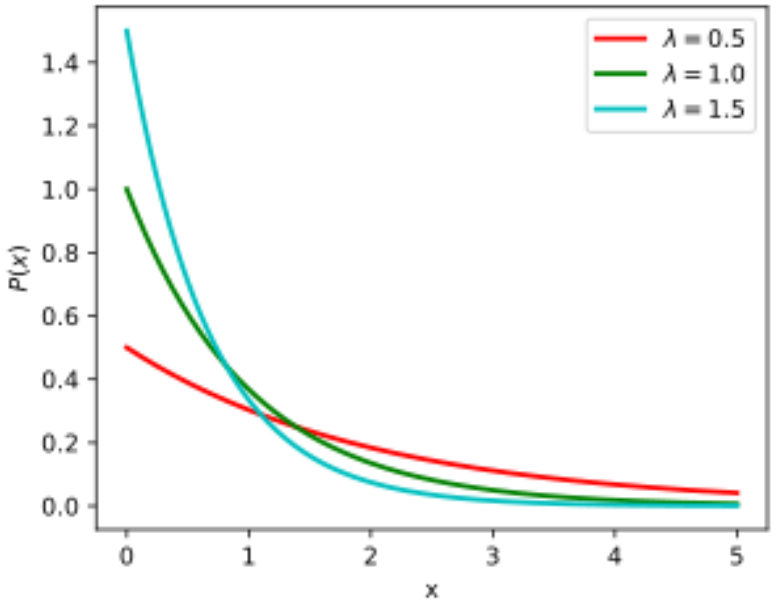
$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(lnx-\mu)^2}{2\sigma^2}}$$

```r
# Lognormal distribution
n_bmi <- sum(complete.cases(nhanes_subset$BMI))

hist(nhanes_subset$BMI, breaks = seq(min(nhanes_subset$BMI, na.rm = TRUE),
max(nhanes_subset$BMI, na.rm = TRUE), length.out = 25), xlab = "BMI [kg/m^2]",
ylab = "Frequency", main = paste("NHANES: BMI (N =", n_bmi,")"))
```

Distributions exponential





$mean: \frac{1}{\lambda}$

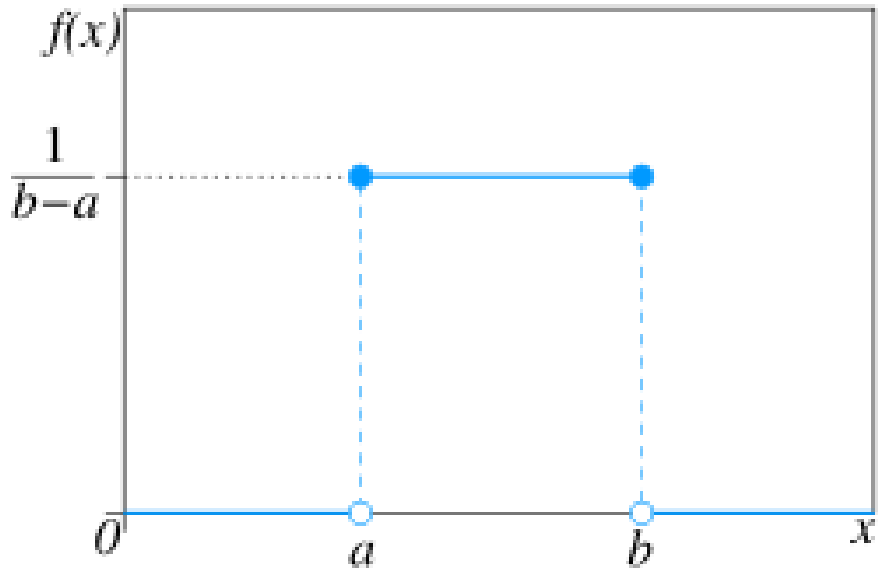
$variance: \frac{1}{\lambda^2}$

$X \sim Exp(\lambda)$

The pdf is:

$f_X(x) = \lambda e^{-\lambda x}$

Distributions uniform (continuous and discrete)



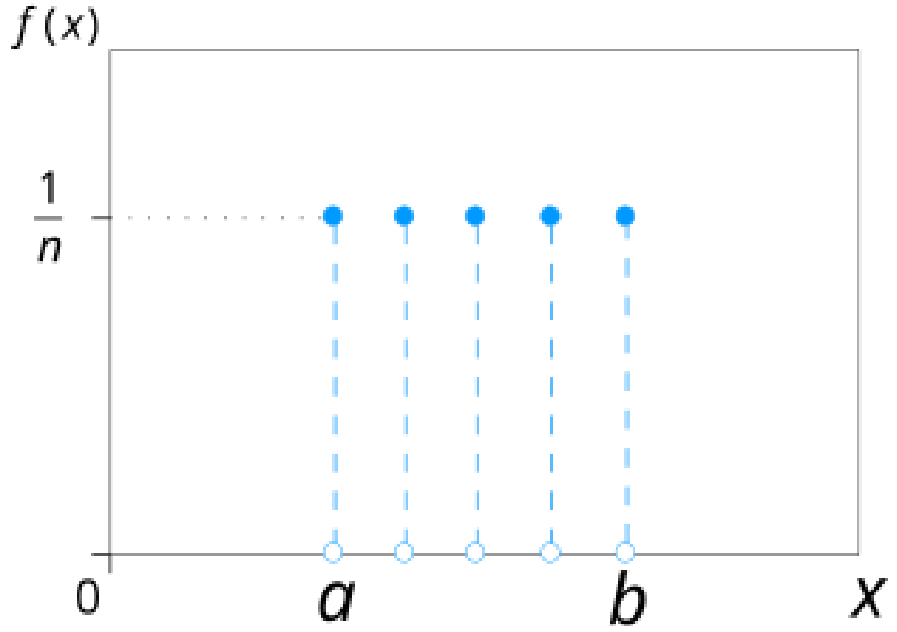
$mean: \frac{1}{2}(a+b)$

$variance: \frac{1}{12}(b-a)^2$

$X \sim u(a,b)$

The pdf is:

$$\begin{cases} \frac{1}{b-a} \; for \; x \in [a,b] \\ 0 \; otherwise \end{cases}$$



$mean: \frac{a+b}{2}$

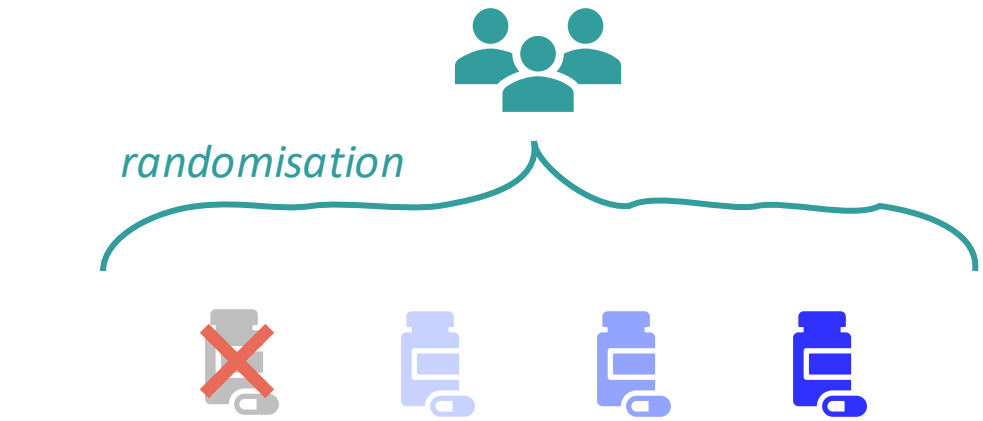
$variance: \frac{(b-a+1)^2-1}{12}$

$X \sim unif\{a,b\}$

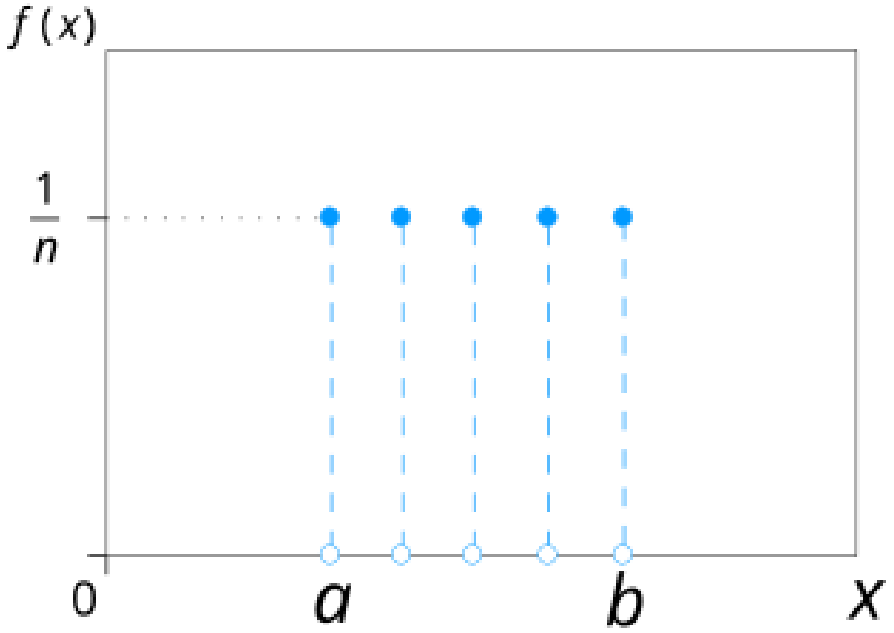
The pmf is:

$f_X(x) = \frac{1}{n}$

Distributions uniform



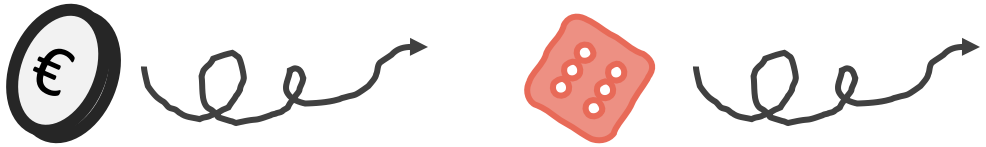
$n = 4$



Distributions binomial (discrete)

Bernoulli process – a sequence of Bernoulli trials

- in a sequence x_1, x_2, x_3… x_i is 1 or 0 (success and failure)
- the probability of $x_i = 1$ is constant across trials.

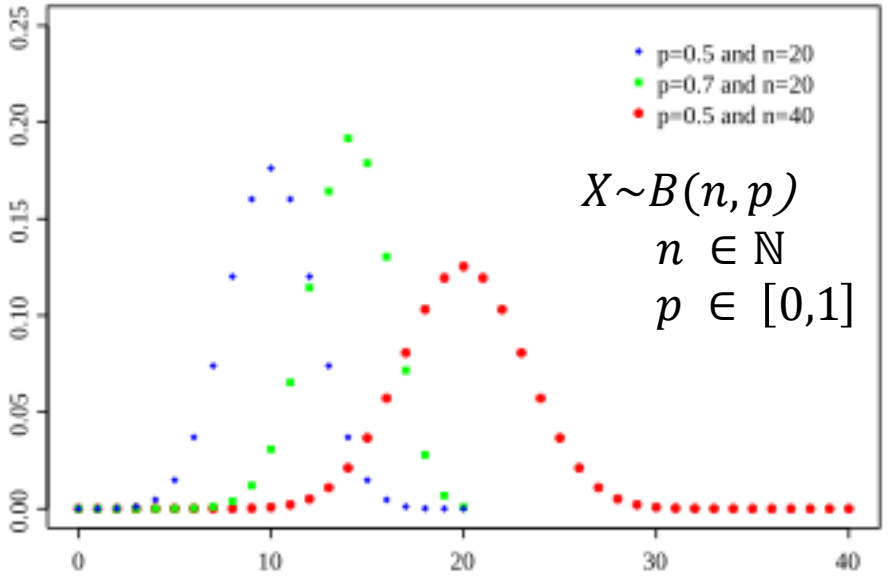


$$X \sim B(n,p)$$

The pmf is:

$$f(k,n,p) = \binom{n}{k} p^k (1-p)^{n-k}$$

k successes
n trials
p probability of success



Interpretation

The probability of *k* successes (y-axis) in a fixed number of trials (*n* trials, x-axis).

Distributions binomial (discrete)

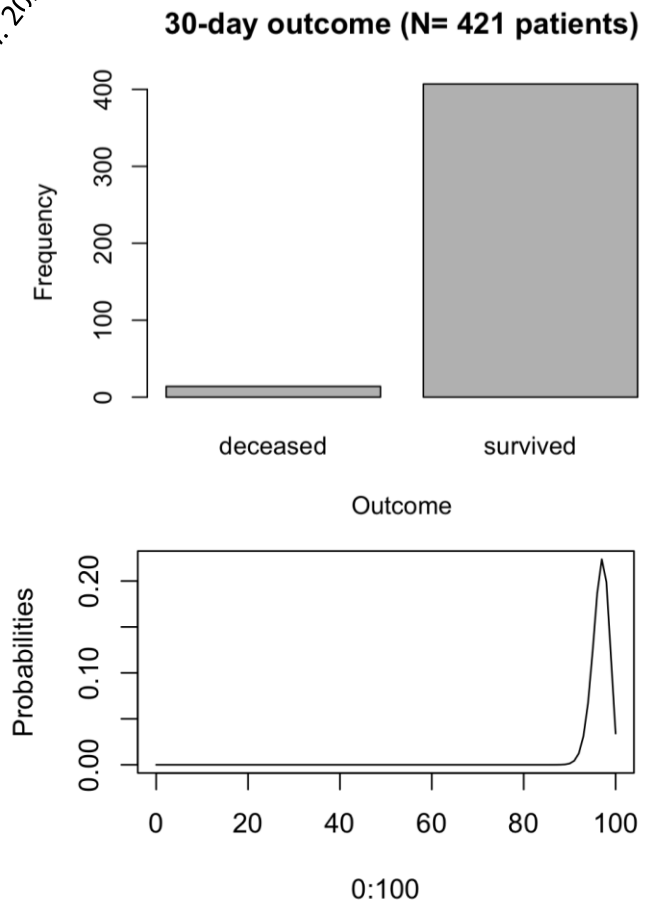


$$X \sim B(n, p)$$

Bernoulli process – a sequence of Bernoulli trials

- in a sequence x_1, x_2, x_3… x_i is 1 or 0 (success and failure)
- the probability of $x_i = 1$ is constant across trials.

Based on: Fazlinovič *et al.* 2021




```r
## Binomial distribution
# we can use the binomial pmf to generate data for the success of a
# hypothetical surgery e.g., p: 0.981, n: 421 (based on Fazlinovic et al. 2021)
surgery_stats <- rbinom(421,1,0.981)
n_count <- length(surgery_stats)

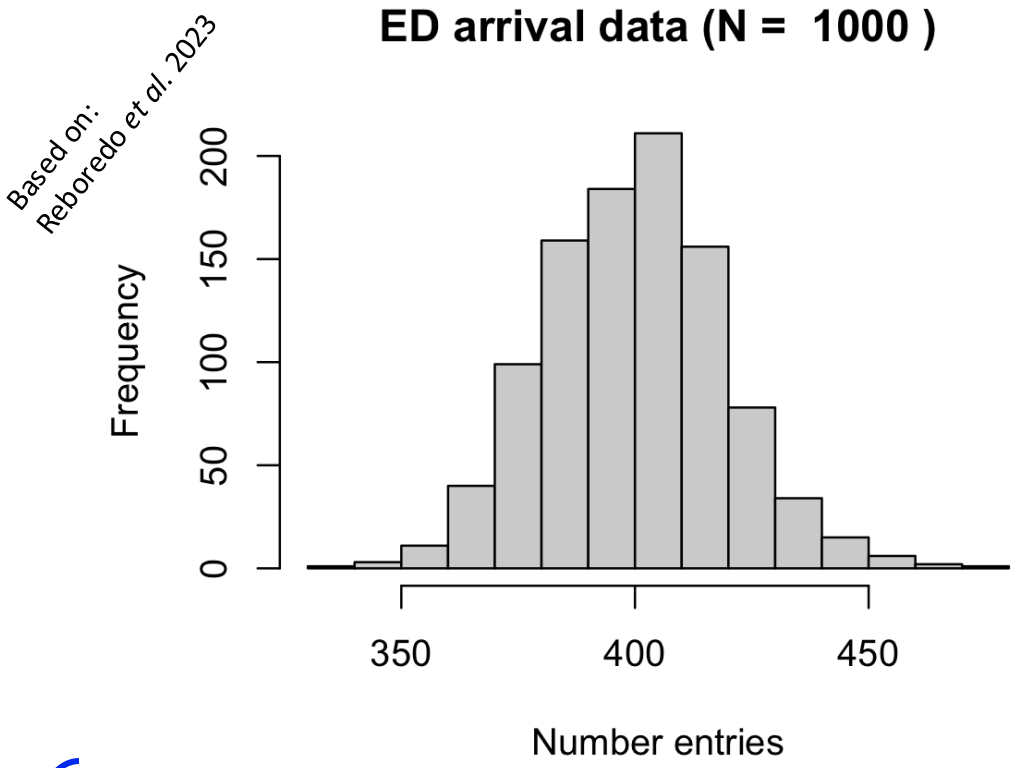
# we can plot the data using for example a bar plot grouped by successes and failures:
count_outcome <- table(surgery_stats)
barplot(count_outcome, names=c("deceased","survived"), main = paste("30-day outcome
(N=",n_count,"patients)"), xlab = "Outcome", ylab = "Frequency")

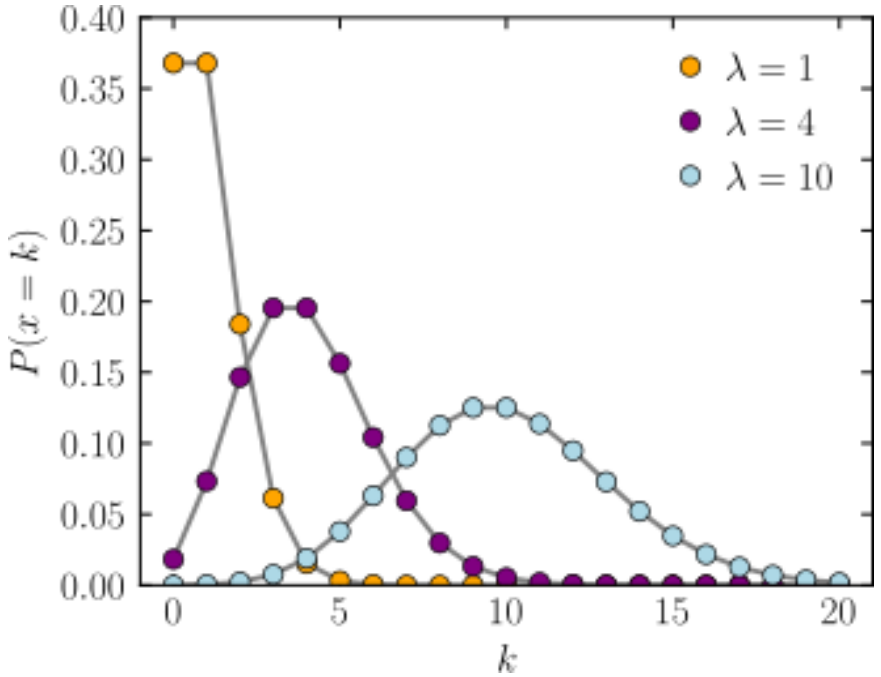
# we can then get the proportions of the two outcomes
probs <- prop.table(count_outcome)

# the probability of survival is then:
p_success <- as.numeric(probs[2])

# we can then plot the probability mass function for a binomial distribution
# in this case, the probability of number of successes given 100 trials
pmf_binom <- dbinom(x = c(0:100), size = 100, prob = p_success)
plot(0:100, pmf_binom, type = "l", ylab ="Probabilities")
```

Distributions Poisson (discrete)





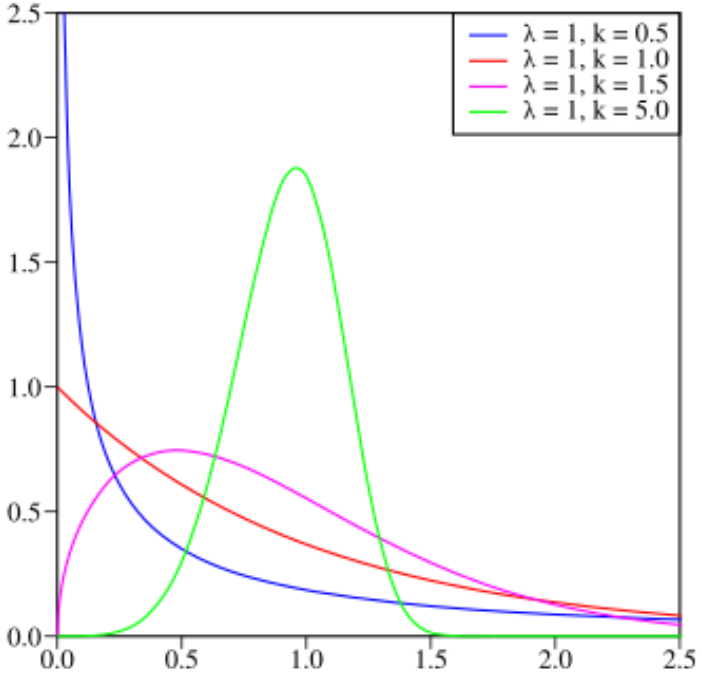
$$Pois(\lambda)$$

$$\frac{\Gamma(\lfloor k+1 \rfloor, \lambda)}{\lfloor k \rfloor !}$$

```r
## Poisson distribution
# we can simulate emergency department arrival data based on Reboredo et al. 2023
# The study indicates mean/gamma arrivals of 400.96 patients during 2015-2020
n_samples <- 1000
dat_arrival <- rpois(n_samples, 400.96)

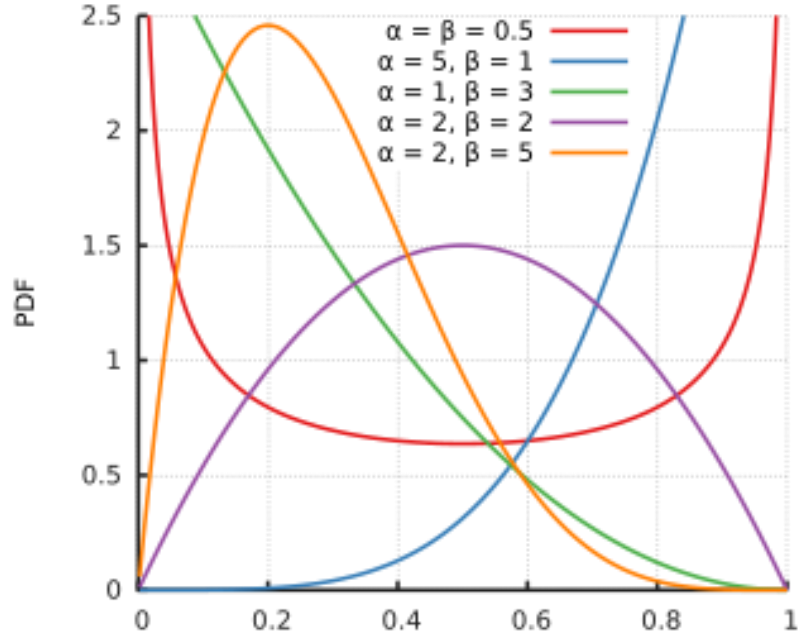
hist(dat_arrival, xlab = "Number entries", ylab = "Frequency",
main = paste("ED arrival data (N = ",n_samples,")"))
```

Distributions Are there other more exotic ones? The Weibull and the beta distribution



The pdf is:

$$\begin{cases} \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} & \\ 0, & x < 0 \end{cases}$$



The pdf is:

$$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$$

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

Distributions transforming variables

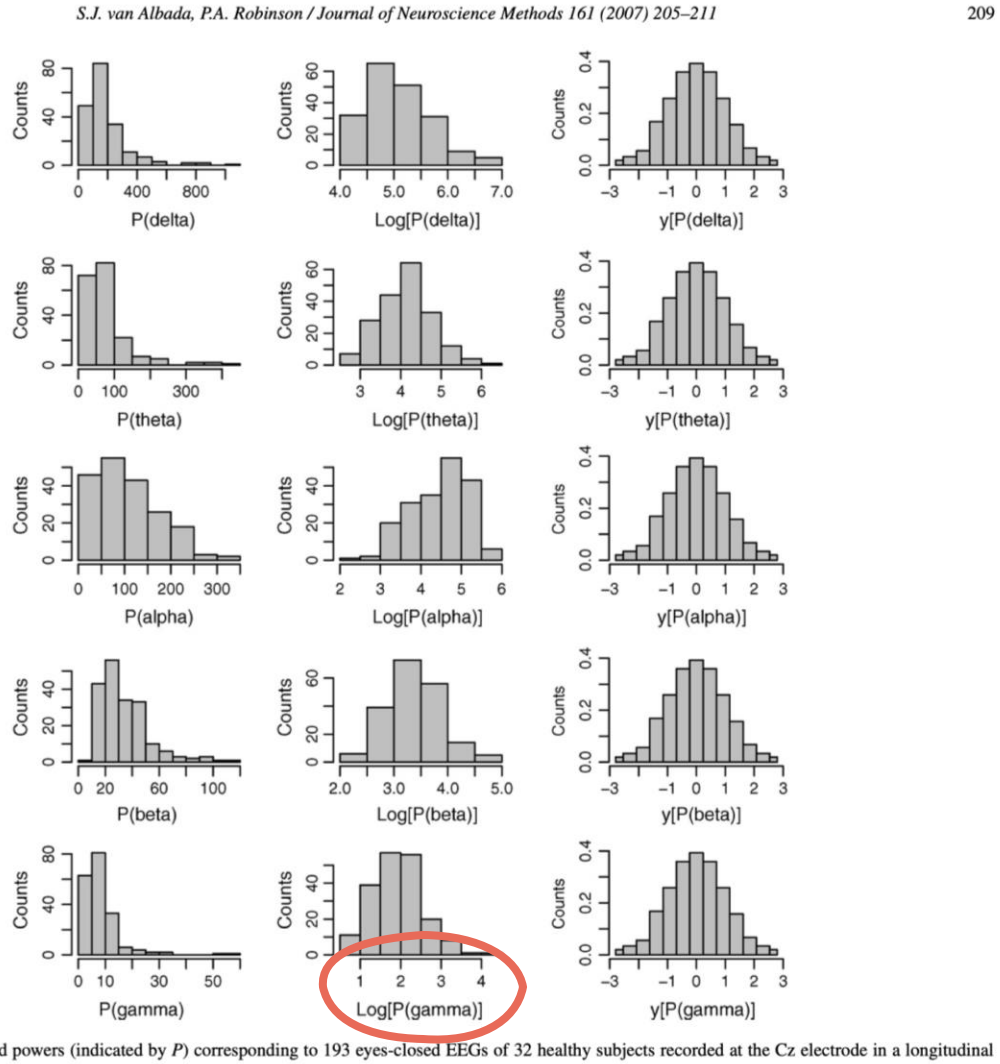


Fig. 2. Histograms for band powers (indicated by P) corresponding to 193 eyes-closed EEGs of 32 healthy subjects recorded at the Cz electrode in a longitudinal study, giving the number of observations in each bin. Untransformed quantities are plotted in the left column, log-transformed data in the middle column, and data transformed with (7) in the right column. The symbol 'y' represents the transformation function.

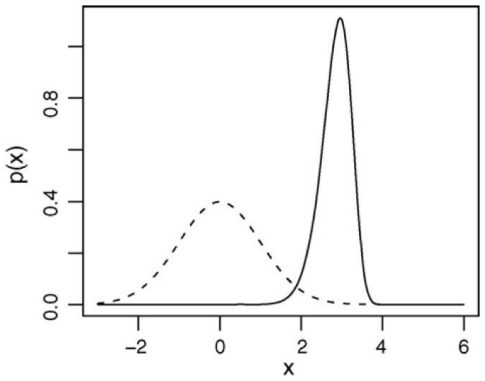


Fig. 1. The Weibull PDF with shape parameter $a = 9$ and scale parameter $b = 1$ (solid) and the standard normal PDF after transformation (dashed).

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda} & (\lambda \neq 0) \\ \log y & (\lambda = 0) \end{cases}$$

for $y > 0$, *or*

$$y^{(\lambda)} = \begin{cases} \dfrac{(y + \lambda^2)^{\lambda_1} - 1}{\lambda_1} & (\lambda_1 \neq 0) \\ \log(y + \lambda_2) & (\lambda_1 = 0) \end{cases}$$

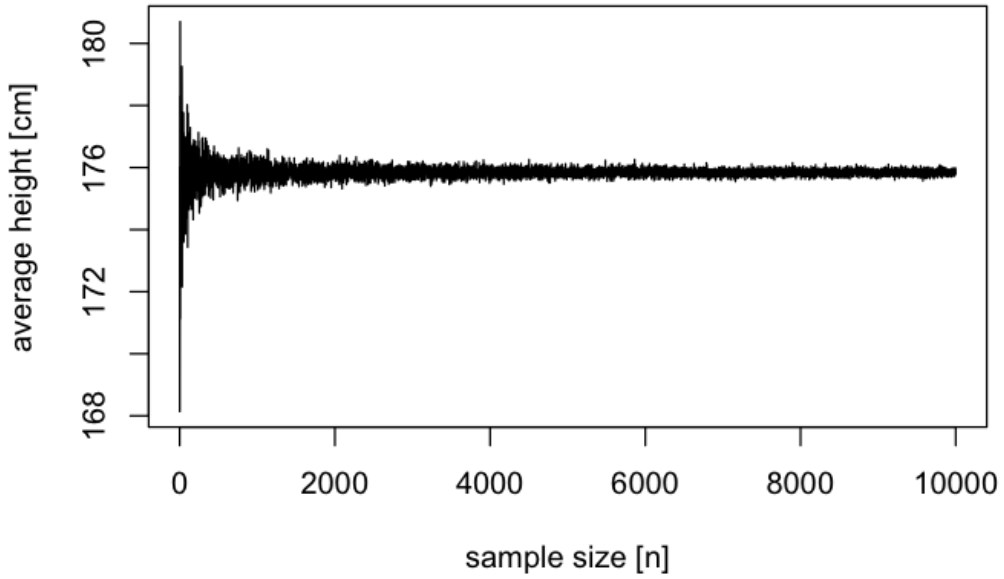
van Albada & Robinson 2007; Box & Cox 1964

Law of Large Numbers

The sample mean approaches the true (expected) mean as sample size increases:

$$X_n \xrightarrow{P} \mu \; as \; n \to \infty \qquad \lim_{n\to\infty} P(|X_n - \mu| < \varepsilon) = 1$$

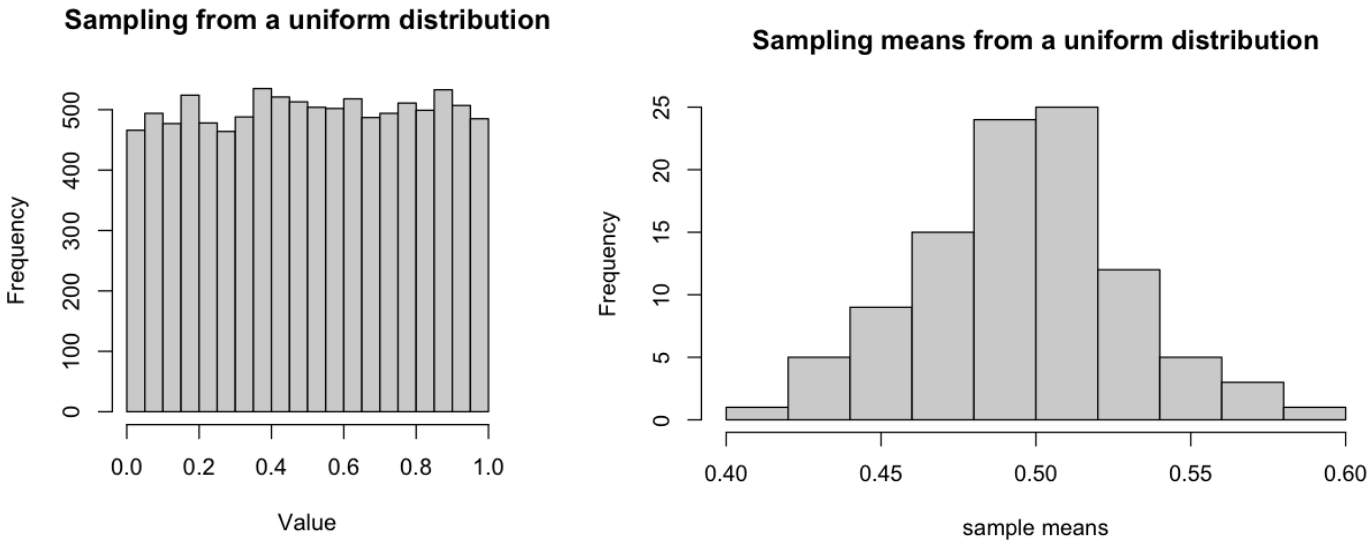
$$P\left(\lim_{n\to\infty} X_n = \mu\right) = 1$$



The Classic Central Limit Theorem

$X_1, X_2, \ldots, X_n$, a sequence of *independent and identically distributed (i.i.d.)* random variables with a finite μ and σ^2, and $\bar{X} = (X_1 + \cdots + X_n)/n$ represents the sample mean. Then, as *n* increases to infinity, the random variable $\left(\sqrt{n}(\bar{X} - \mu)\right)/\sigma$ converges to an approximately normal variable of a standard normal distribution.

The distribution of sums and means **Approach** normality as n increases.



Zhang *et al*. 2023; Lavine 2013

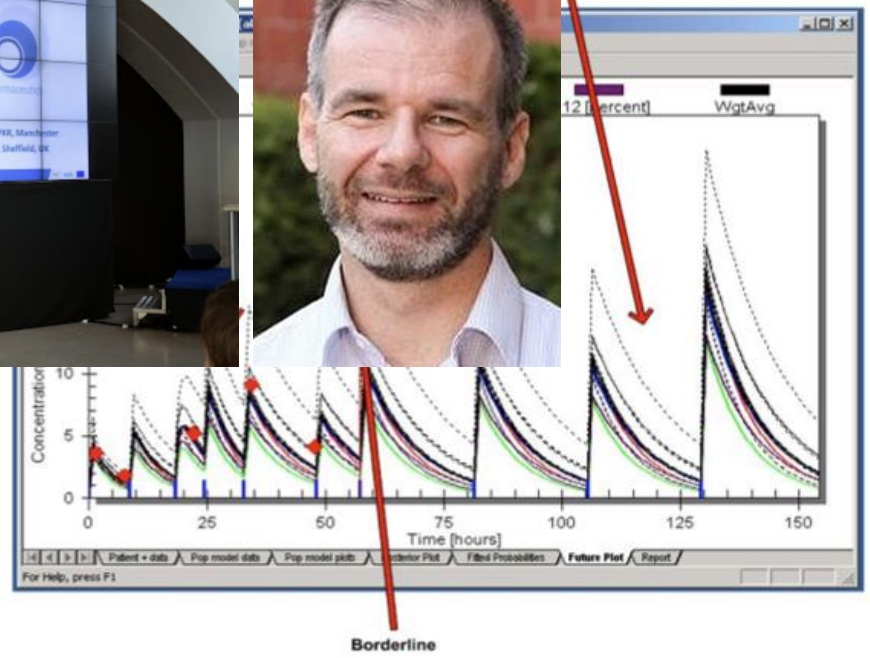
Nonparametric distributions









Nonparametric distributions

Table 1. Examples of symmetrical kernel functions [11].

Kernel	Definition
Epanechnikov	$K(t)=\begin{cases}\frac{3}{4\sqrt{5}}(1-\frac{1}{5}t)^2 & \text{for } \lvert t\rvert<\sqrt{5}\\ 0 & \text{for } \lvert t\rvert\geq\sqrt{5}\end{cases}$
Biweight	$K(t)=\begin{cases}\frac{15}{16}(1-t^2)^2 & \text{for } \lvert t\rvert<1\\ 0 & \text{for } \lvert t\rvert\geq 1\end{cases}$
Triangular	$K(t)=\begin{cases}1-\lvert t\rvert & \text{for } \lvert t\rvert<1\\ 0 & \text{for } \lvert t\rvert\geq 1\end{cases}$
Gaussian	$K(t)=\frac{1}{\sqrt{2\pi}}e^{-t^2/2}$
Rectangular	$K(t)=\begin{cases}\frac{1}{2} & \text{for } \lvert t\rvert<1\\ 0 & \text{for } \lvert t\rvert\geq 1\end{cases}$

Table 2. Examples of asymmetrical kernel functions. Symbol b denotes the smoothing parameter.

Kernel	Definition
Gamma 1 [18]	$K_{GAM1}(x,b;t)=\frac{t^{x/b}e^{-t/b}}{b^{x/b+1}\Gamma(x/b+1)}$
Gamma 2 [18]	$K_{GAM2}(\rho_b(x),b;t)=\frac{t^{\rho_b(x)-1}e^{-t/b}}{b^{\rho_b(x)}\Gamma(\rho_b(x))}$ $\rho_b(x)=\begin{cases}x/b & \text{for } x\geq 2b\\ \frac{1}{4}(x/b)^2+1 & \text{for } x\in[0,2b)\end{cases}$
Inverse Gaussian [19]	$K_{IG}(x,b;t)=\frac{1}{\sqrt{2\pi bt^3}}e^{-\frac{1}{2bx}\left(\frac{t}{x}-2+\frac{x}{t}\right)}$
Reciprocal Inverse Gaussian [19]	$K_{RIG}(x,b,t)=\frac{1}{\sqrt{2\pi bt}}e^{-\frac{x-b}{2b}\left(\frac{t}{x-b}-2+\frac{x-b}{t}\right)}$
Lognormal [20]	$K_{LN}(x,b;t)=\frac{1}{\sqrt{8\pi\ln(1+b)t}}e^{-\frac{(\ln t-\ln x)^2}{8\ln(1+b)}}$

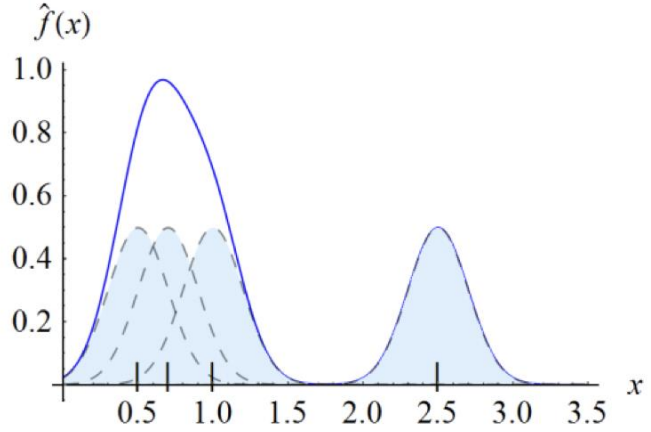


Fig. 1. Construction of kernel density estimator (1) (continuous line) with a symmetric kernel (dashed lines) for a 4-element sample (vertical segments).

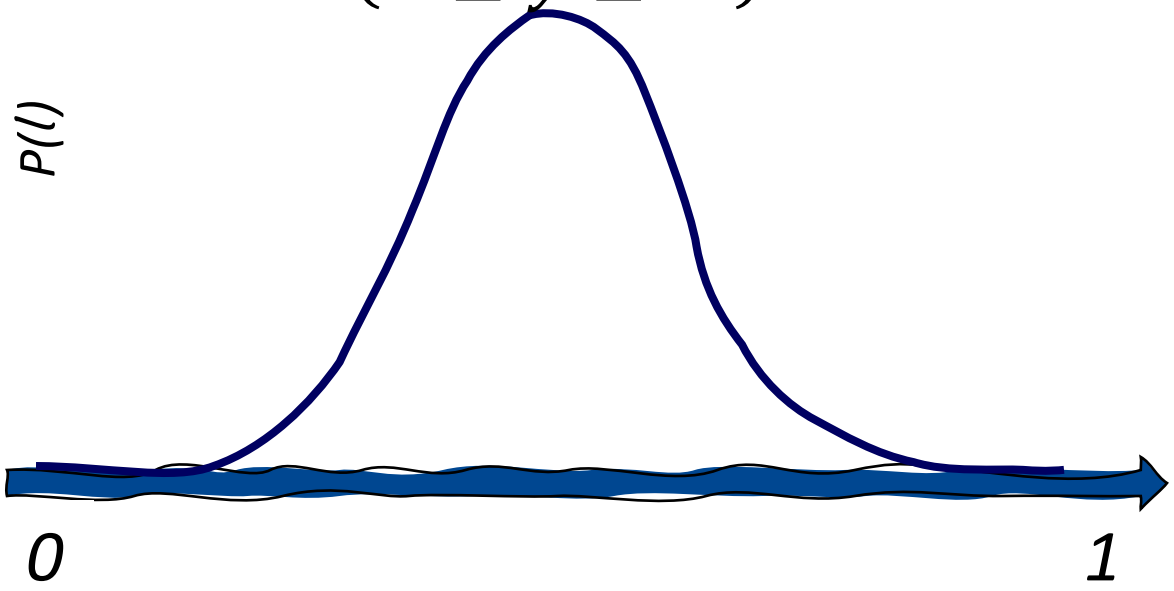
Węglarczyk 2018

L1-4 Sampling from Distributions

Continuous probabilities

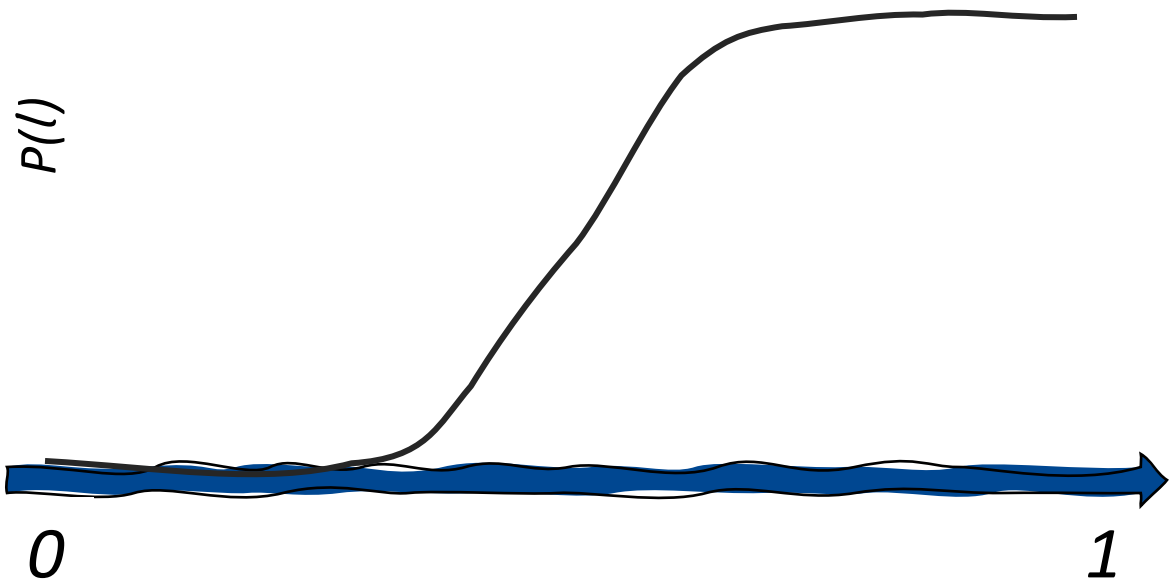
Probability density function (pdf)

$$P(A \leq y \leq B)$$



Cumulative density function (cdf)

$$P(y \leq B)$$



pdf is obtained by differentiating the cdf:

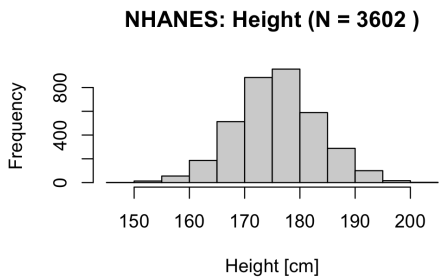
$$f(x) = \frac{d}{dx}[F(x)]$$

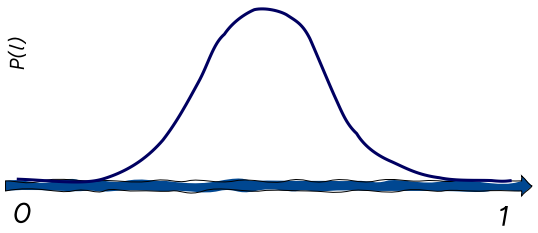
cdf is obtained by integrating the pdf:

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

Lavine 2013; Kuter 2024

Distributions estimation





closed-form maximum likelihood estimates
for the mean (*mu*) and variance (*sigma squared*)

$$y = \{y_1, \dots, y_n\}$$

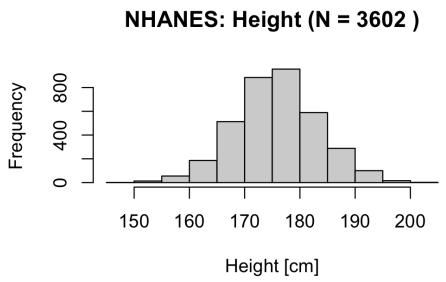
$$y_i \sim \mathrm{N}(\mu, \sigma^2), i = 1, \dots, n$$

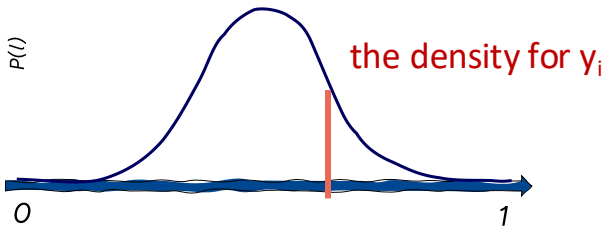
$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (y_i - \bar{y})^2$$

```
# R uses packages for specialised functions
and datasets these can be installed using the
following script:
install.packages("name")
# packages are then called using:
library(name)
# MASS is one of the most popular packages for
statistical analysis it can be called through:
library(MASS)
# Packages have vignettes (manuals), for example:
https://cran.r-project.org/web/packages/MASS/MASS.pdf
```

Distributions estimation





1. The likelihood function for the normal/Gaussian probability density function is:

$$\prod_{i=1}^{n} p(y_i|\mu,\sigma^2) = \sqrt{\frac{1}{(2\pi\sigma^2)^n} e^{\left[-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{y_i-\mu}{\sigma}\right)^2\right]}}$$

2. Calculate the log-likelihood (*ll*) function:

$$\ell\ell(\mu,\sigma^2) = \log p(y|\mu,\sigma^2) = -\frac{n}{2} log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y^T y - 2n\bar{y}\mu + n\mu^2)$$

3. Maximise the *ll* (or minimise the negative *ll*) by searching for the optimal value of *mu* and *sigma-squared*.
This can be done using, for example a direct search method or a gradient search method (In MASS, Nelder-Mead or BFGS).

Ripley *et al*. 2025; Soch 2024

Distributions estimation

fitdistr *Maximum-likelihood Fitting of Univariate Distributions*

Description

Maximum-likelihood fitting of univariate distributions, allowing parameters to be held fixed if desired.

Usage

```
fitdistr(x, densfun, start, ...)
```

Arguments

x	A numeric vector of length at least one containing only finite values.
densfun	Either a character string or a function returning a density evaluated at its first argument. Distributions "beta", "cauchy", "chi-squared", "exponential", "gamma", "geometric", "log-normal", "lognormal", "logistic", "negative binomial", "normal", "Poisson", "t" and "weibull" are recognised, case being ignored.
start	A named list giving the parameters to be optimized with initial values. This can be omitted for some of the named distributions and must be for others (see Details).
...	Additional parameters, either for densfun or for optim. In particular, it can be used to specify bounds via lower or upper or both. If arguments of densfun (or the density function corresponding to a character-string specification) are included they will be held fixed.

Details

For the Normal, log-Normal, geometric, exponential and Poisson distributions the closed-form MLEs (and exact standard errors) are used, and start should not be supplied.

For all other distributions, direct optimization of the log-likelihood is performed using optim. The estimated standard errors are taken from the observed information matrix, calculated by a numerical approximation. For one-dimensional problems the Nelder-Mead method is used and for multi-dimensional problems the BFGS method, unless arguments named lower or upper are supplied (when L-BFGS-B is used) or method is supplied explicitly.

For the "t" named distribution the density is taken to be the location-scale family with location m and scale s.

For the following named distributions, reasonable starting values will be computed if start is omitted or only partially specified: "cauchy", "gamma", "logistic", "negative binomial" (parametrized by mu and size), "t" and "weibull". Note that these starting values may not be good enough if the fit is poor: in particular they are not resistant to outliers unless the fitted distribution is long-tailed.

There are print, coef, vcov and logLik methods for class "fitdistr".

Value

An object of class "fitdistr", a list with four components,

estimate	the parameter estimates,
sd	the estimated standard errors,
vcov	the estimated variance-covariance matrix, and
loglik	the log-likelihood.

pp 47-48 in Ripley *et al*. 2025 – Package 'MASS' Vignette:
https://cran.r-project.org/web/packages/MASS/MASS.pdf

Distributions estimation

```r
## Distribution Estimation & Sampling
#  Let's use adult male height from NHANES:
library(NHANES)
library(dplyr)
library(MASS)

# Filter the data for adult males
nhanes_subset <- NHANES %>%
  filter(Age > 18 & Gender == "male")
n_height <- sum(complete.cases(nhanes_subset$Height))

nhanes_height <- na.omit(nhanes_subset$Height)

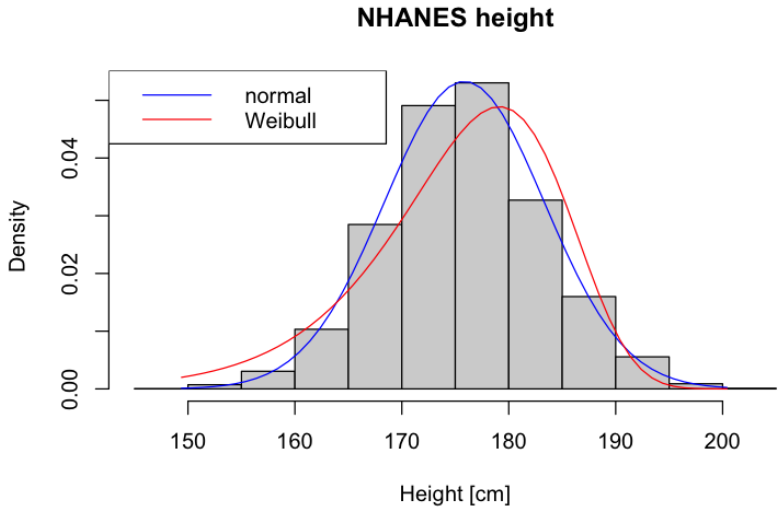
# Use the package MASS to fit a normal distribution to the data
# This will make use of the closed form solution
normal_params <- fitdistr(nhanes_height, "normal")

# Distributions like the Weibull require numerical optimisation
# we can test this out as well (note that this may be sensitive to initial estimates)
weibull_params <- fitdistr(nhanes_height, "weibull", list(shape = 20, scale = 200))

# We can then produce a plot of the data and the two pdfs of the fitted distributions
# First we need to specify a few things...
# set inputs for pdfs
height_min <- min(nhanes_height)
height_max <- max(nhanes_height)
x_pdf <- seq(height_min, height_max, 1)

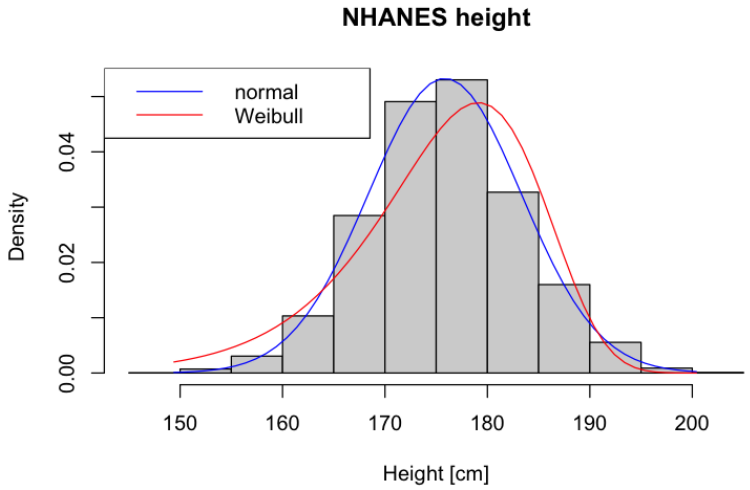
# Then we sample from the pdfs using the parameter estimates
y_norm <- dnorm(x_pdf, normal_params$estimate["mean"], normal_params$estimate["sd"])
y_weib <- dweibull(x_pdf, weibull_params$estimate["shape"], weibull_params$estimate["scale"])

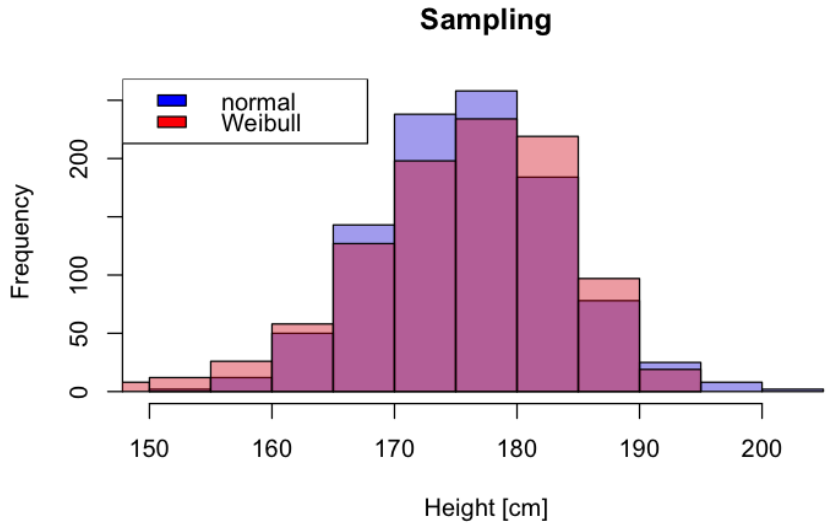
# we start with a histogram, specifying to plot the density (freq = FALSE)
hist(nhanes_height, freq = FALSE, xlab = "Height [cm]", ylab = "Density", main = "NHANES height")
lines(x_pdf, y_norm, col="blue")
lines(x_pdf, y_weib, col="red")
legend(x = "topleft", c("normal","Weibull"), col=c("blue","red"), lty = 1, cex = 1)
```



Parameters	Estimates (SE)
Normal Distribution	
mean (*mu*)	176.8 (0.1)
standard deviation (*sigma*)	7.5 (0.1)
Weibull Distribution	
shape (*k*)	23.8 (0.3)
scale (*gamma*)	179.4 (0.1)

Distributions sampling





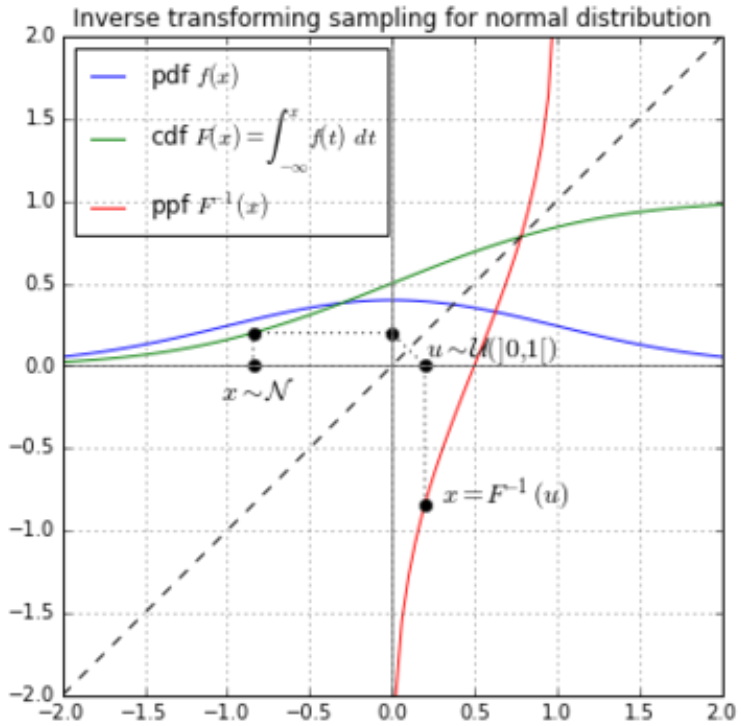

```r
## Distribution Sampling
# to sample from distributions we can use the rnorm and rweibull functions, super easy:
n_samples <- 1000
norm_sample <- rnorm(n_samples, normal_params$estimate["mean"], normal_params$estimate["sd"])
weib_sample <- rweibull(n_samples, weibull_params$estimate["shape"], weibull_params$estimate["scale"])

# we can then plot these as two overlaying histograms
color1 <- rgb(20, 23, 219, max = 255, alpha = 100) # blue
color2 <- rgb(219, 20, 20, max = 255, alpha = 100) # red

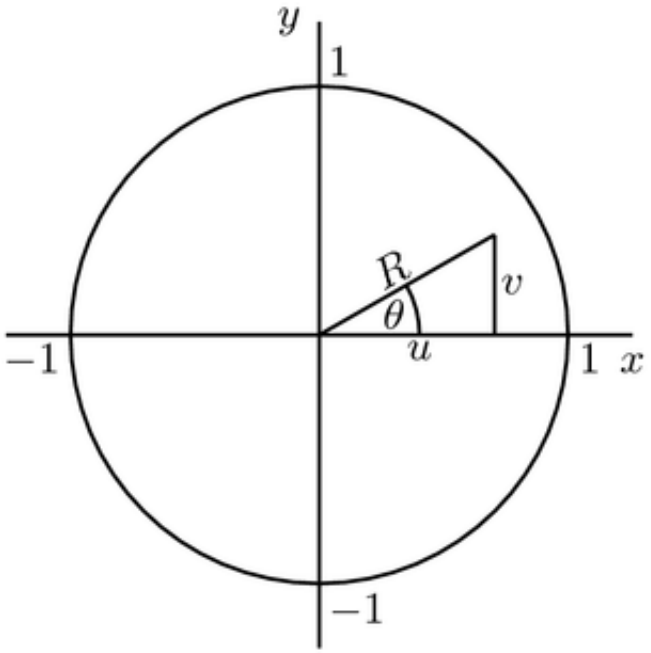
h1 <- hist(norm_sample, plot = FALSE)
h2 <- hist(weib_sample, plot = FALSE)

plot(h1, col = color1, xlab = "Height [cm]", ylab = "Frequency", main = "Sampling")
plot(h2, col = color2, add = TRUE)
legend(x = "topleft", c("normal","Weibull"), fill=c(”blue”,”red”))
```

Distribution sampling



Box-Muller transformation



$$R^2 = u^2 + v^2$$

$$\cos\theta = \frac{u}{R}$$

$$\sin\theta = \frac{v}{R}$$

L1-5 Hypothesis testing

The Scientific Method & Hypothesis testing

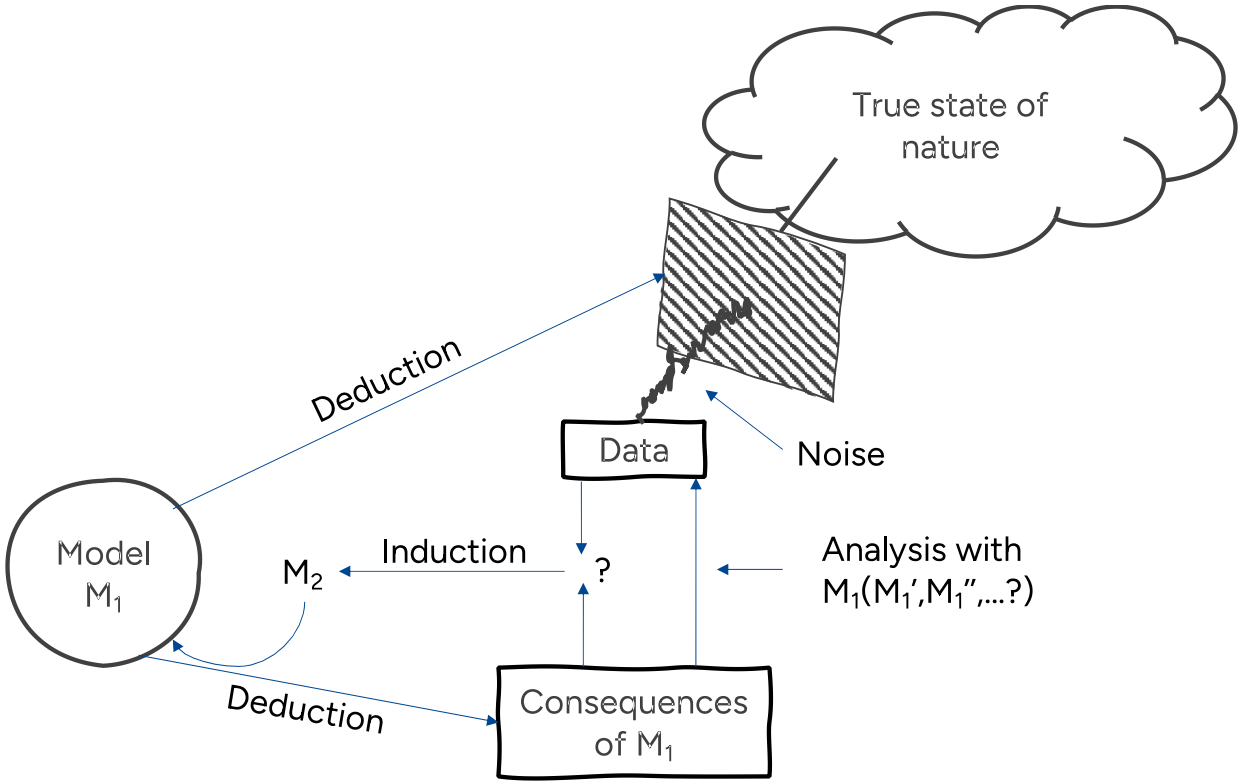
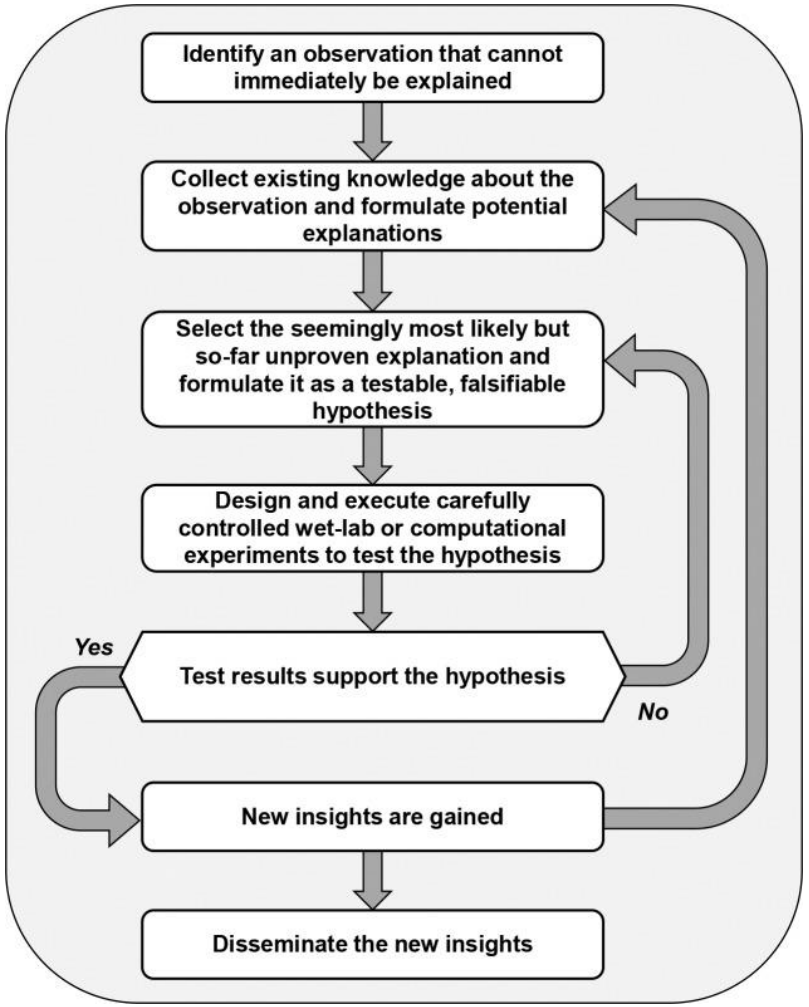


Figure adapted from Box GEP, *et al*. 2005



Voit 2019; Box *et al*. 2005

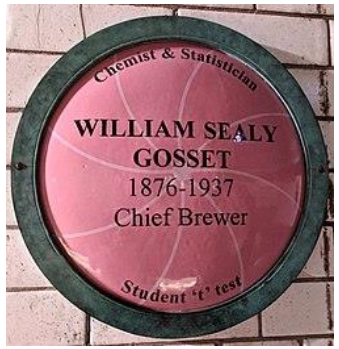
Hypothesis testing Student's t-test

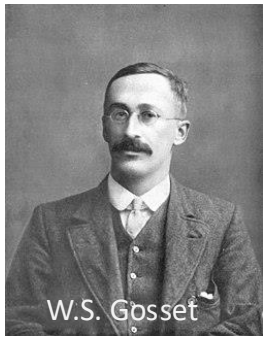


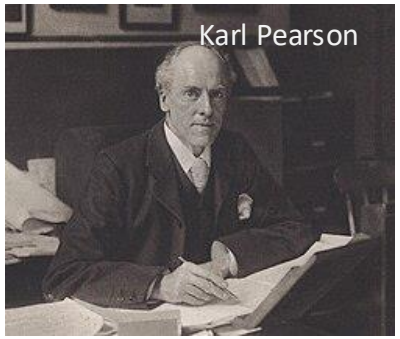




We want to make an inference about a population based on a small sample.




W.S. Gosset


Karl Pearson


Ronald Fisher

	lbs. head corn per acre			Price of head corn in shillings per quarter			cwts. straw per acre			Value of crop per acre in shillings *		
	N. K. D.	K. D.	Diff.	N. K. D.	K. D.	Diff.	N. K. D.	K. D.	Diff.	N. K. D.	K. D.	Diff.
1899	1903	2009	+106	26½	26½	0	19¼	25	+5¾	140½	152	+11½
	1935	1915	− 20	28	26½	−1½	22¾	24	+1¼	152½	145	−7½
	1910	2011	+101	29½	28½	−1	23	24	+1	158½	161	+2½
	2496	2463	− 33	30	29	−1	23	28	+5	204½	199½	−5
	2108	2180	+ 72	27½	27	−½	22½	22½	0	162	164	+2
	1961	1925	− 36	26	26	0	19¾	19⅞	−¼	142	139½	−2½
	2060	2122	+ 62	29	26	−3	24¼	22¼	−2¼	168	155	−13
1900	1444	1482	+ 38	29½	28½	−1	15½	16	+½	118	117½	−½
	1612	1542	− 70	28½	28	−½	18	17¼	−¾	128½	121	−7½
	1316	1443	+127	30	29	−1	14¼	15¾	+1½	109½	116½	+7
	1511	1535	+ 24	28½	28	−½	17	17¼	+¼	120	120½	+½
Average	1841·5	1875·2	+33·7	28·45	27·55	−·91	19·95	21·05	+1·10	145·82	144·68	+1·14
Standard Deviation	—	—	63·1	—	—	·79	—	—	2·25	—	—	6·67
Standard Deviation ÷√8	—	—	22·3	—	—	·28	—	—	·80	—	—	2·40

* Straw being valued at 15*s*. per ton.

Student 1908; Murtagh 2024 – Scientific American

Student's t-test

Research Question Is there an advantage to kiln drying barely seeds before sowing?

H_0: the mean barley crop yield does not differ for non-kiln dried (NKD) barley seeds and kiln dried (KD) barley seeds. $\mu_{Yield,NKD} = \mu_{Yield,KD}$

H_1: the mean barley crop yield differs for NKD and KD barley seeds. $\mu_{Yield,NKD} \neq \mu_{Yield,KD}$

Significance level: *alpha* = 0.05

Experimental data:

NKD: Lbs. head corn per acre	KD: Lbs. head corn per acre
1903	2009
1935	1915
1910	2011
2496	2463
2108	2180
1961	1925
2060	2122
1444	1482
1612	1542
1316	1443
1511	1535

Student 1908; Murtagh 2024 – Scientific American

Student's t-test

```r
# Student's t-test Guinness data
yield_nkd = c(1903, 1935, 1910, 2496, 2108, 1961, 2060, 1444, 1612, 1316, 1511)
yield_kd = c(2009, 1915, 2011, 2463, 2180, 1925, 2122, 1482, 1542, 1443, 1535)

n_nkd <- length(yield_nkd)
x_nkd <- mean(yield_nkd)
sigma_nkd <- sd(yield_nkd)

n_kd <- length(yield_kd)
x_kd <- mean(yield_kd)
sigma_kd <- sd(yield_kd)

df <- n_kd + n_nkd - 2

var_tot <- ((n_nkd - 1)*sigma_nkd^2 + (n_kd - 1)*sigma_kd^2) / (n_nkd + n_kd - 2)

tstat <- (x_nkd - x_kd) /(sqrt(var_tot*((1/n_nkd)+(1/n_kd))))

# at the degrees of freedom (df=20) we get a t-statistics of 0.234
# compared to the t-distribution table at 95% two-sided, 0.234 is smaller than
# 2.086. In fact, 0.234 is smaller than 50%. So, the p-value is larger than
# 0.50. Therefore, we do not reject the null hypothesis.
# table: https://en.wikipedia.org/wiki/Student%27s_t-distribution

# we can also retrieve the p-value using the following function:
p_value <- 2 * pt(abs(tstat), df, lower.tail = FALSE)

# Let's verify our calculations using R's built-in function (or the other way around :-) ):
stat_res <- t.test(yield_kd,yield_nkd, alternative = "two.sided", paired = FALSE,
var.equal = TRUE, conf.level = 0.95)
# comment: indeed, we get a p-value of 0.8173 and can therefore not reject the null hypothesis.
```

$$H_0: \mu_{NKD} = \mu_{KD}$$
$$H_1: \mu_{NKD} \neq \mu_{KD}$$

$$\sigma_p^2 = \frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2 - 2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_p^2}{n_1} + \frac{\sigma_p^2}{n_2}}} = \frac{difference\ in\ means}{standard\ error}$$

$t = 0.331$, look up in t − table at df = 20:

$$p > 0.50, \qquad p = 0.8173$$

We do not reject the null hypothesis at alpha = 0.05.

Student 1908; BMJ 2025

The t-distribution

Formulate H_0 and H_1

Study design and data collection

Carry out t-test

t-statistic: difference in group means over standard error

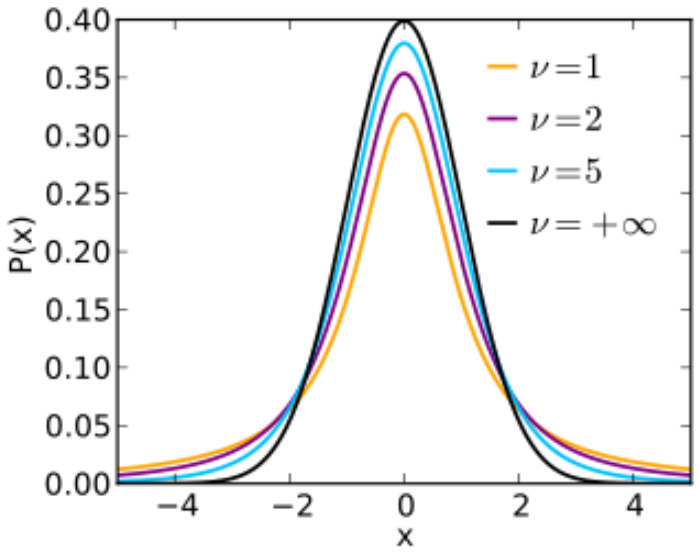
t is n SEs away from the mean.

t-table

p-value: the probability of obtaining the results (or larger difference) if H0 is true

Draw conclusions with regards to H_0 and H_1

The t-distribution



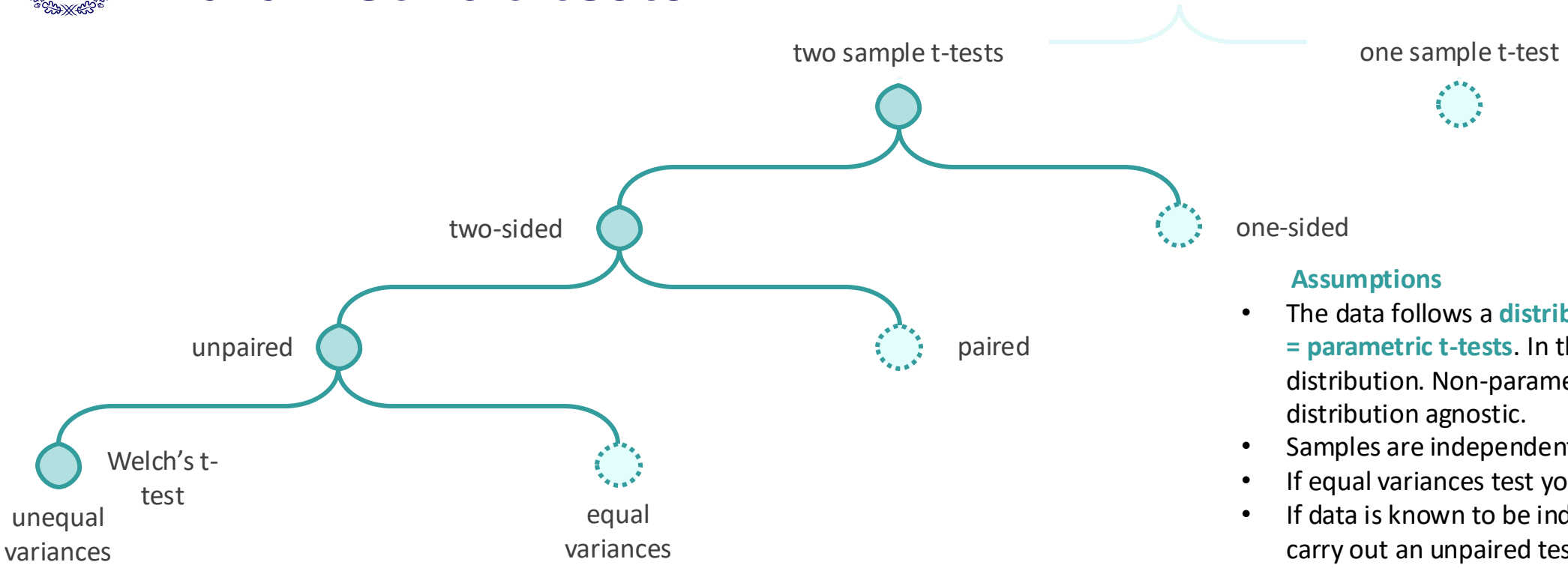
Probability density function:

$$f(t) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{\pi v}\Gamma\left(\frac{v}{2}\right)}\left(1+\frac{x^2}{v}\right)^{\frac{v+1}{2}}$$

v $degrees\ of\ freedom\ (n-1)$

Γ gamma function

Parametric t-tests



Assumptions

- The data follows a **distribution = parametric t-tests**. In this case, normal distribution. Non-parametric tests are distribution agnostic.
- Samples are independent.
- If equal variances test you assume so.
- If data is known to be independent carry out an unpaired test.
- T-tests are robust to moderate deviations in these assumptions.

```
t.test(x1, x2, alternative = c("two.sided",
"less", "greater"), paired = TRUE/FALSE,
var.equal = TRUE/FALSE, conf.level = 0.95)

t.test(yield ~ group, …)
```

The normality assumption data is approximately normally distributed

Recommendation

- **Prior knowledge of the data and its type (The CDA)**
- Visual assessments
  - **q-q plot**
  - histogram
  - boxplot
- Assessment of **skewness** (inference on means) and kurtosis (inference on variance)

Statistical tests

H_0: data is normally distributed.

Shapiro-Wilk test, Kolmogorov-Smirnov test, Anderson-Darling test, and more.

R example: `shapiro.test(data)`

When to use the statistical tests

Some claim that these are useful for low sample sizes where additional data collection is expensive. The problem is that they then lack power. In large datasets these methods are highly sensitive to deviations in normality. Be mindful that each test carried out increases the risk of false positive tests. Engineering, material strengths, failure times, component lifetimes, reliability models, failure rates. Pharmaceutics, manufacturing, milling and formulation development. Economics, risk management models, and more.

The normality assumption visual assessment

```r
# To better understand data in an exploratory fashion we can
visualise it:
## normality
#  Let's take a look at the height data in NHANES again
library(NHANES)

# Filter the data for adult males
nhanes_subset <- NHANES %>%
  filter(Age > 18 & Gender == "male")
n_height <- sum(complete.cases(nhanes_subset$Height))

nhanes_height <- na.omit(nhanes_subset$Height)

# q-q plot: there are many functions for these, for example:
library(car)
qqPlot(nhanes_height, xlab = "normal quantiles",
ylab ="NHANES: height [cm]")

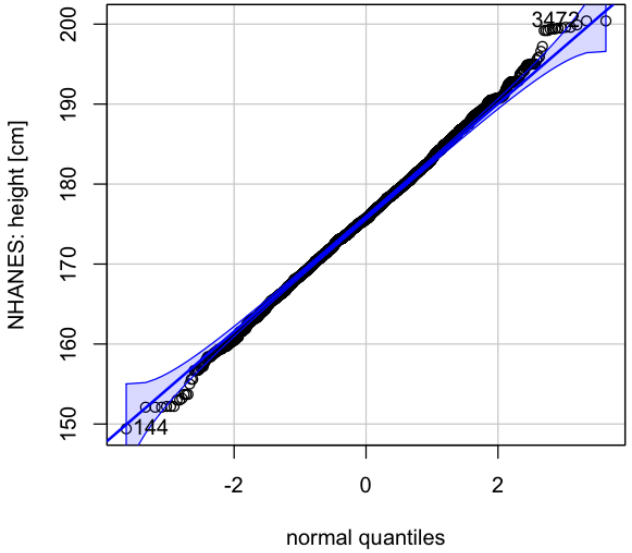
# boxplot
boxplot(nhanes_height, xlab = "NHANES: adult males",
ylab = "Height [cm]")

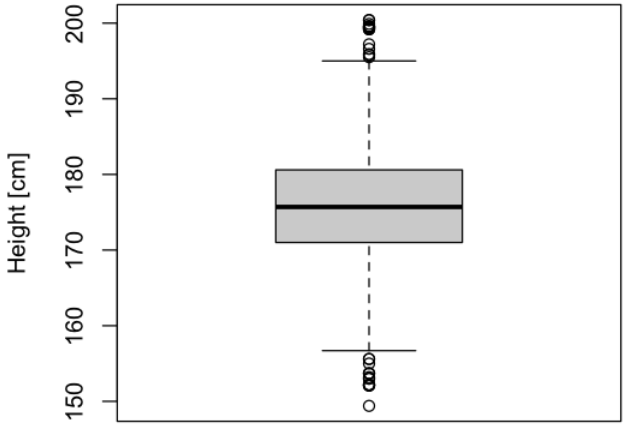
# histogram
normal_params <- fitdistr(nhanes_height, "normal")

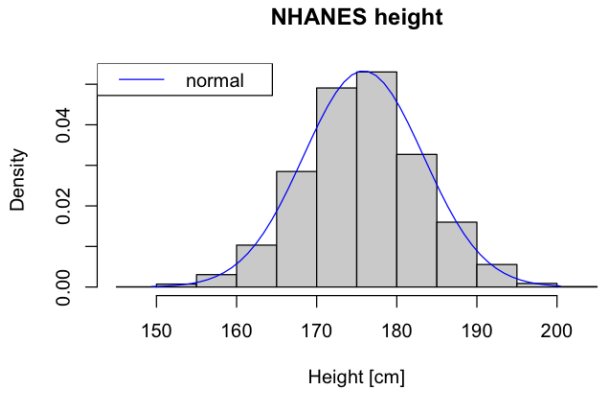
height_min <- min(nhanes_height)
height_max <- max(nhanes_height)
x_pdf <- seq(height_min, height_max, 1)

y_norm <- dnorm(x_pdf, normal_params$estimate["mean"],
normal_params$estimate["sd"])

hist(nhanes_height, freq = FALSE, xlab = "Height [cm]",
ylab = "Density", main = "NHANES height")
lines(x_pdf, y_norm, col="blue")
legend(x = "topleft", c("normal"), col=c("blue"),
lty = 1, cex = 1)
```

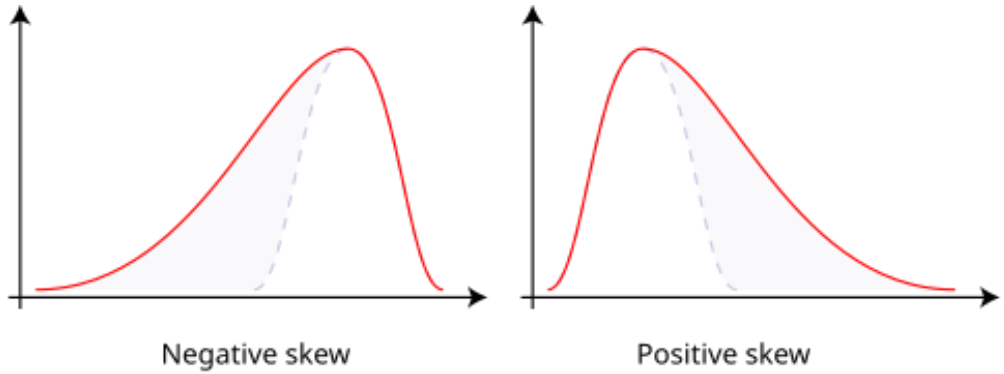


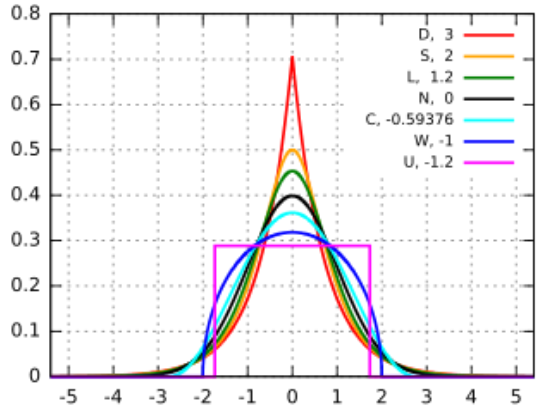





```
> summary(nhanes_height)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  149.4   171.0   175.7   175.8   180.6   200.4
>
```

The normality assumption skewness and kurtosis





$$Skewness = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3}{(n-1)\sigma^3}$$

σ *standard deviation,*
$\bar{x}$ *mean,*
n *sample size.*

Symmetry: -0.5 to 0.5

$$Kurtosis = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^4}{(N-1)\sigma^4}$$

σ *standard deviation,*
$\bar{x}$ *mean,*
n *sample size.*

Excess kurtosis: -2 to 2

R example: `library(moment), skewness(data), kurtosis(data)`

The equal variances assumption

Assumption

Variance is equal across groups.

Recommendation

- Assume unequal variance unless we know otherwise to be true.
  See Welch's t-test.

Statistical tests

H_0: variances in the two groups are equal.

F-test, Levene's test.

R example: `library(car), leveneTest(y ~ group, data)`

When to use the statistical tests

Testing hypotheses related to homogeneity of variances across groups. Be mindful that each test carried out increases the risk of false positive tests.

Ruxton et al. 2006; Delacre *et al*. 2017

Paired or unpaired tests

- H_0: $\mu_x = \mu_y$ *with independent samples*
- H_0: $\mu_D = 0$ *with dependent samples*

A paired t-test accounts for the reduced variability in the paired experiment.

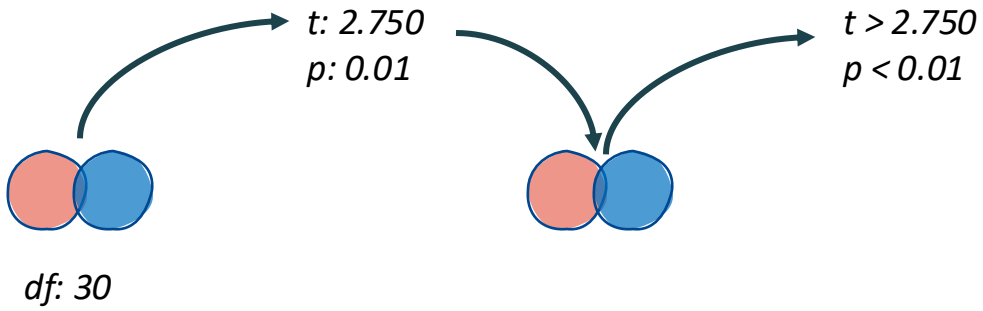
The meaning of the *p*-value

***P*-value** is the probability of obtaining the test statistic (*t*) or a more extreme value if the null hypothesis is true (Fife & Rogers 2021).

***P*-value** is a measure of the discrepancy of the fit of a model or null hypothesis H_0 to data *y* (Gelman 2013).

$$P(T(y^{rep}) > T(y)|H_0)$$

y^{rep} is hypothetical replication under H_0.
T is the test statistic (deviation from the mean).



It is convenient to take this point [p=0.05] as a limit in judging whether a deviation ought to be considered significant or not.


Ronald Fisher

Alpha and power and being in the wrong

Power=1 – Beta
Power: 80%, Type II error: 20%

	Do not reject H_0	Reject H_0
H_0 is true	Correct outcome	Incorrect outcome: Type I error *Alpha*
H_0 is false	Incorrect outcome: Type II Error *Beta*	Correct outcome

A false positive
False significant outcome

A false negative
False nonsignificant outcome

Significance level=Alpha
Significance level: 0.05, Type I error: 5%

Multiple comparisons

The probability of at least one Type I error
Family-Wise Error Rate (FWER) for multiple testing:

$$FWER = P(TypeIError \geq 1)$$

$$P(TypeIError \geq 1) = 1 - P(TypeIError = 0)$$
$$= 1 - 0.95^{ntests}$$

$$Example\ at\ \propto = 0.05\ for\ 10\ tests:$$
$$= 1 - 0.95^{10}$$
$$= 0.40$$

Bonferroni correction:

$$Reject\ H_{0,i}\ if\ p_i \leq \frac{\alpha}{m}$$

May increase risk of type II error.
Many other less conservative corrections exist.

R for example:

```
library(stats)
p_corrected <- p.adjust(p_values,
method = ”bonferroni”)
```

Concluding remarks and advice

Given the problems described and concerns raised by Pernerger[12] what is appropriate statistical advice for authors contemplating multiple statistical testing? As stated by Streiner and Norman[3], to correct or not to correct depends on the circumstances of the study.

1. No correction would be advised in the following circumstances:
   - if the study is restricted to a small number of planned comparisons.[3, 61]
   - if a study is exploratory involving *post-hoc* testing of unplanned comparisons which are regarded as hypotheses for further investigation.
   - if multiple usage of a simple test such as '*t*' or '*r*' is envisaged, if it is the results of the *individual* tests that are important. Instead, the exact *p* values for each individual test should be quoted and discussed appropriately.
   - if it is imperative to avoid a type II error.
2. A Bonferroni correction should be considered if:
   - a single test of the 'universal null hypothesis' (H_o) that *all* tests are not significant is required.
   - it is imperative to avoid a type I error.
   - a large number of tests are carried out without preplanned hypotheses in an attempt to establish any results that may be significant.[3]
3. If a correction is required but the original Bonferroni procedure is regarded as too conservative, then a possible alternative is to use the Bonferroni-Holm[62] or Hochberg[63] methods. Both correct for the family-wise error rate, employ sequential testing, and are less conservative than the original Bonferroni method. In both methods, *p* values for the various tests are ranked from low to high, i.e., representing the most to the least significant difference. For the Bonferroni-Holm method with comparisons 1 to *T* and $p = 0.05$ as the 'α' level, if the most significant $p_1 < \alpha/T$ then the first H_o is rejected. If the first H_o is rejected the analysis is terminated, otherwise the next highest *p* (p_2) is tested. If $p_2 < \alpha/T\text{-}1$ then the second H_o is rejected. The analysis then proceeds to subsequent steps, terminating when a H_o is not rejected.[64] By contrast, the Hochberg method tests the largest *p* value (p_T) first. If $p_T < \alpha$, then all H_o up to and including p_T are rejected and the analysis terminated. If the first H_o is not rejected, the analysis proceeds to test the second highest $p_{T\text{-}1}$ and if $p_{T\text{-}1} < \alpha/2$, all H_o up to and including $p_{T\text{-}1}$ are rejected. The analysis then proceeds to subsequent steps, terminating when a H_o in the sequence is rejected.[64]
4. In all studies involving multiple testing, investigators should clearly describe the design of their study including whether specific hypothesis testing or hypothesis generation is envisaged, provide a rationale for their choice of adjusting or not adjusting *p* values, justify the method selected if *p* values are adjusted, and quote adjusted *p* values correctly.

Armstrong 2014; Larsen & Marx 2006

Effect size

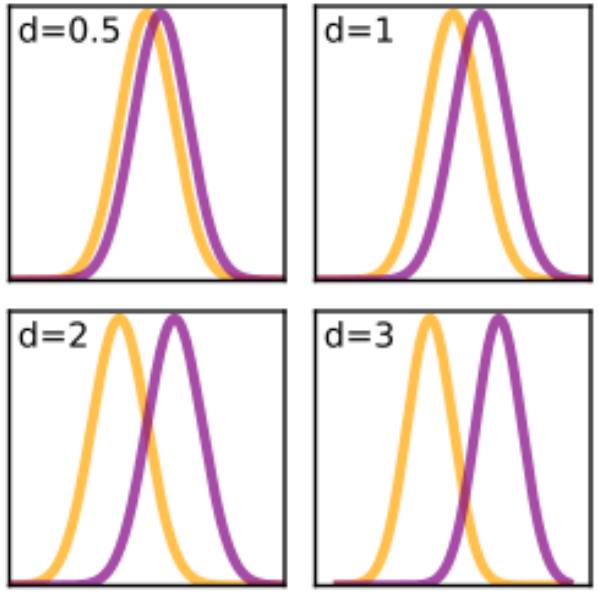
Estimate of the effect in a standardised form

$$Cohen's\ d = \frac{\bar{x}_T - \bar{x}_C}{\sigma_{pooled}}$$

Interpretation: d=0.2 small, d=0.5 moderate, d=0.8 large effect (Cohen 1988).

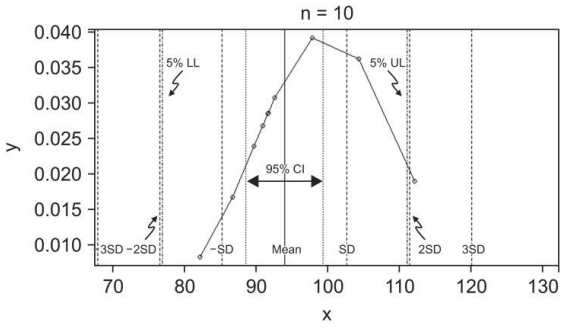
R example:

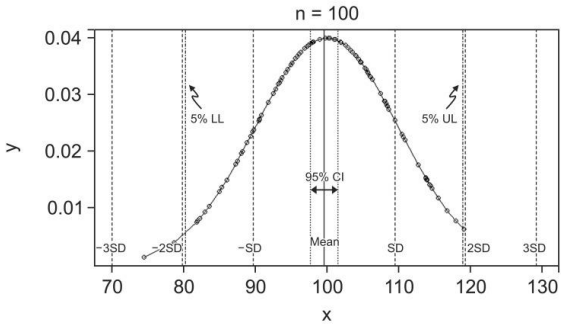
```
library(lsr)
effect_size <- cohensD(x, y, data)
```

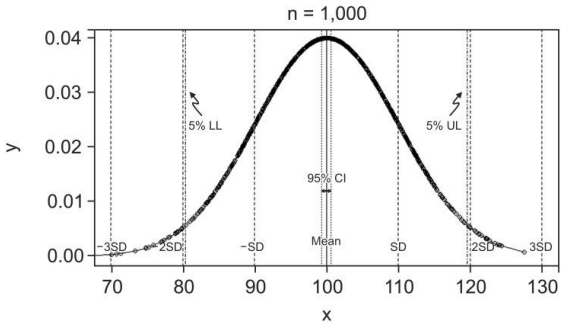


ES	Standardizer	Use
Cohen's d_{pop}	σ (population)	Independent groups, use in power analyses when population σ is known, σ calculated with *n*
Cohen's d_s	Pooled *SD*	Independent groups, use in power analyses when population σ is unknown, σ calculated with *n*-1
Hedges' *g*	Pooled *SD*	Independent groups, corrects for bias in small samples, report for use in meta-analyses
Glass's Δ	*SD* pre measurement or control condition	Independent groups, use when experimental manipulation might affect the *SD*
Hedges' g_{av}	$(SD_1 + SD_2)/2$	Correlated groups, report for use in meta-analyses (generally recommended over Hedges' g_{rm})
Hedges' g_{rm}	*SD* difference scores corrected for correlation	Correlated groups, report for use in meta-analyses (more conservative then Hedges' g_{av})
Cohen's d_z	*SD* difference scores	Correlated groups, use in power analyses

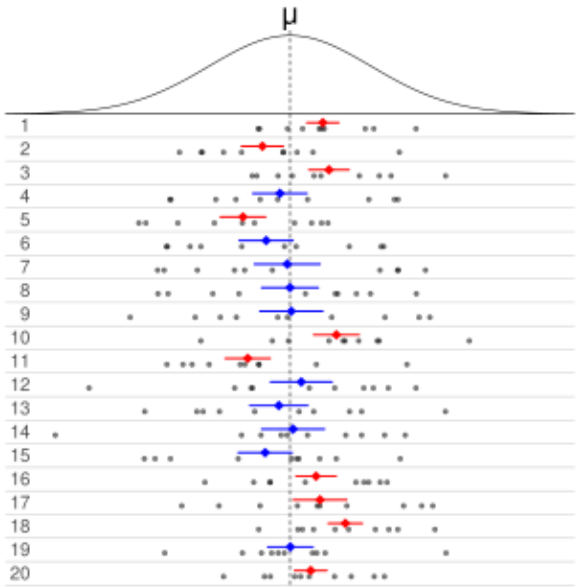
Confidence intervals







n	10	100	1,000
Mean	104.8	99.6	99.9
SD	7.1	9.84	10
SEM	2.2	1	0.31
95% CI	100.4−109.2	97.7−101.6	99.3−100.6
5% alpha	90.9−118.7	80.3−118.9	80.3−119.5



Upon repetition of an experiment the true population mean would fall within the 95% confidence interval of the mean 95% of the times.

For a continuous, normally distributed variable CI of the population mean:

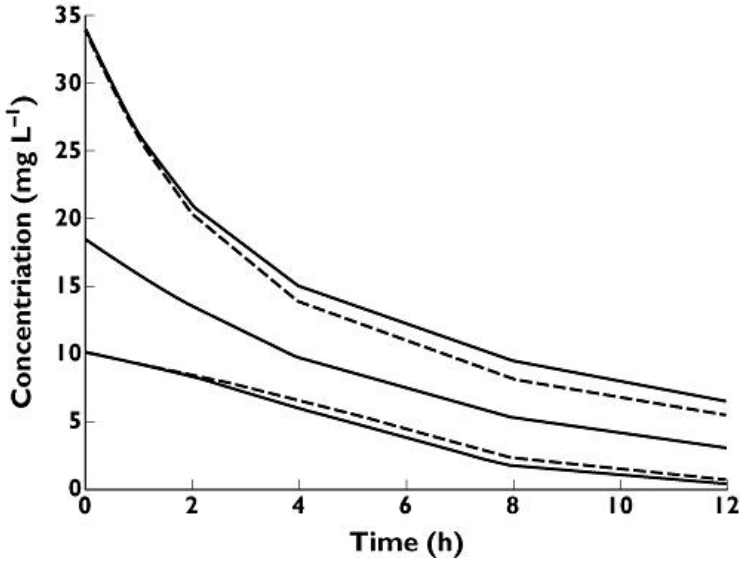
$$CI = \bar{x} \pm z\frac{\sigma}{\sqrt{n}}$$

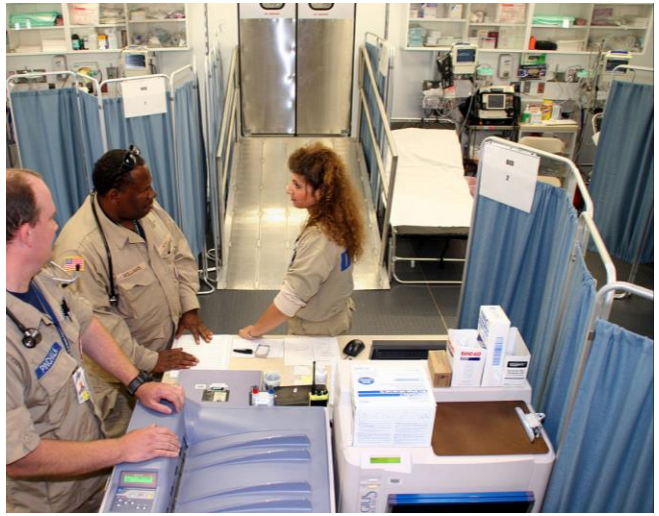
CI $confidence\ interval$
$\bar{x}$ $sample\ mean$
σ $sample\ standard\ deviation$
n $sample\ size$
z $z\ score, CI: 95\%, 1.96$
$z = (x_i - \mu)/\sigma$

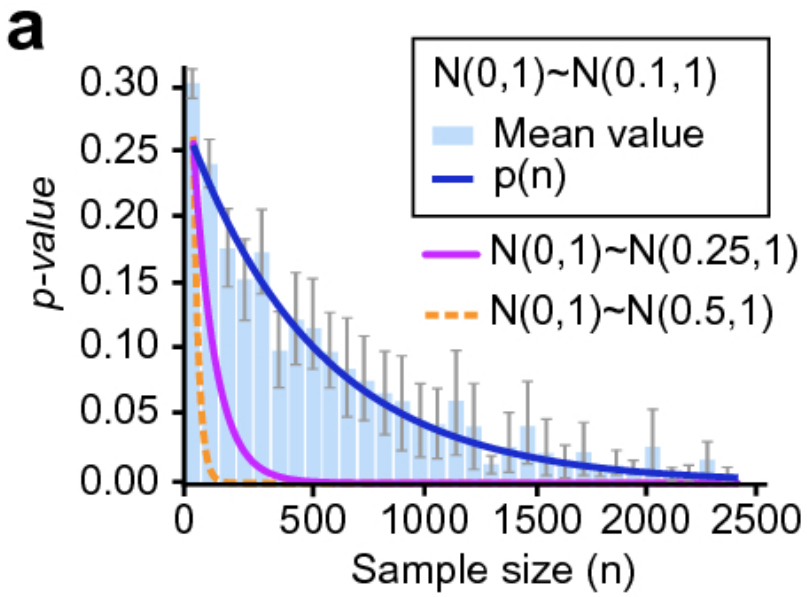
R example:
many statistical tests will output CIs.

```
CIs <- confint(model, level=0.95)
```

Clinical significance

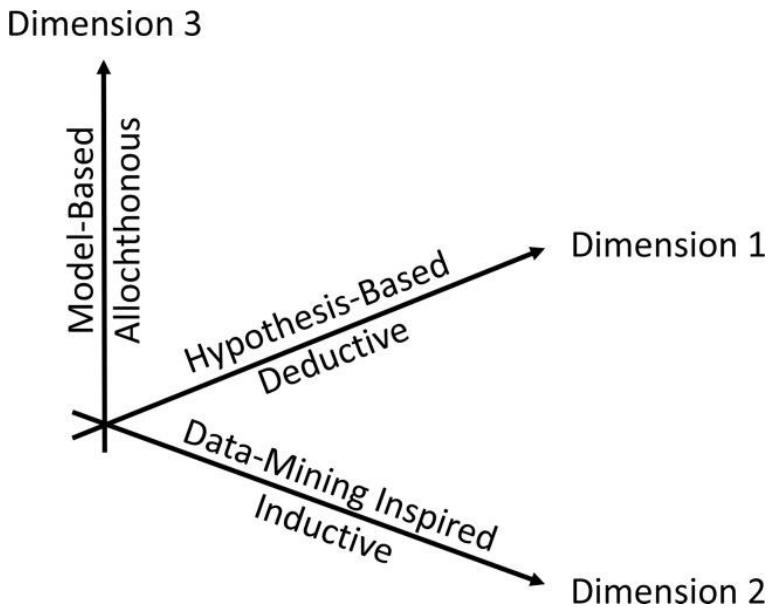


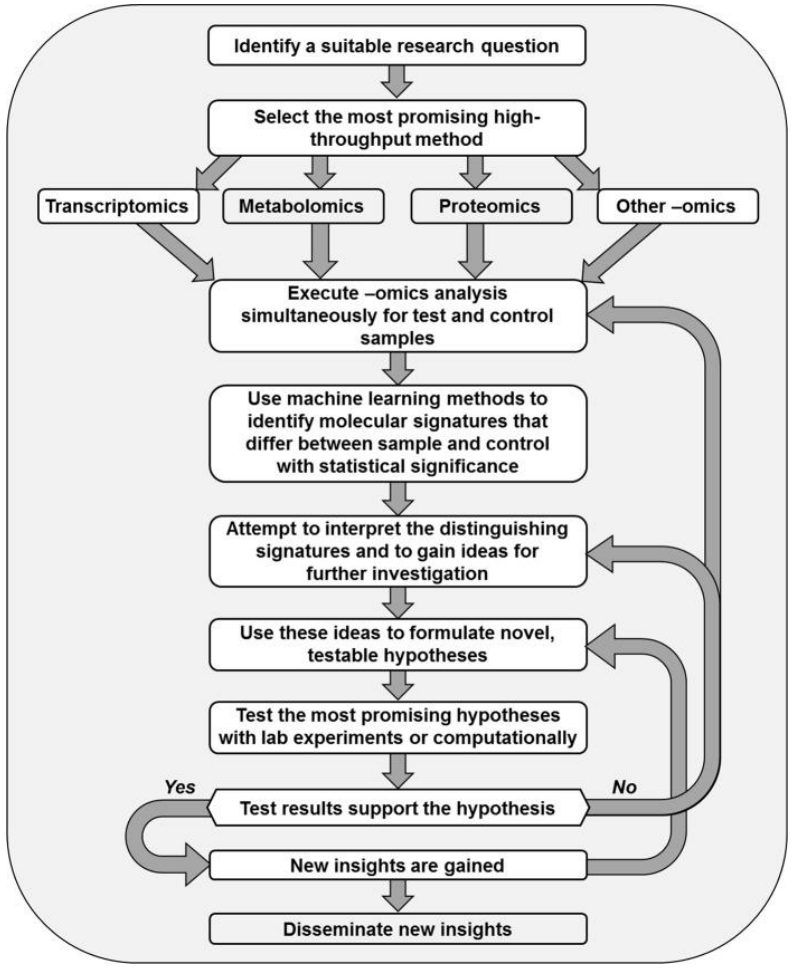


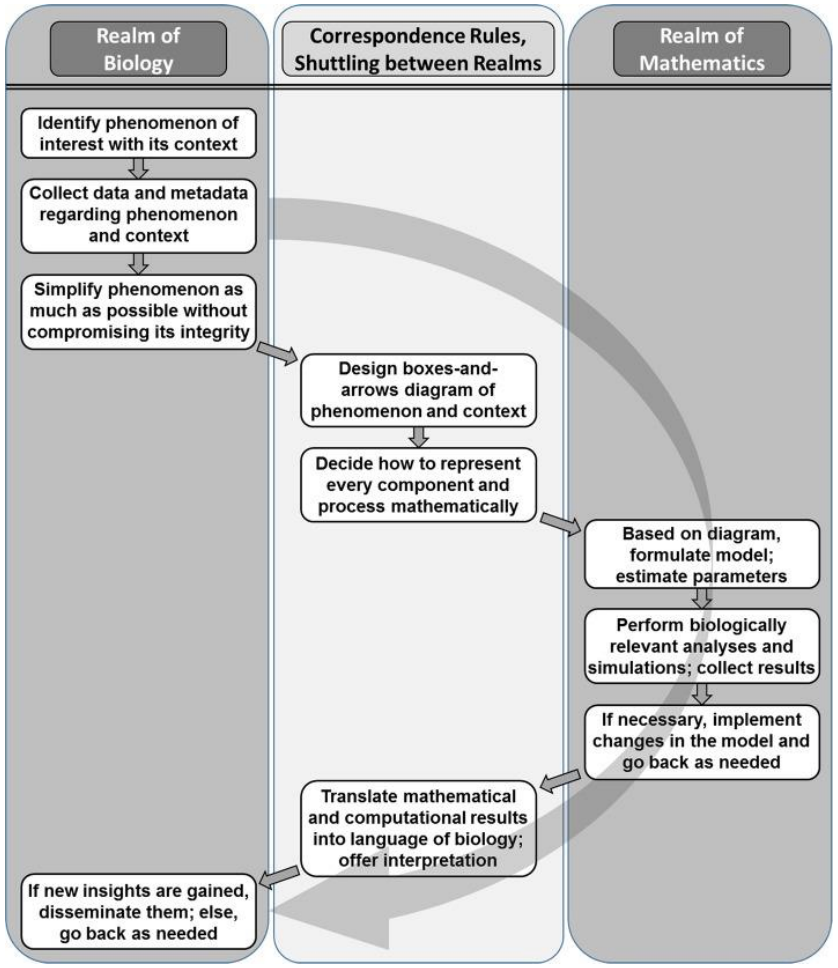


Duffull *et al*. 2011; Gómez-de-Mariscal et al. 2021

The Scientific Method & Estimation Statistics







Voit 2019

L1-6 Nonparametric statistical tests (I)

What are nonparametric tests?

Distribution-free tests ≠ parametric tests assuming other than the normal distribution

A set of methods rely on:

sign:

$$\operatorname{sgn} x := \begin{cases} -1 & if\ x < 0 \\ 0 & if\ x = 0 \\ +1 & if\ x > 0 \end{cases}$$

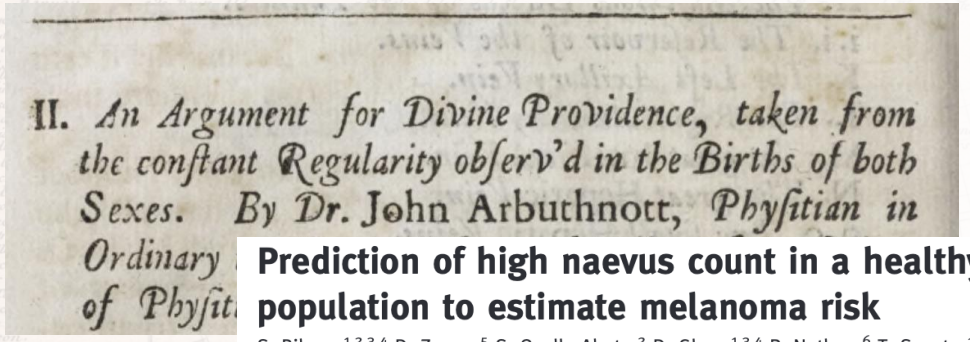
sort & rank:

x	*rank*
23	*2*
105	*4*
32	*3*
14	*1*

But methods are by no means limited to the above.

The Sign Test

One sample or paired test of medians



II. *An Argument for Divine Providence, taken from the conſtant Regularity obſerv'd in the Births of both Sexes. By Dr.* John Arbuthnott, *Phyſitian in Ordinary of Phyſit*

Prediction of high naevus count in a healthy U.K. population to estimate melanoma risk

S. Ribero,[1,2,3,4] D. Zugna,[5] S. Osella-Abate,[2] D. Glass,[1,3,4] P. Nathan,[6] T. Spector[1] and V. Bataille[1,7]

[1]Department of Twin Research and Genetic Epidemiology, King's College London, St Thomas' Campus, Westminster Bridge Road, London SE1 7EH, U.K.
[2]Section of Dermatology and [5]Unit of Cancer Epidemiology – CERMS; Department of Medical Sciences, University of Turin, Torino, Italy
[3]Imperial College London, London, U.K.
[4]Department of Dermatology, London North West Healthcare NHS Trust, London, U.K.
[6]Mount Vernon Cancer Network and [7]Department of Dermatology; West Herts NHS Trust, Herts, U.K.

Linked Comment: Naldi and Cazzaniga, *Br J Dermatol* 2016; **174**: 261–262

H_0: median mole count on the left arm equals the right arm

$$\eta_{moles,left} = \eta_{moles,right}$$

H_1: median mole count on the left arm differs from the right arm

$$\eta_{moles,left} \neq \eta_{moles,right}$$

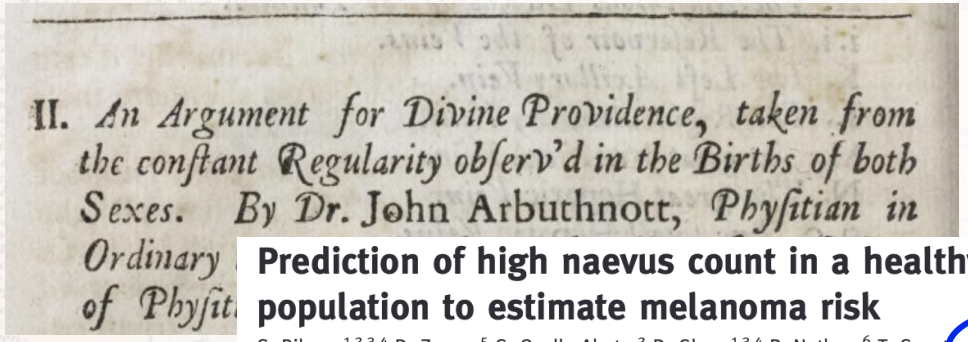
Right	*Left*	*Diff.*
6	7	+
6	6	0
1	1	0
1	2	+
5	6	+
1	2	+
3	3	0
4	5	+
5	4	-
4	4	0
3	3	0
2	3	-
3	3	0
5	4	+
7	9	-

+: 6
-: 3
0: 6

Binomial test against probability p: 0.5

The Sign Test

One sample or paired test of medians



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[2]Section of Dermatology and [5]Unit of Cancer Epidemiology – CERMS; Department of Medical Sciences, Universit
[3]Imperial College London, London, U.K.
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Linked Comment: Naldi and Cazzaniga, *Br J Dermatol* 2016; **174**: 261–262

Right	*Left*	*Diff.*
6	*7*	*+*
6	*6*	*0*
1	*1*	*0*

+: 6
-: 3
0: 6

H_0: median mole count on the left arm equals the right ar

$\eta_{moles,left} = \eta_{moles,right}$

H_1: median mole count on the left arm differs from the rig

$\eta_{moles,left} \neq \eta_{moles,right}$

```
binom.test(n_sucesses, n_trials, alternative =
"two.sided, conf.level = 0.95)
>      Exact binomial test

data:  6 and n_samples
number of successes = 6, number of trials = 15,
p-value = 0.6072
alternative hypothesis: true probability of
success is not equal to 0.5
95 percent confidence interval:
 0.1633643 0.6771302
sample estimates:
probability of success
                   0.4
```

R

The Wilcoxon Signed Rank Test

Prediction of high naevus count in a healthy U.K. population to estimate melanoma risk

S. Ribero,[1,2,3,4] D. Zugna,[5] S. Osella-Abate,[2] D. Glass,[1,3,4] P. Nathan,[6] T. Spector[1] and V. Bataille[1,7]

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[6]Mount Vernon Cancer Network and [7]Department of Dermatology; West Herts NHS Trust, Herts, U.K.

Linked Comment: Naldi and Cazzaniga, *Br J Dermatol* 2016; **174**: 261–262

$$T = \sum_{i=1}^{N} \operatorname{sgn}(X_i) R_i$$

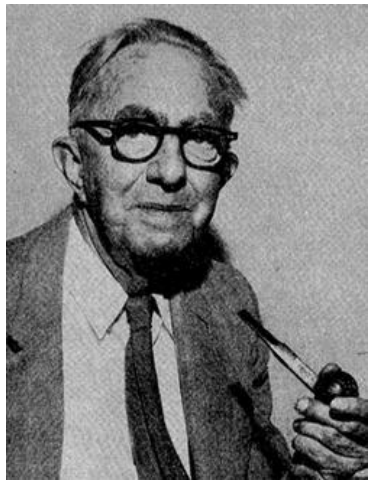
X $data$

R $rank$

```
wilcox.test(right_arm, left_arm, paired =
TRUE, exact = TRUE)
>      Wilcoxon signed rank test with
continuity correction

data:  right and left
V = 16, p-value = 0.1314
alternative hypothesis: true location shift
is not equal to 0
```

R



H_0: median of pairwise difference in mole count on the left arm equals the right arm does not differ from zero.

$\eta_{moles,left} = \eta_{moles,right}$

H_1: median mole count on the left arm differs from the right arm

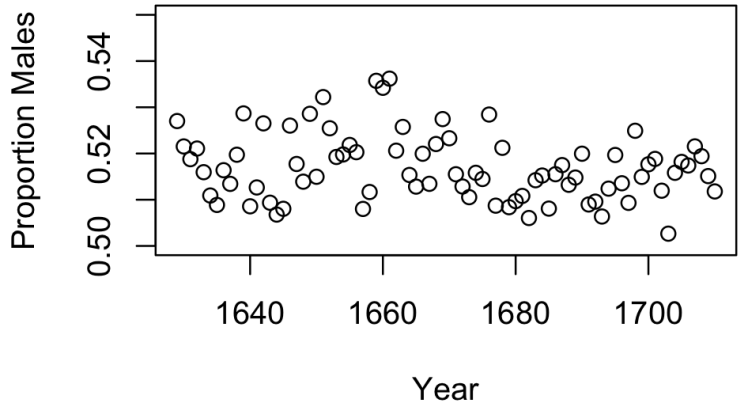
$\eta_{moles,left} \neq \eta_{moles,right}$

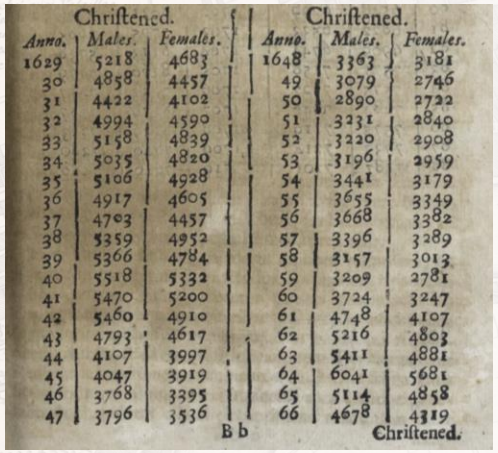
Wilcoxon Rank Sum/Mann-Whitney U Test
Unpaired data

Tests of Proportion (parametric)

II. *An Argument for Divine Providence, taken from the conſtant Regularity obſerv'd in the Births of both Sexes. By Dr.* John Arbuthnott, *Phyſitian in Ordinary to Her Majeſty, and Fellow of the College of Phyſitians and the Royal* Society.

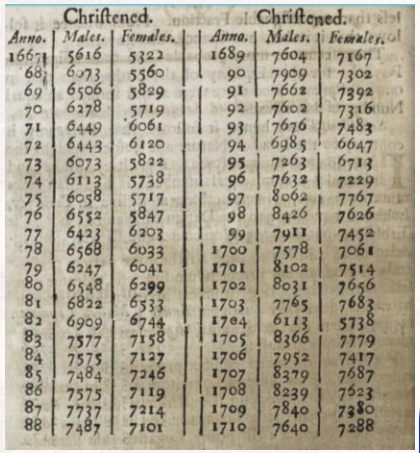
Provident nature, by the proposal of its wise Creator, brings forth more Males than Females, and that in a almost constant proportion.





Anno.	Males.	Females.	Anno.	Males.	Females.
1629	5218	4683	1648	3363	3181
30	4858	4457	49	3079	2746
31	4422	4102	50	2890	2722
32	4994	4590	51	3231	2840
33	5158	4839	52	3220	2908
34	5035	4820	53	3196	2959
35	5106	4928	54	3441	3179
36	4917	4605	55	3655	3349
37	4703	4457	56	3668	3382
38	5359	4952	57	3396	3289
39	5366	4784	58	3157	3013
40	5518	5332	59	3209	2781
41	5470	5200	60	3724	3247
42	5460	4910	61	4748	4107
43	4793	4617	62	5216	4803
44	4107	3997	63	5411	4881
45	4047	3919	64	6041	5681
46	3768	3395	65	5114	4858
47	3796	3536	66	4678	4319

Bb Christened.



Anno.	Males.	Females.	Anno.	Males.	Females.
1667	5616	5322	1689	7604	7167
68	6073	5560	90	7909	7302
69	6506	5829	91	7662	7392
70	6278	5719	92	7602	7316
71	6449	6061	93	7676	7483
72	6443	6120	94	6985	6647
73	6073	5822	95	7263	6713
74	6113	5738	96	7632	7229
75	6058	5717	97	8062	7767
76	6552	5847	98	8426	7626
77	6423	6203	99	7911	7452
78	6568	6033	1700	7578	7061
79	6247	6041	1701	8102	7514
80	6548	6299	1702	8031	7656
81	6822	6533	1703	7765	7683
82	6909	6744	1704	6113	5738
83	7577	7158	1705	8366	7779
84	7575	7127	1706	7952	7417
85	7484	7246	1707	8379	7687
86	7575	7119	1708	8239	7623
87	7737	7214	1709	7840	7380
88	7487	7101	1710	7640	7288

H_0: the observed proportion (p_{obs}) of male births is equal to the expected proportion, p_e: 0.5.

$$p_{obs} = p_e$$

H_1: the observed proportion of male births is not equal (is different) to the expected proportion, 0.5.

$$p_{obs} \neq p_e$$

```
n_successes <- sum(sign(dat_birth$males - dat_birth$females) > 0)
n_trials <- length(dat_birth$year)

# proportion test
prop.test(n_successes, n_trials, alternative = "two.sided", correct = TRUE)

>            1-sample proportions test with continuity correction

data:  n_successes out of n_trials, null probability 0.5
X-squared = 80.012, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.9442369 1.0000000
sample estimates:
p
1
```

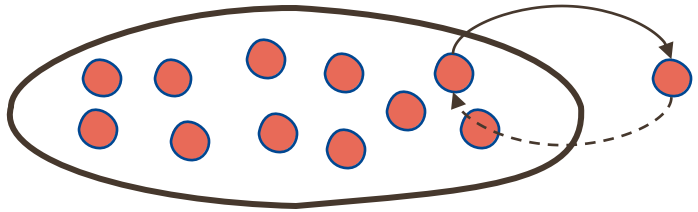
R

Nonparametric bootstrapping in short
We will return to this topic later

Let $\hat{\theta}$ be the estimate of a test statistic (mean, median, etc) based on sample $x_1 \ldots x_n$.

$$\hat{\theta}, \qquad \hat{\sigma} = \sigma(F)$$

We have a point estimate of a statistic from a probability distribution and want to estimate its standard error (or its accuracy).



```
data_boot <- replicate(10000, sample(data, replace = TRUE)
library(boot)
```

For the samples, calculate the test statistic.

Then, calculate the confidence interval across replicates.

Efron & Tibshirani 1994

Permutation Test

- Specify hypotheses and test statistic
- Assuming H_0 to be true: sample across the data (permutations) to establish a distribution for the test statistic (*e.g.*, 10,000 times)
- Calculate and assess the p-value:

$$p = \frac{n\ (test\ statistics > observed)}{m\ permutations}$$

- Assumes that observations are exchangeable (randomised experiment), or shape of or symmetric distributions to allow exchangeability

H0: median CGAS does not differ in the treatment group compared to the controls.
H1: median CGAS differs in the treatment group compared to controls.
Test statistic: absolute difference in median

```
library(perm)
```

Lorentzen *et al. BMC Psychology* (2020) 8:25
https://doi.org/10.1186/s40359-020-0393-x

BMC Psychology

RESEARCH ARTICLE Open Access

A randomized controlled trial of a six-session cognitive behavioral treatment of emotional disorders in adolescents 14–17 years old in child and adolescent mental health services (CAMHS)

Veronica Lorentzen[1,2*], Kenneth Fagermo[2], Bjørn Helge Handegård[3], Ingunn Skre[1,4] and Simon-Peter Neumer[5,3]

p=0.00114

```r
# calculate the test statistic of observed
tstat_obs <- abs(median(data_cgas$CGAS[data_cgas$Group == "treat"])
-
      median(data_cgas$CGAS[data_cgas$Group == "control"]))
# carry out permutation test in R
np_sample <- length(data_cgas$Group)
n_perm <- 100000
dv_perm <- data_cgas$CGAS
# permutation data out
data_permutation <- matrix(0, nrow = np_sample, ncol = n_perm)
# generate permutation samples
for(i in 1:n_perm){
  data_permutation[,i] <- sample(dv_perm, size = np_sample, replace
= FALSE)
}
# calculate test-statistics
tstat_perm <- rep(0,n_perm)

for(j in 1:n_perm){
  tstat_perm[j] <- abs(median(data_permutation[data_cgas$Group ==
"treat",j]) -
      median(data_permutation[data_cgas$Group == "control",j]))
}
# p-value calculation
p_value <- mean(tstat_perm >= tstat_obs)
```

R

Bootstrapping the median difference

Test Statistic	*2.5% confidence interval (CI)*	*Median*	*97.5% CI*
Median difference	*6*	*10*	*13*

```r
## Bootstrapping
#  OK, we get a p-value of 0.00114, let's estimate the confidence
interval
#  of the difference in CGAS score across the groups.

# number of bootstrap samples
n_boot <- 10000

# We calculate our test statistic, for example:
tstatb_obs <- (median(data_cgas$CGAS[data_cgas$Group == "treat"])
               - median(data_cgas$CGAS[data_cgas$Group ==
"control"]))

n_control <- length(data_cgas$CGAS[data_cgas$Group == "control"])
n_treat <-  length(data_cgas$CGAS[data_cgas$Group == "treat"])

# then we can carry out the bootstrap
boot_control <- matrix(sample(data_cgas$CGAS[data_cgas$Group ==
"control"],
                              size = n_boot*n_control, replace =
TRUE),
                       ncol = n_boot, nrow = n_control)
boot_treat <-  matrix(sample(data_cgas$CGAS[data_cgas$Group ==
"treat"],
                              size = n_boot*n_treat, replace =
TRUE),
                       ncol = n_boot, nrow = n_treat)

median_bcontrol <- apply(boot_control, 2, median)
median_btreat <- apply(boot_treat, 2, median)
boot_mediandif <- median_btreat - median_bcontrol

# Construct the 95% confidence intervals based on percentiles
b_ci95 <- quantile(boot_mediandif, c(0.025,0.5,0.975))
```

Efron & Tibshirani 1994

Wilcoxon Mann Whitney test

Wilcoxon Rank Sum/Mann-Whitney U Test

Two unpaired datasets (X and Y) that are at least ordinal.

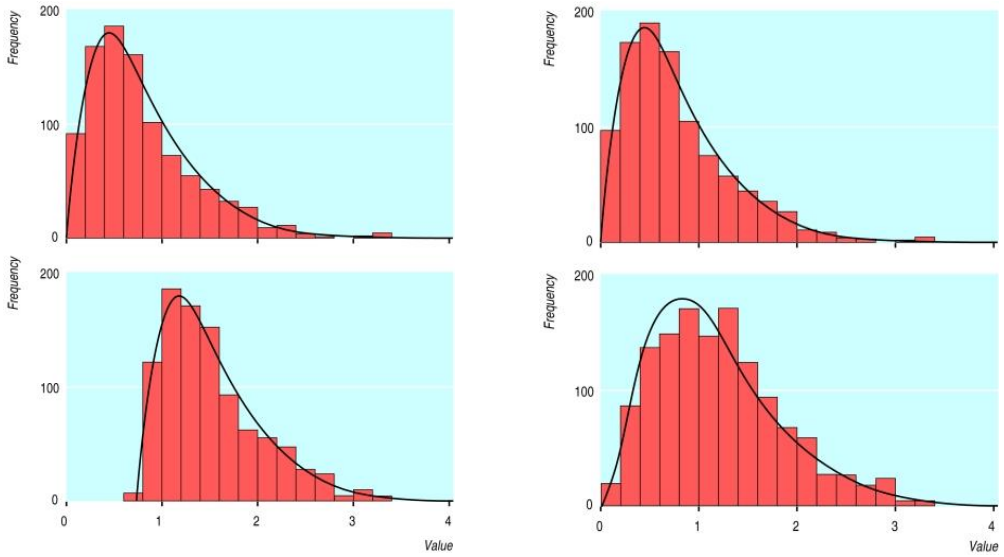
Carries out pair-wise comparisons for $n_x \cdot n_y$, then:

$\mathbf{H_0}$: $P(x_i > y_i) = ½$

$\mathbf{H_1}$: $P(x_i > y_i) \neq ½$

A test of medians if distributions are the same.

If not, a test of distributions.



Lorentzen et al. BMC Psychology (2020) 8:25
https://doi.org/10.1186/s40359-020-0393-x

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```
wilcox.test(CGAS ~ Group, data = data_cgas)
```

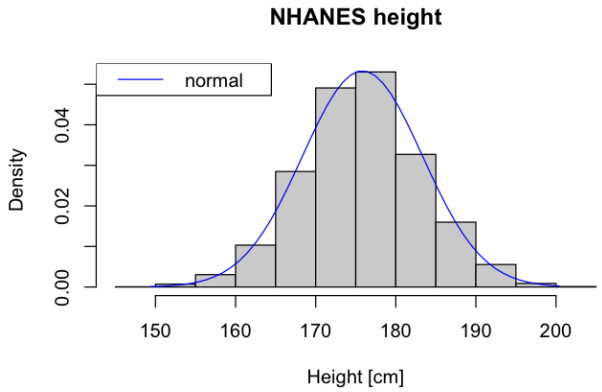
```
        Wilcoxon rank sum test with continuity correction

data:  CGAS by Group
W = 129, p-value = 0.0003652
alternative hypothesis: true location shift is not equal to 0
```

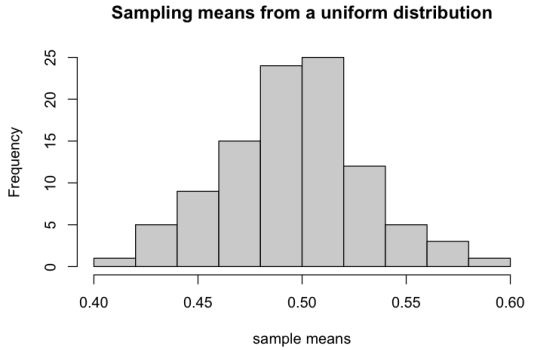
Hart 2001

L1-7 Some thoughts on selection of statistical tests

Nonparametric vs. parametric t-tests
some often-noted considerations



Nonparametric tests are: "Always valid but not always efficient"



If data is normal, then use a parametric t-test that assumes normality.

Rarely is real-world data normally distributed. However, t-tests are very robust to violations of this assumption. Approximately normal is good enough.

note: using data to inform analysis is not compatible with a confirmatory data analysis (CDA) approach.

If data is normal, a nonparametric test will still be valid. May compromise some power and robustness.

Likewise, if data deviates from normality, nonparametric alternatives may be more robust and have more powerful.

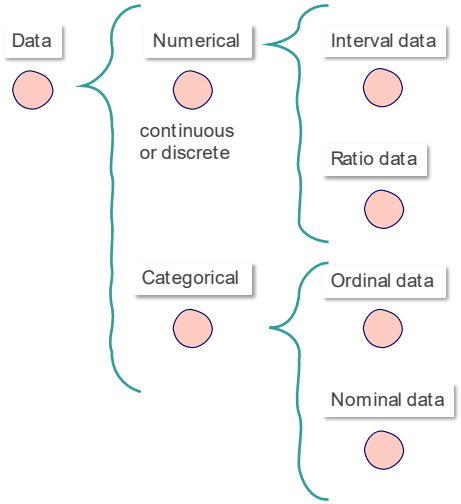
Nonparametric when deviating from normality, skewness, low sample size.

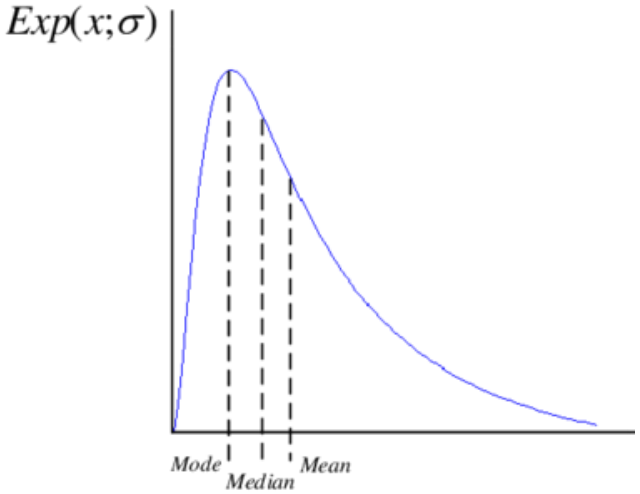
According to the central limit theorem parametric t-tests are generally robust for $n > 30$. Regardless of data.

This is a simplification invoked by many researchers and textbook authors. Might hold true for some distributions with defined means and variances. The exact *n* value will differ depending on the departure from normality and may be implausibly large (Zhang et al. 2023).

Nahm 2016; Zhang *et al*. 2023

Nonparametric vs. parametric t-tests
what is a reasonable approach?





	Do not reject H_0	Reject H_0
H_0 is true	Correct outcome	Incorrect outcome: Type I error *Alpha*
H_0 is false	Incorrect outcome: Type II Error *Beta*	Correct outcome

Consider the data type.

Generally, we consider normal parametric t-tests for continuous data.
The Wilcoxon Mann Whitney test is suitable for ordinal or continuous data.

Consider the test statistic.

- What is most representative or relevant?
- It could be reasonable to transform random variables, *e.g.*, testing the geometric mean of a log-normal distribution using a normal t-test.
- Paired t-test (equivalent to a one sample t-test on the difference) assumes that the difference is approximately normal.

Consider prior data. What is the robustness of the proposed test? What is the impact on significance level and power?

- Examine prior data (the exploratory data analysis approach).
- Examine the literature for prior data and/or evaluations of candidate tests.
- Carry out simulation experiments on candidate tests.
- T-tests are quite robust if sample size is not too small and skewness not too big.

Nunez-Alvarez & Crespo Cintas 2010; Nahm 2016; Zhang *et al*. 2023

Nonparametric vs. parametric t-tests
Robustness

Power=1 – Beta
Power: 80%, Type II error: 20%

	Do not reject H_0	Reject H_0
H_0 is true	Correct outcome	Incorrect outcome: Type I error *Alpha*
H_0 is false	Incorrect outcome: Type II Error *Beta*	Correct outcome

A false positive
False significant outcome

A false negative
False nonsignificant outcome

Significance level=Alpha
Significance level: 0.05, Type I error: 5%

Robustness
The extent to which violation of its assumptions does not significantly affect or change the probability of its Type I error. Also relates to Type II error.

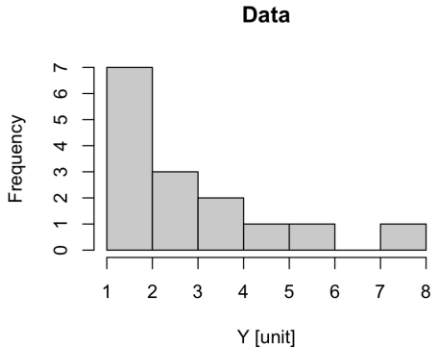
One way of looking at this is relative power of two tests.

Simulating the relative power of two tests

i. **specify a model and effect** distributional assumption and biologically/clinically/etc. relevant effect one would regret missing.
ii. **generate observations**
iii. **Test H_0**
iv. **Repeat and calculate power** Power is the proportion of times where the test actually gave a significant result.

Hunter & May 1993; Sawilowsky 1990; Larsen & Marx 2007; Fadeyi 2021; Wickelmaier 2022; Tian 2022;

Nonparametric vs. parametric t-tests
Simulating the relative power of two tests



Simulating the relative power of two tests

i. **specify a model and effect** The variable Y is best described assuming a log-normal distribution. We would like to detect a difference in geometric means of 0.55 [units] given the variability in Y in two groups Y_C and Y_T.

ii. **generate observations** We generate 15·2 observations per test.

iii. **Test H_0** We test two models.
Unpaired t-test H_0: $\mu_{Y,C} = \mu_{Y,T}$
Wilcoxon Mann Whitney test H_0: $med_{Y,C} = med_{Y,T}$

iv. **Repeat and calculate power** We repeat 10,000 times and calculate the power of the two tests. The proportion of significant outcomes per test is the power $(1 - \beta)$.

```
set.seed(34567)
# Let's test a hypothetical uniform dataset
# We would like to test mean/median difference using:
# t-test & Wilcoxon Mann Whitney test
# log-normal distribution
x_contr <- 0.4
sd_contr <- 0.49
n_control <- 15

x_exp <- 0.95
sd_exp <- 0.51
n_experiment <- 15

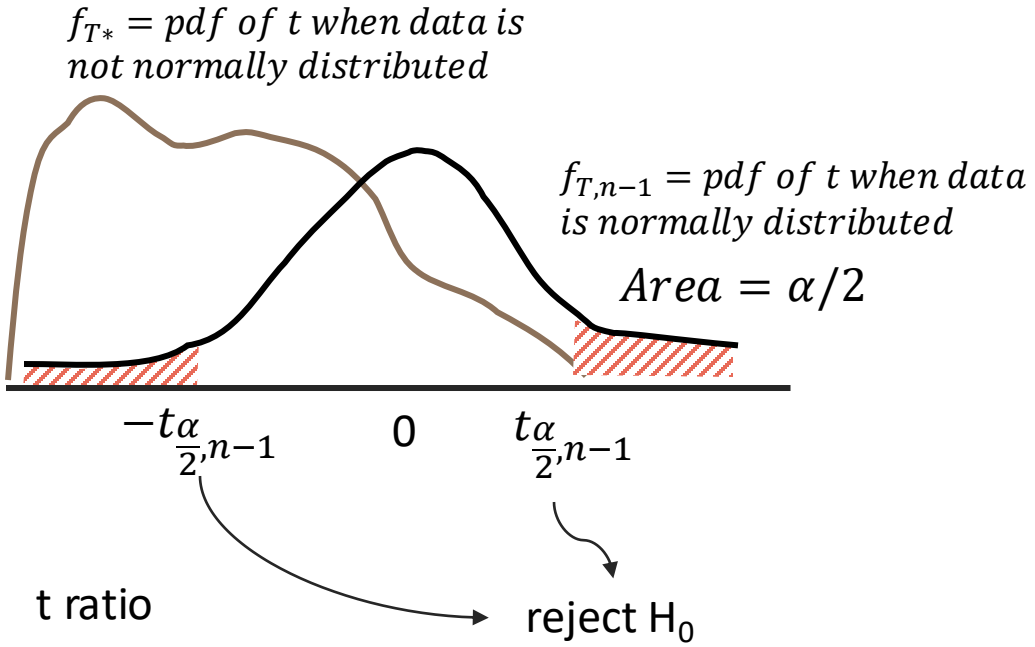
pval_wmw <- replicate(10000, {
  y_contr <- rlnorm(n_control, x_contr, sd_contr)
  y_exp <- rlnorm(n_experiment, x_exp, sd_exp)
  wilcox.test(y_contr, y_exp)$p.value
})
mean(pval_wmw < 0.05)
set.seed(34567)
pval_ttst <- replicate(10000, {
  y_contr <- rlnorm(n_control, x_contr, sd_contr)
  y_exp <- rlnorm(n_experiment, x_exp, sd_exp)
  t.test(y_contr,y_exp, alternative = "two.sided", paired = FALSE,
var.equal = FALSE)$p.value
})
mean(pval_ttst < 0.05)
```

Power $(1 - \beta)$

Wilcoxon Mann Whitney: 0.80

t-test: 0.78

The normality assumption
Simulation experiments of α



$$t = \frac{\bar{Y} - \mu_0}{S/\sqrt{n}}$$

If the normality assumption is not true:

$$P\left(t \leq -t_{\frac{\alpha}{2},n-1}\right) + P\left(t \geq t_{\frac{\alpha}{2},n-1}\right) \neq \alpha$$

$$\alpha = \int_{-\infty}^{-t_{\frac{\alpha}{2},n-1}} f_{T*}(t)dt + \int_{t_{\frac{\alpha}{2},n-1}}^{\infty} f_{T*}(t)dt$$

If alphas differs a lot, then significance may not hold true.

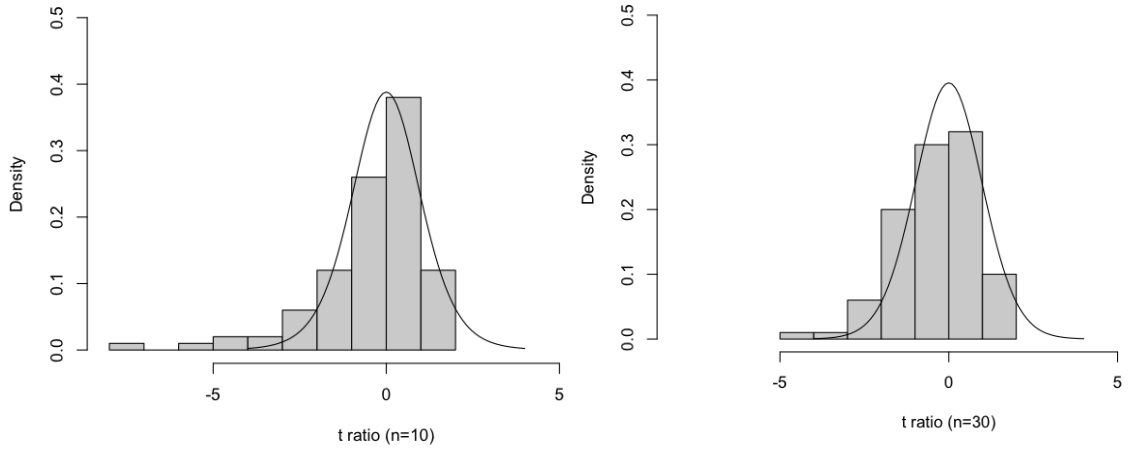
Larsen & Marx 2007

The normality assumption
Simulation experiments of α

Simulating the sensitivity of the t–ratio to violations of the normality assumtion

We have an idea of an approximate distributional model

i. **specify a model** The variable *Y* is best described assuming a given distributional model (we do not know this to be true).
ii. **generate data** Generate n samples from the distributional model.
iii. **Calculate the t-ratio** Calculate for the n samples of Y.
iv. **Plot histogram** Plot the histogram of the t-ratios.
v. **Plot pdf** Plot the pdf of f_{Tn-1}.
vi. **Compare** Compare the deviation of the histogram and pdf.




```r
## SIGNIFICANCE
set.seed(34567)
# Let's say we have a variable that we assume to be of approximately
an exponential
# distribution. We can sample from the exponential:
n_repeat <- 100
n_samples <- 30
gamma_y <- 1

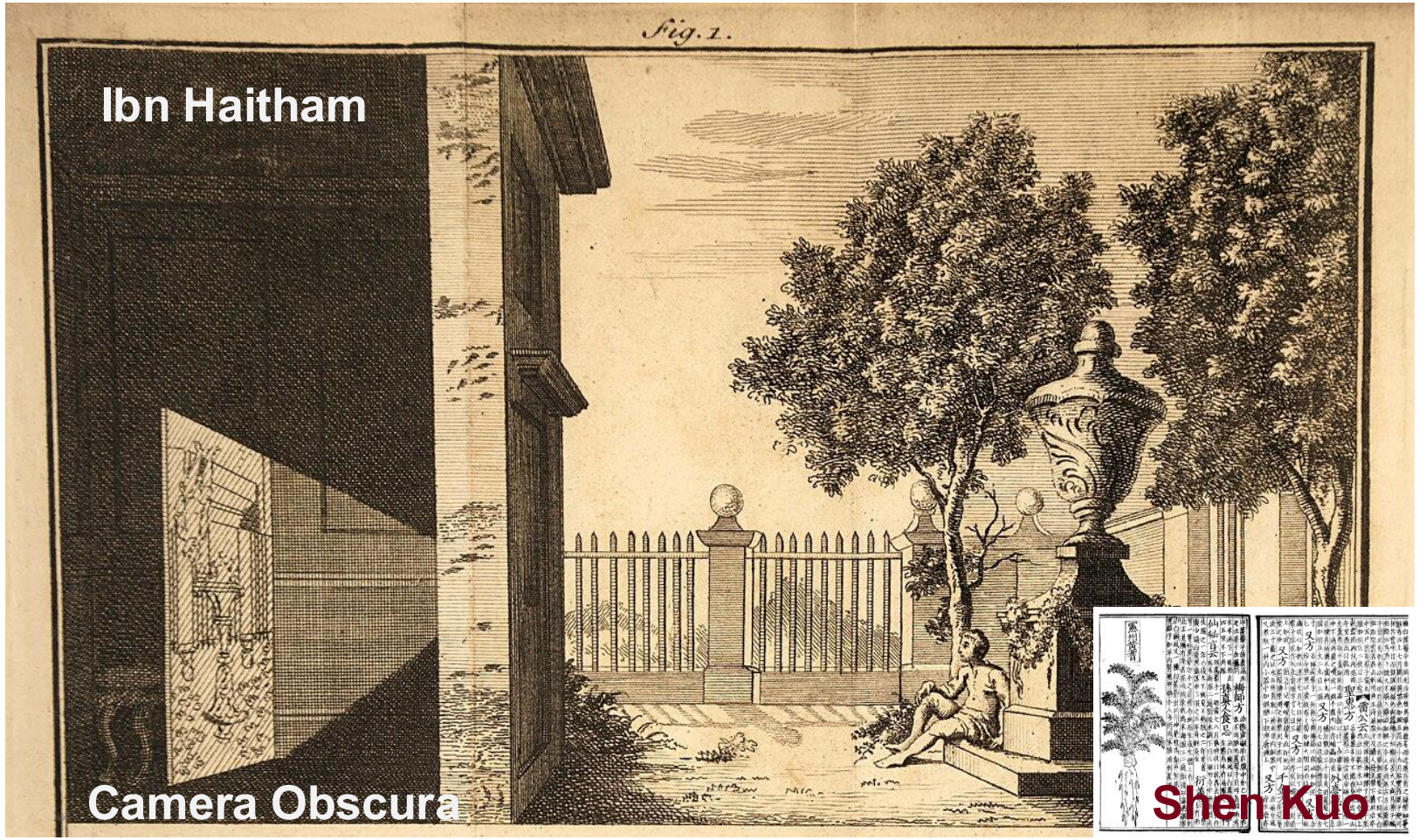
# we calculate the t-ratio of 100 replicates
t_y <- replicate(n_repeat, {
  y_dat <- rexp(n_samples, rate = gamma_y)
  x_y <- mean(y_dat)
  sd_y <- sd(y_dat)
  t_y <-(x_y - gamma_y)/(sd_y/sqrt(n_samples))
})

# Plot the histogram and overlay the theoretical pdf
df_y <- n_samples - 1
x_tdist <- seq(-4,4,0.1)
tdist_pdf <- dt(x_tdist, df_y)
hist(t_y, freq = FALSE, xlab = "t ratio (n=30)", ylab = "Density",
main = "", xlim = c(-8,8), ylim = c(0,0.5))
lines(x_tdist, tdist_pdf)
```

Alpha (α)

The t-test is robust.
The t distribution is relatively insensitive as long as *n* is not too small and the data is not too skewed.

L1-8 Study Design (I)



BRITISH MEDICAL JOURNAL

LONDON SATURDAY SEPTEMBER 30 1950

SMOKING AND CARCINOMA OF THE LUNG

PRELIMINARY REPORT

BY

RICHARD DOLL, M.D., M.R.C.P.

Member of the Statistical Research Unit of the Medical Research Council

AND

A. BRADFORD HILL, Ph.D., D.Sc.

Professor of Medical Statistics, London School of Hygiene and Tropical Medicine; Honorary Director of the Statistical Research Unit of the Medical Research Council

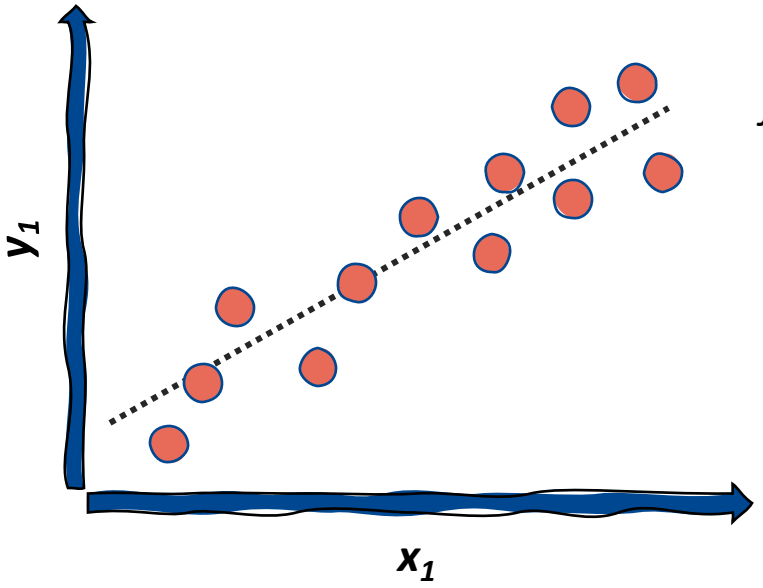
In England and Wales the phenomenal increase in the number of deaths attributed to cancer of the lung provides one of the most striking changes in the pattern of mortality recorded by the Registrar-General. For example, in the quarter of a century between 1922 and 1947 the annual number of deaths recorded increased from 612 to 9,287, or roughly fifteenfold. This remarkable increase is, of course, out of all proportion to the increase of population—both in total and, particularly, in its older age groups. Stocks (1947), using standardized death rates to allow for these population changes, shows the following trend: rate per 100,000 in 1901–20, males 1.1, females 0.7; rate per 100,000 in 1936–9, males 10.6, females 2.5. The rise seems to have been particularly rapid since the end of the first world war; between 1921–30 and 1940–4 the death rate of men at ages 45 and over increased sixfold and of women of the same ages approximately threefold. This increase is still continuing. It has occurred, too, in Switzerland, Denmark, the U.S.A., Canada, and Australia, and has been reported from Turkey and Japan.

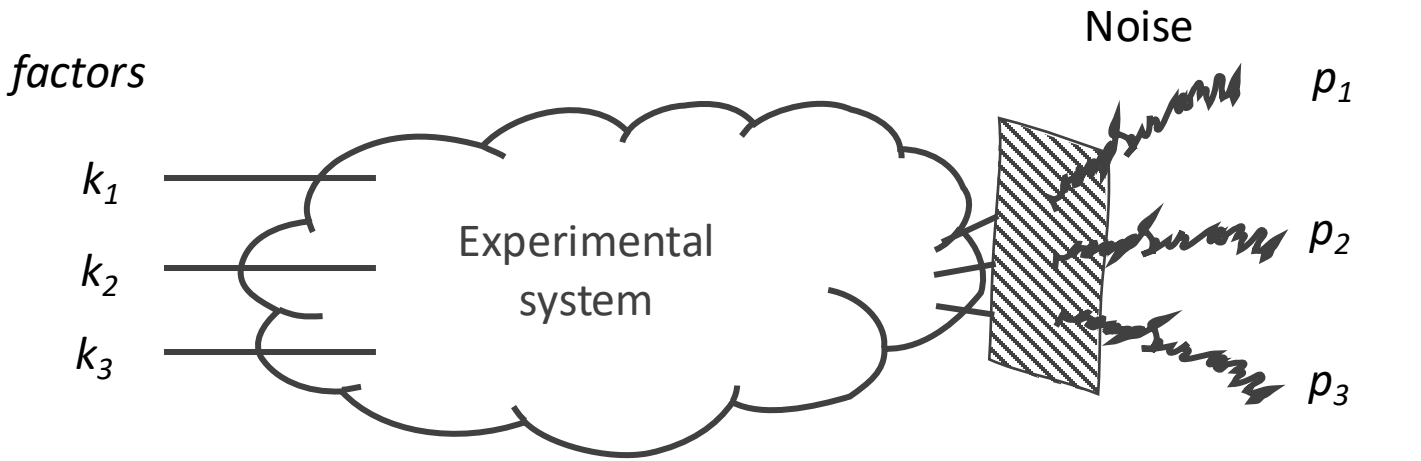
Many writers have studied these changes, considering whether they denote a real increase in the incidence of the disease or are due merely to improved standards of diagnosis. Some believe that the latter factor can be regarded as wholly, or at least mainly, responsible—for example, Willis (1948), Clemmesen and Busk (1947), and Steiner whole explanation, although no one would deny that it may well have been contributory. As a corollary, it is right and proper to seek for other causes.

Possible Causes of the Increase

Two main causes have from time to time been put forward: (1) a general atmospheric pollution from the exhaust fumes of cars, from the surface dust of tarred roads, and from gas-works, industrial plants, and coal fires; and (2) the smoking of tobacco. Some characteristics of the former have certainly become more prevalent in the last 50 years, and there is also no doubt that the smoking of cigarettes has greatly increased. Such associated changes in time can, however, be no more than suggestive, and until recently there has been singularly little more direct evidence. That evidence, based upon clinical experience and records, relates mainly to the use of tobacco. For instance, in Germany, Müller (1939) found that only 3 out of 86 male patients with cancer of the lung were non-smokers, while 56 were heavy smokers, and, in contrast, among 86 "healthy men of the same age groups" there were 14 non-smokers and only 31 heavy smokers. Similarly, in America, Schrek and his co-workers (1950) reported that 14.6% of 82 male patients with cancer of the lung were non-smokers, against 23.9% of 522 male patients admitted with cancer of sites other than the upper respiratory and digestive

The experimenter's problem and statistical methods
implications on study design





n entities $= k_3 \times p_3$

some model function of $y \leftarrow f(x)$

$$y_1 \sim \beta_{1,1}\, x_1 + \mathcal{E}$$

$$y_2 \sim \beta_{2,1}\, x_1 + \beta_{2,3}\, x_3 + \mathcal{E}$$

β coefficient

Box GEP, *et al*. 2005

Modelling: Problem Description → Research Question → PICO(S) & Hypothesis → Experimental Design

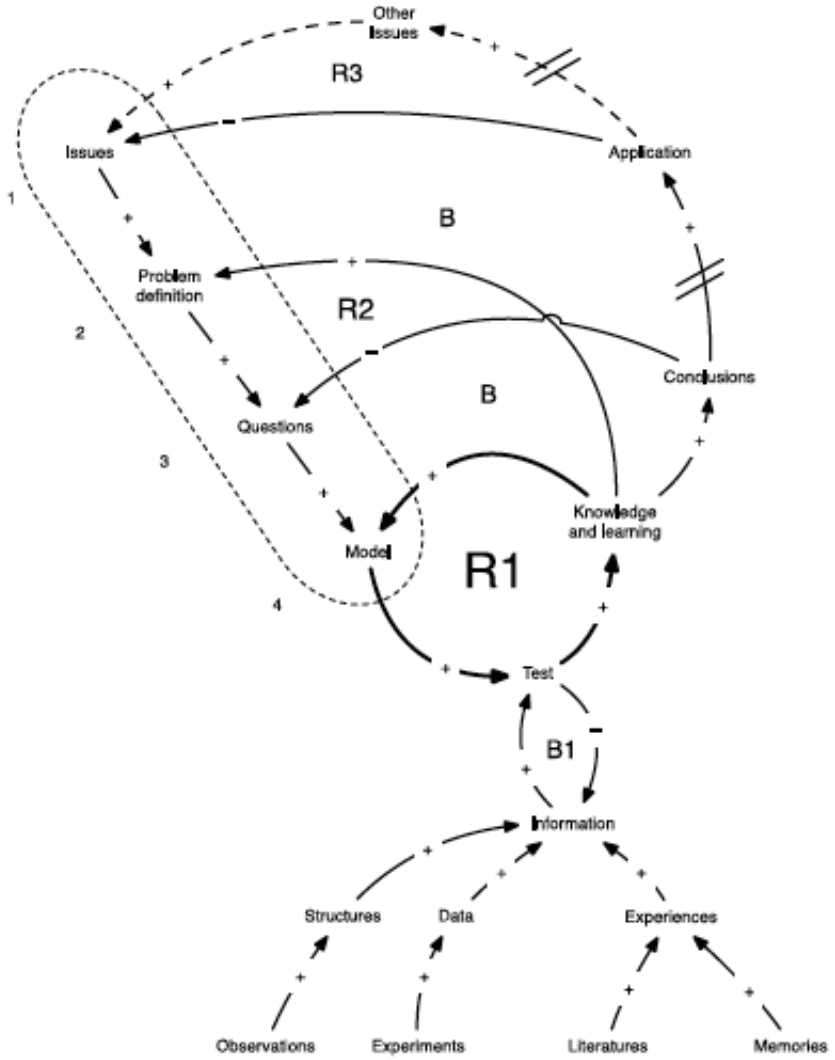
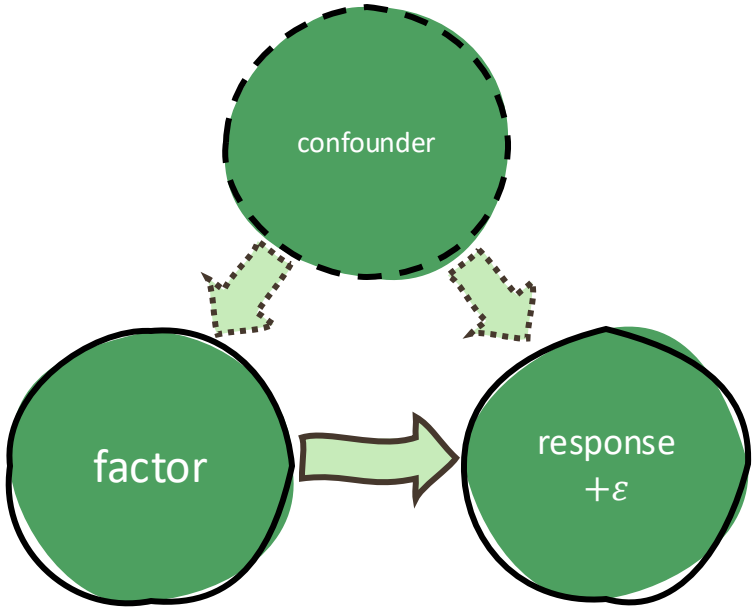


Figure 3.9. The use of extended learning loop model results in adaptive learning.



CM2014 Simulation Methods
Good modelling practices from RQ to intended use.
HL2014 Safe Medical Devices
Clinical Evidence generation.

Population/**P**atient Group

Intervention

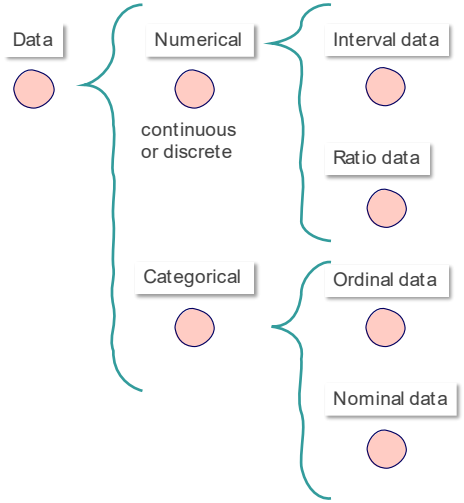
Control

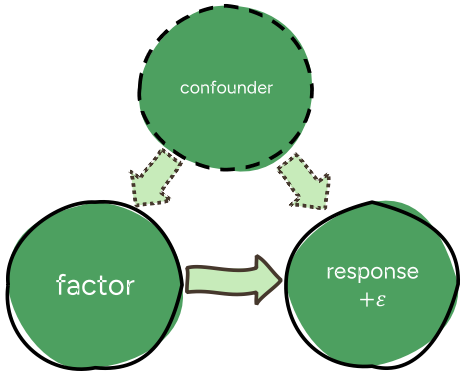
Outcome Measures

(S)tudy Design

Haraldsson & Sverdrup 2021; Darwich *et al*. 2025

Some basics of statistics-informed study design



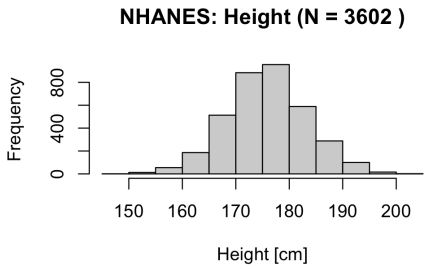


H_0: median mole count on the left arm equals the right arm

$\eta_{moles,left} = \eta_{moles,right}$

H_1: median mole count on the left arm differs from the right arm

$\eta_{moles,left} \neq \eta_{moles,right}$

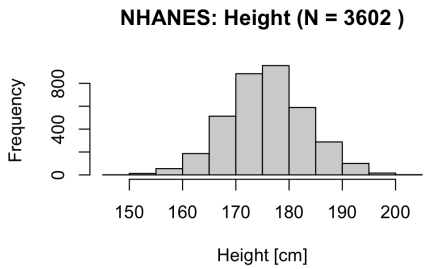


Consider the data type and the test statistic

Reduce the risk of confounders by controlling and randomising

Hypothesis testing is a framework for study design

Reasoning, prior data and statistics can save experiments and studies



Reasoning, prior data and statistics can save experiments and studies

- Approximate distribution of the response variable.
- Reason around what constitutes a meaningful change in the response.
- Propose statistical analysis.
- Investigate the ability to detect change.
- Investigate the impact of study power and sample size.

Clozapine-induced tachycardia

EudraVigilance - European database of suspected adverse drug reaction reports

Contacts | English

Home | About | Understanding reports | Search | Medicine safety | Switch to Veterinary

Online access to suspected side-effect reports



On this website you can view data on suspected side-effects, also known as suspected adverse drug reactions, for authorised medicines in the European Economic Area (EEA).

For centrally authorised medicines, access to reports is granted both by the name of the medicine or the name of the active substance. For non-centrally authorised medicines, access is granted based on the name of the active substance only.

Search for a report

Search here for suspected adverse drug reaction reports

COVID-19 vaccines important messages

To consult the reports for COVID-19 vaccines, follow this link, then click on the letter 'C' and scroll down until "COVID-19".

How to report a side-effect

Key information

- The information on this website relates to ***suspected* side effects** , i.e. medical events that have been observed following the use of a medicine, but which are **not necessarily related to or caused by the medicine**.
- Information on suspected side effects **should not be interpreted** as meaning that the medicine or the active substance causes the observed effect or is **unsafe to use**. Only a detailed evaluation and scientific assessment of all available data allows for robust conclusions to be drawn on the benefits and risks of a medicine.
- The European Medicines Agency publishes these data so that its stakeholders, including the general public, can access information that European regulatory authorities use to review the safety of a medicine or active substance. **Transparency** is a key guiding principle of the Agency.

Home | Contacts | Browser compatibility and Javascript | © 2012 - 2024

EUROPEAN MEDICINES AGENCY SCIENCE MEDICINES HEALTH

EudraVigilance

NHS

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Health A to Z | NHS services | Live Well | Mental health

Home > Health A to Z > Conditions A to Z

Supraventricular tachycardia (SVT)

Supraventricular tachycardia (SVT) is a condition where your heart suddenly beats much faster than normal. It's not usually serious, but some people may need treatment.

Causes of supraventricular tachycardia (SVT)

Supraventricular tachycardia (SVT) happens when the electrical system that controls your heart rhythm is not working properly.

This causes your heart to suddenly beat much faster. It can then slow down abruptly.

A normal resting heart rate is 60 to 100 beats per minute (bpm). But with SVT your heart rate suddenly goes above 100bpm. This can happen when you're resting or doing exercise.

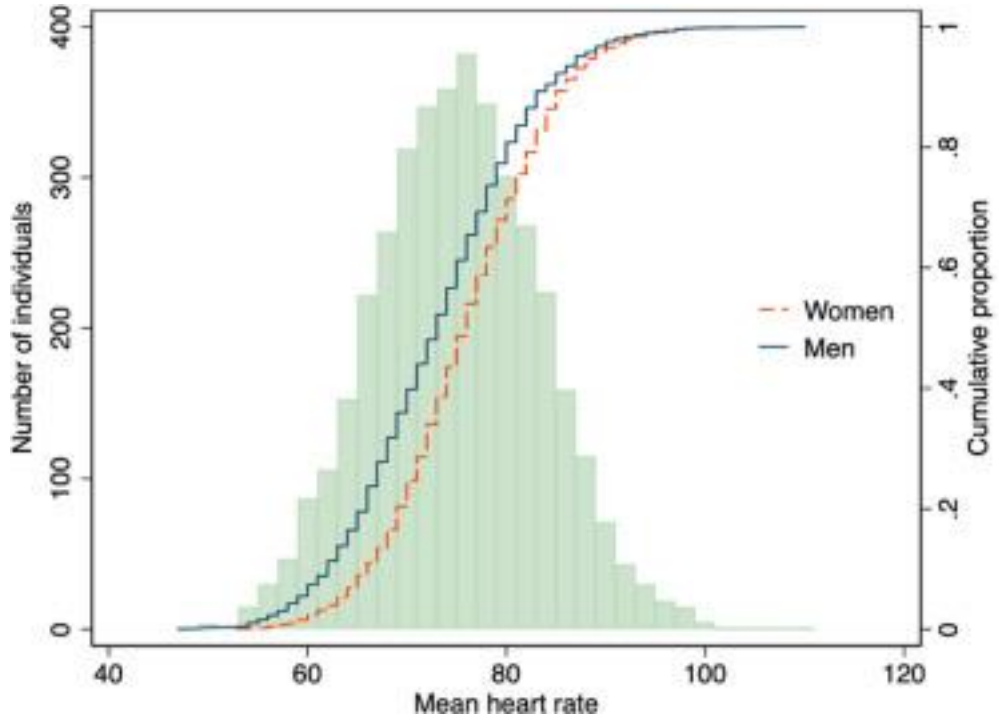
Symptoms of supraventricular tachycardia (SVT)

Having supraventricular tachycardia (SVT) means your heart suddenly beats faster.

This:

- usually lasts for a few minutes, but can sometimes last for several

Clozapine-induced tachycardia



Statistic	*Value*
Mean (beats/min)	*73*
SD (beats/min)	*9*

Persson *et al*. 2024

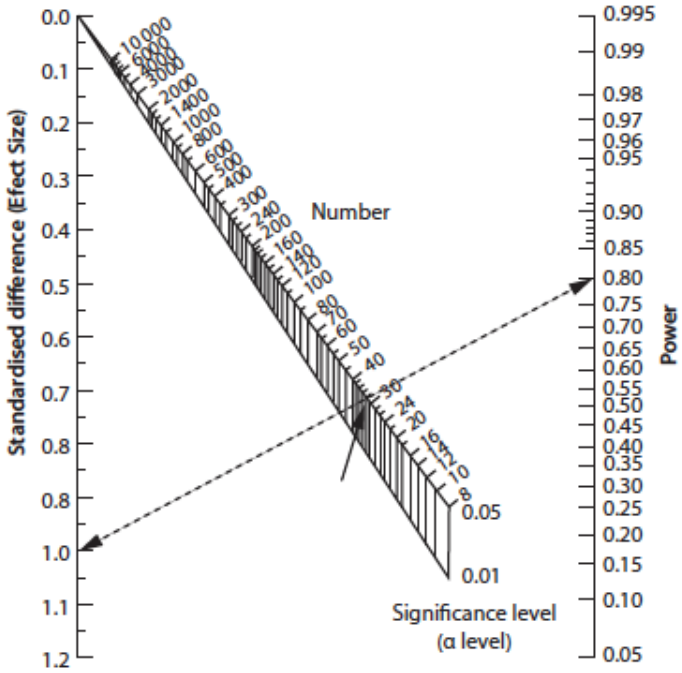


FIGURE 2. Nomogram for sample size and power, for comparing two groups of equal size. Gaussian distributions assumed. Standardized difference (effect size) and aimed power values are initially selected on the nomogram. The line connecting these values cross the significance level region of the nomogram. The intercept at the appropriate significance value presents the required sample size for the study. In the above example, for effect size = 1, power = 0.8 and alpha value = 0.05, the sample size is found to be 30. (Adapted from reference 16).

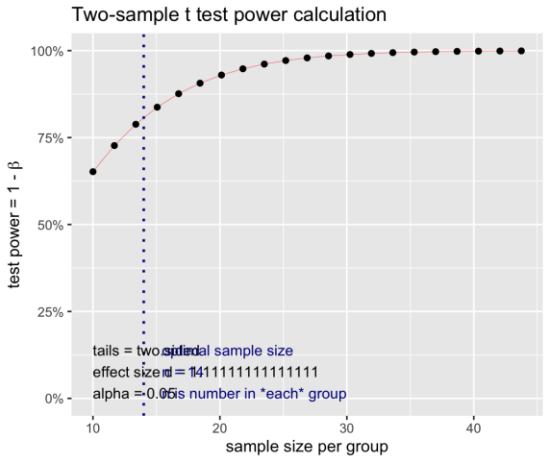
Serdar *et al*. 2021

Clozapine-induced tachycardia
effect size, power and sample size

Group	Mean hear rate (X) bpm	Standard Deviation (SD) bpm	Sample size (N)
Control	73	9	20
Clozapine	80	9	20

$$\sigma_{pooled} = \sqrt{\frac{\sigma_T^2 + \sigma_C^2}{2}} = 9$$

$$Cohen's\ d = \frac{\bar{x}_T - \bar{x}_C}{\sigma_{pooled}} = 1.1$$



Or we can use a simulation-based approach.

```r
set.seed(34567)
library(pwr) # load the library pwr: https://cran.r-project.org/web/packages/pwr/pwr.pdf

## Clozapine-induced tachycardia
# According to Persson et al. 2024, heart rate as bpm is approximately normal
# The mean (sd) heart rate is 73 (9) bpm. Under that assumption we can
# sample from the theorical distribution.

# Normal sampling
x_contr <- 73
sd_contr <- 9
nc_samples <- 1000
bpm_contr <- rnorm(nc_samples, x_contr, sd_contr)

# Data is continuous and approximately normal, suggesting a t-test.
# Tachycardia means a bpm of 100, but not the whole day.
# For simplicity, let's say we want to detect a difference 10 bpm.

x_treat <- 83
sd_treat <- 9

# calculate Cohen's D effect size
sd_pooled <- sqrt((sd_treat^2+sd_contr^2)/2)
es_d <- (x_treat - x_contr)/sd_pooled

# calculate power over sample size
p.t.two <- pwr.t.test(d = es_d, power = 0.80, type = "two.sample", alternative = "two.sided")
plot(p.t.two, xlab = "sample size per group")
```

Champely *et al*. 2020

A Frank Statement to Cigarette Smokers

RECENT REPORTS on experiments with mice have given wide publicity to a theory that cigarette smoking is in some way linked with lung cancer in human beings.

Although conducted by doctors of professional standing, these experiments are not regarded as conclusive in the field of cancer research. However, we do not believe that any serious medical research, even though its results are inconclusive should be disregarded or lightly dismissed.

At the same time, we feel it is in the public interest to call attention to the fact that eminent doctors and research scientists have publicly questioned the claimed significance of these experiments.

Distinguished authorities point out:

1. That medical research of recent years indicates many possible causes of lung cancer.

2. That there is no agreement among the authorities regarding what the cause is.

3. That there is no proof that cigarette smoking is one of the causes.

4. That statistics purporting to link cigarette smoking with the disease could apply with equal force to any one of many other aspects of modern life. Indeed the validity of the statistics themselves is questioned by numerous scientists.

We accept an interest in people's health as a basic responsibility, paramount to every other consideration in our business.

We believe the products we make are not injurious to health.

We always have and always will cooperate closely with those whose task it is to safeguard the public health.

For more than 300 years tobacco has given solace, relaxation, and enjoyment to mankind. At one time or another during those years critics have held it responsible for practically every disease of the human body. One by one these charges have been abandoned for lack of evidence.

Regardless of the record of the past, the fact that cigarette smoking today should even be suspected as a cause of a serious disease is a matter of deep concern to us.

Many people have asked us what we are doing to meet the public's concern aroused by the recent reports. Here is the answer:

1. **We are pledging aid and assistance to the research effort into all phases of tobacco use and health. This joint financial aid will of course be in addition to what is already being contributed by individual companies.**

2. **For this purpose we are establishing a joint industry group consisting initially of the undersigned. This group will be known as TOBACCO INDUSTRY RESEARCH COMMITTEE.**

3. **In charge of the research activities of the Committee will be a scientist of unimpeachable integrity and national repute. In addition there will be an Advisory Board of scientists disinterested in the cigarette industry. A group of distinguished men from medicine, science, and education will be invited to serve on this Board. These scientists will advise the Committee on its research activities.**

This statement is being issued because we believe the people are entitled to know where we stand on this matter and what we intend to do about it.

TOBACCO INDUSTRY RESEARCH COMMITTEE

500 EMPIRE STATE BUILDING, NEW YORK 1, N. Y.

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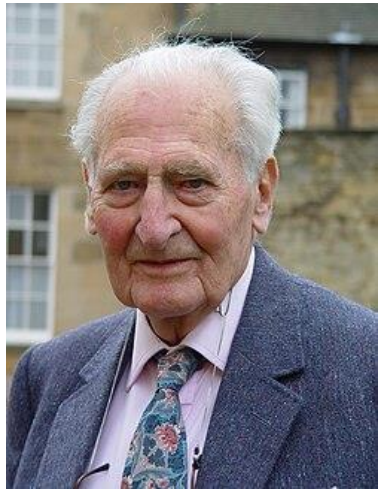
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BRITISH MEDICAL JOURNAL

LONDON SATURDAY SEPTEMBER 30 1950

SMOKING AND CARCINOMA OF THE LUNG

PRELIMINARY REPORT

BY

RICHARD DOLL, M.D., M.R.C.P.

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In England and Wales the phenomenal increase in the number of deaths attributed to cancer of the lung provides one of the most striking changes in the pattern of mortality recorded by the Registrar-General. For example, in the quarter of a century between 1922 and 1947 the annual number of deaths recorded increased from 612 to 9,287, or roughly fifteenfold. This remarkable increase is, of course, out of all proportion to the increase of population—both in total and, particularly, in its older age groups. Stocks (1947), using standardized death rates to allow for these population changes, shows the following trend: rate per 100,000 in 1901–20, males 1.1, females 0.7; rate per 100,000 in 1936–9, males 10.6, females 2.5. The rise seems to have been particularly rapid since the end of the first world war; between 1921–30 and 1940–4 the death rate of men at ages 45 and over increased sixfold and of women of the same ages approximately threefold. This increase is still continuing. It has occurred, too, in Switzerland, Denmark, the U.S.A., Canada, and Australia, and has been reported from Turkey and Japan.

Many writers have studied these changes, considering whether they denote a real increase in the incidence of the disease or are due merely to improved standards of diagnosis. Some believe that the latter factor can be regarded as wholly, or at least mainly, responsible—for example, Willis (1948), Clemmesen and Busk (1947), and Steiner whole explanation, although no one would deny that it may well have been contributory. As a corollary, it is right and proper to seek for other causes.

Possible Causes of the Increase

Two main causes have from time to time been put forward: (1) a general atmospheric pollution from the exhaust fumes of cars, from the surface dust of tarred roads, and from gas-works, industrial plants, and coal fires; and (2) the smoking of tobacco. Some characteristics of the former have certainly become more prevalent in the last 50 years, and there is also no doubt that the smoking of cigarettes has greatly increased. Such associated changes in time can, however, be no more than suggestive, and until recently there has been singularly little more direct evidence. That evidence, based upon clinical experience and records, relates mainly to the use of tobacco. For instance, in Germany, Müller (1939) found that only 3 out of 86 male patients with cancer of the lung were non-smokers, while 56 were heavy smokers, and, in contrast, among 86 "healthy men of the same age groups" there were 14 non-smokers and only 31 heavy smokers. Similarly, in America, Schrek and his co-workers (1950) reported that 14.6% of 82 male patients with cancer of the lung were non-smokers, against 23.9% of 522 male patients admitted with cancer of sites other than the upper respiratory and digestive